Slot-Chain: A Preconfirmation Protocol

Audited Design Candidate — v2.25

August 2026

Abstract

Slot-Chain assigns one-second L2 slots to bonded builders. Builders sign parent-linked blocks off chain; any party can prove and submit a batch to L1. Normal candidates compete for a fixed settlement interval. After an objective finality breach, or when a forced L1 message becomes due, anyone can prove an unsigned deterministic escape block. Recovery is an episode of renewable, objectively expiring rounds, so a prover outage does not make its target stale; as with every validity rollup, producing the next proof still requires a reconstructible canonical prestate. This revision separates proved outputs from L1 close metadata, replaces the unauthenticatable queue hash chain with a range-provable Merkle vector, makes forced-message execution Ethereum-valid, and specifies bounded admission, data, reward and migration state. It is an audited architecture candidate, not yet an implementation-ready specification or production authorization: the missing executable profile and the proving, gas, economics and adversarial gates in §14 remain mandatory.

Contents

1 Status, claims and exclusions

Normative words MUST, MUST NOTand SHOULDhave their usual RFC meaning. Where prose and either executable model disagree, deployment is blocked until the disagreement is resolved; neither source silently overrides the other.

1.1 Claims

Subject to L1 safety/liveness, a sound validity-proof system, availability of a canonical prestate reconstruction package as defined below, and—for kind-1 credits only—safety/finality of the supported source domain and its profile-authenticated, append-only Bridge registry, the protocol claims:

1. **Finalized-state safety.** L1 accepts only a proved state transition extending the stored canonical tuple. Builder signatures never substitute for execution validity.
2. **Bounded protocol recovery.** A builder cartel, an absent aggregator, an empty builder registry, and missing standbys cannot prevent an unsigned escape candidate from advancing the canonical state while the canonical prestate and any unexpired forced bytes remain reconstructible.
3. **Forced transaction inclusion.** A correctly prepaid L1 envelope at the queue head is consumed by the next canonical block after it becomes due; kind 0 is executed if valid bytes are supplied before its bounded validity or consumed-and-discarded otherwise, while durable kind 1 is always applied idempotently.
4. **Permissionless landing and recovery.** No allowlist, builder key, aggregator seat or payment recipient is required to submit a proof or an escape candidate.
5. **Deterministic lookahead.** Every implementation derives the same schedule from an authenticated finalized snapshot and the byte encoding in §4.

1.2 Non-claims

- A preconfirmation is not finality or insurance. Before normal close it may be revoked by builder equivocation, an unprovable skip, a withheld body, or an escape commit. Applications must price this as probabilistic/economic assurance and publish their own exposure limit.
- A bond cannot cap aggregate off-chain promises: the protocol has no reservation ledger and a builder can sign mutually exclusive promises. `L_LEASE` is deterrence and a recovery source, not guaranteed coverage.
- Protocol liveness is a capability claim. Economic liveness additionally assumes at least one actor can fund proof generation and an L1 transaction. A payment failure never invalidates an otherwise valid canonical transition.
- Censorship by L1 for longer than `T_INCLUDE_MAX_SECONDS`, failure of L1 finality, or a broken proof system is outside the model.
- A supported source domain finality failure or unauthorized change to its profile-pinned Bridge registry is outside the kind-1 safety claim. It cannot authorize a kind-0 user envelope.
- A source message is not yet an admitted forced envelope. Any preparer must supply its full Message preimage and queue deposit to the same-L1 adapter; source emission alone is

not reported as forced-queue acceptance. If nobody enqueues before the record's bounded `enqueueBy`, permissionless source cancellation returns its ETH escrow and permanently forbids later enqueue. Launch V2 is DIRECT-only. ERC20, ERC721, ERC1155 and every official Vault flow remain byte-for-byte V1 and cannot originate a V2 credit. A future asset protocol requires a new kind, domain, release, codec and audit.

- The immutable V2 custody kernel is not self-healing. A defect in its compiled SourceBridge, DestinationBridge, inbox store, accumulator, history router, quota manager, liquidity pool or fixed Merkle verifier cannot be patched for an already-live credit without introducing an authority able to forge delivery/failure and take reserved value. This design chooses fail-closed immutability; production status therefore requires the compiled-code, formal-invariant and differential-test gates in §14, not merely this document.
- A state root is a binding commitment, not an erasure code. If every copy of the canonical state trie, executed bodies, runtime-bytecode preimages and required witness data is destroyed after Ethereum's applicable data-retention horizon, a new prover cannot reconstruct the preimage from L1 roots. Arbitrary-duration recovery after total canonical-data loss is therefore not claimed.

2 Actors and trust model

Builder.
A registered address eligible for scheduled slots. Builders may equivocate, skip parents, withhold bodies, collude, or disappear.

Aggregator.
The current auction seat, expected to collect data, prove and submit normal candidates. It has no exclusive consensus authority.

Prover/lander.
Any address. It supplies a validity proof and names an optional reward beneficiary in the proof statement. Proof validity, not identity, authorizes transition.

Canonical archive.
Any untrusted, replaceable source of canonical L2 bodies, state-trie nodes and runtime-bytecode preimages sufficient to reconstruct the current prestate. Each bytecode object is addressed by its legacy-Keccak `codeHash`; this includes code reached through proxy, beacon or delegation indirection. Content is verified against the L1-canonical block/state roots, so correctness does not require trusting the source; liveness requires at least one complete copy to remain reachable.

Forced-message sender.
Any L1 account that submits a gas-bounded, prepaid L2 transaction envelope.

Bridge source domain.
For kind 1 only at launch, settlement Ethereum, identified by chain ID, genesis hash, fresh immutable non-proxy SourceBridge, non-proxy credit registry, immutable terminal verifier and namespace. The adapter direct-reads its append-only credit record on the same L1. Registry governance is an external safety dependency; a current resolver lookup never substitutes for the emission record.

Protocol governance.
Delayed governance selects each validity verifier, execution profile and Settlement implementation. It is already a safety root: an authorized malicious verifier can finalize an

arbitrary state transition. V2 recovery adds no stronger writer. Each immutable Settlement records its own proof-bound canonical history internally; the protocol-lifetime router only registers exact version/code identities and reads those histories. It has no governor, canonical-write callback or caller-supplied-root setter.

Bridge message archive.
Any untrusted, replaceable source of the full Message bytes emitted on L1. Correctness follows from `msgHash`; successful destination execution needs one copy until processing. Loss before `enqueueBy` leads to source cancellation; loss after the L2 pin leads to permissionless destination expiry and failure. Source DIRECT ETH remains recoverable by credit ID from the SourceBridge's pull-refund record, without reconstructing Message bytes.

L1.
Ethereum provides canonical ordering, timestamps, EIP-4788 beacon roots, EIP-2935 historical block hashes, blob commitments and the KZG point-evaluation precompile.

The safety threshold contains no honest-builder assumption. Honest builders improve normal throughput and preconfirmation reliability; the unsigned escape lane provides state and forced-queue progress when all of them collude. The L1-liveness assumption includes transaction inclusion and at least 64 canonical L1 blocks becoming available within `T_DEPTH_MAX=900s`; a longer L1 stall suspends, rather than violates, the L2 recovery clock. Archive operators have no protocol key, seat or exclusive API. Anyone may mirror or serve the same content-addressed data. This is an availability assumption, not an authority assumption; it limits long-horizon cold-start recovery but does not weaken proof soundness or permissionless submission.

3 Clock domains and canonical state

Before activation, Settlement is in `PREACTIVE`; candidate submission, session open/post/seal and every other target-local state mutation are disabled, and slot arithmetic saturates. Off-chain data preparation may proceed but cannot create a target session. L2 slot zero is fixed by `GENESIS_TIMESTAMP`, with

`currentL2Slot=max(0,block.timestamp-GENESIS_TIMESTAMP).`

L2 slot and every deadline are timestamp seconds. L1 confirmation depth is counted in L1 block numbers. Beacon slot is used only in schedule authentication. Implementations MUST NOTsubstitute one clock for another. `settlementChainId` is the L1 chain executing Settlement and domains every L1 contract commitment; `l2ChainId` is the EVM chain ID enforced for L2 transactions and execution. They are independent immutable launch constants and are both explicit in every cross-domain proof statement and builder header.

The L1 contract stores a proved core and local L1 metadata:

```
CanonicalCore = (l2BlockNumber, tipHash, tipSlot, stateRoot, messageCursor,
                 winningDataCommitment, nextBaseFee, nextExcessBlobGas,
                 terminalRoot, terminalCount)
Canonical = (CanonicalCore core, uint64 canonicalizedAtBlock)
```

The proof binds `baseCanonicalHash`, the hash of both stored objects, but outputs only a new `CanonicalCore`. Settlement compares every base field with storage, verifies the proof, writes

the explicit proved core, and only then sets `canonicalizedAtBlock=block.number`. A prover is therefore never asked to predict the L1 block in which its transaction will land. No code attempts to read or store `blockhash(block.number)`, which does not exist during that block.

`l2BlockNumber` is the canonical EVM height, distinct from the one-second slot because slot gaps are legal. A candidate with `blockCount=n` proves `endL2BlockNumber=base.l2BlockNumber+n`; every executed header uses the next consecutive EVM block number. While the deployed `ICheckpointStore` uses `uint48`, launch and every transition additionally require `endL2BlockNumber<=UINT48_MAX`; changing that interface and bound is a protocol-version upgrade. The canonical event always emits `(l2BlockNumber,tipHash,stateRoot,terminalRoot,terminalCount)`. The proof constrains the latter pair to the exact post-state slots of the protocol-lifetime `TerminalAccumulatorV2`; Settlement never accepts them as unverified auxiliary calldata. In the same internal storage transition that writes `Canonical`, Settlement increments a checked `uint64canonicalSequence` and writes the complete sequence-tagged canonical read tuple to its 256-cell history ring. A second canonical transition requires `block.number>lastCanonicalCommitBlock`; it cannot overwrite two cells in one L1 block. PREACTIVE and FROZEN instances cannot execute any canonical path. Recording this ring is an internal write, not an external router call, reward dependency or best-effort publication. `nextBaseFee` and `nextExcessBlobGas` are the exact fork-rule outputs needed to construct the next header without retrieving a prunable historical parent body/header. The proof derives them from the last executed header; Settlement stores them but does not calculate fork arithmetic.

A candidate has one prior-block anchor. Normal activation stores the hash of its earlier arm block using native `blockhash`; candidate submissions hash the supplied execution header to that stored value and use MPT proofs, including the historical forced-queue root/count. EIP-2935 is used later to authenticate that arm hash for equivocation evidence. A recovery round instead stores `(block.number-1,blockhash(block.number-1))`, directly stores the current queue root/count, and requires that anchor to reach depth `F_L1`. The EVM does not expose the parent timestamp. At candidate submission Settlement hashes and parses the supplied RLP parent header, requiring its number/hash to equal the frozen pair; its state root and timestamp become verifier-statement fields but are not separately frozen scalars. Current queue state is never claimed to be part of the parent's historical state root. An L1 reorg rolls the stored canonical state and anchor back together. Initial values and bridge cursors are migration inputs in §12.

Every canonical or ordinary new-work state-mutating entry point begins with `sync()`, except the exact DataSession open/post/seal sidecars, claims, surplus sweep and mode-specific maintenance frozen in §6. If `sync()` changes mode or commits a mature normal candidate, a wrapper that uses this rule MUSTreturn `SYNCED` successfully before parsing or validating the caller's candidate. The caller resubmits against the new version. This rule is necessary because a later Solidity revert would otherwise roll back the sync transition in the same transaction.

4 Builder registry and exact lookahead

4.1 Registry lifecycle

A registration contains a nonzero address, a `uint192` bond not exceeding `BOND_MAX`, a monotonic `uint64registrationIndex`, an `effectiveL2Slot`, and separately reserved `L_LEASE[window]` tranches. Address zero is permanently `VACANT`. Duplicate live addresses

and reuse while any prior generation remains active or liable are forbidden. Once the old generation's last evidence/reorg deadline has passed, all tranches are terminal, its liability leaf is safely cleared and no evidence can still land, the address may register again with a fresh `registrationIndex`. No permanent address tombstone mapping exists. With `currentWindow=floor(currentL2Slot/384)`, a reservation is accepted only for `currentWindow<=W<=currentWindow+MAX_TRANCHE_AHEAD_WINDOWS`, where the initial ahead bound is 16; an exit request permanently disables new reservations for that generation. Only a present *active* generation with no exit request and no tombstone may execute `FREE->RESERVED`; a replacement or exit atomically closes reservations before moving the old generation into liability. Liability records may only be slashed or released, never extended with a new reservation.

The active table has exactly 64 indexed cells. On accepted registration, `effectiveL2Slot= currentL2Slot+384*ENTRY_DELAY_WINDOWS`; the stored value, not a later recomputation from a window boundary, is authoritative. Thus a registration becomes effective only after `ENTRY_ DELAY_WINDOWS=8`. If the table is full it may replace only the smallest-bond effective live entry, breaking ties by greatest registration index, and only with a strictly greater bond. At most `MAX_ GENERATION_MOVES_PER_WINDOW=4` replacements or effective exits are processed. Exit requests receive a monotonic sequence and mature at `requestWindow+EXIT_DELAY_WINDOWS`. Every replacement path first processes the oldest matured exits by `(matureWindow,exitSequence)` within the window's remaining movement budget; if any matured exit remains, replacement rejects. Anyone, including the exiting builder, may run this bounded maintenance. Thus transaction ordering cannot starve an eligible exit. A displaced generation is moved, with its tranche root and bond, into the fixed `MAX_LIABILITY_GENERATIONS=1,072` liability ring; an occupied ring cell is never overwritten before its release predicate holds. Registration therefore rejects, rather than losing slashable history, when the replacement budget or ring is full.

The six-level `registryRoot` over the 64 active cells commits each cell's mutable tranche root and therefore changes on reservation transitions. It is used only by schedule sealing. A separate eleven-level `admissionRoot` over 2,048 fixed positions contains stable generation identity for the 64 active and 1,072 liability records plus canonical empty padding; its leaves do *not* contain tranche roots. Only generation admission, movement, tombstoning, slashing or release increments `admissionVersion` and updates `admissionRoot`. A `FREE/RESERVED/LIABLE` tranche transition updates the tranche and registry roots but cannot invalidate signed blocks or proofs by changing the admission pair. The admission root retains displaced scheduled keys. Normal activation and every recovery round freeze the pair `(admissionVersion,admissionRoot)`. Each scheduled signer carries an inclusion proof for a non-tombstoned record effective at that L2 slot; a slash cannot retroactively invalidate an already frozen window. Outside an already-open normal window or live recovery round, a tombstone makes sealed future tickets at or after its effective slot return `VACANT`; inside that frozen candidate context the stored admission pair remains authoritative until close/commit/expiry.

Admission positions are canonical: active cell i is always position i for $0 \leq i < 64$; liability ring slot j is always position $64 + j$ for $0 \leq j < 1,072$. A generation move uses `j= movementSequencemod1072`. The transaction first proves/releases the old occupant, writes the moved generation at that same slot, clears its active cell or writes the replacement, increments `movementSequence`, then updates both affected roots and increments `admissionVersion`; the new admission pair is published atomically. Reservation-only mutations publish only the registry root and leave that pair byte-for-byte unchanged. There is no first-free search, compaction or caller-chosen liability position.

Eligibility is evaluated before ranking. A cell's effective L2 slot must be no later than the source-derived L2 slot, its fully reserved tranche for W must be proven, and its tombstone-effective L2 slot must be later than the source. All 64 registry cells are supplied; only present cells supply tranche-ring proofs at `Wmod512`. A canonical `EMPTY` tranche leaf is valid input and makes the cell ineligible; it need not claim window W. A nonempty leaf with a different stored window is likewise ineligible, not a seal revert. Only an exact `RESERVED` leaf with stored window W and the full `L_LEASE` amount is eligible. The 64 eligible entries with greatest bond, then smallest registration index, form the snapshot set. Supplying proofs only for purported winners is invalid because it cannot prove omission-free ranking.

An exit request becomes eligible for processing only after `EXIT_DELAY_WINDOWS=MAX_LIVE_WINDOWS`; it may wait behind the same bounded FIFO of older matured exits, and the bond remains escrowed until every generation and tranche is releasable. A tranche follows `FREE->RESERVED(W)->LIABLE(W)->RELEASED` or `RESERVED/LIABLE->SLASHED`. Every candidate block slot must be at least the normal window's frozen minimum (or the recovery round start), not merely its tip. Let `windowEndSlot(W)=384*(W+1)-1`. Release requires `windowEndSlot(W)<globalMinReferencedSlot`, no stored best or active round references it, and the absolute evidence and L1-reorg deadlines have expired. At that point maintenance atomically marks the schedule cell `EXPIRED` and may release its tranches; an unexpired sealed record is never overwritten. `evidenceDeadline(W)=GENESIS_TIMESTAMP+384*(W+1)+EVIDENCE_DELAY_SECONDS`; `evidenceReplayDeadline(W)=evidenceDeadline(W)+REORG_MARGIN_SECONDS` is both the contract's last admissible evidence time and the tranche's absolute release boundary. Evidence intended to survive a reorg must first be submitted by `evidenceDeadline(W)`. `globalMinReferencedSlot` is the minimum of the current dynamic floor, an open normal window's frozen floor, an active round's start and the first slot of any stored best (absent values are ignored). Slot indices are never compared with timestamps or window numbers.

The liability-capacity proof is a launch invariant, not an assertion by inspection. A generation moved in window C can have no reservation beyond `C+MAX_TRANCHE_AHEAD_WINDOWS`. Therefore

```
MAX_LIABILITY_RESIDENCE_WINDOWS =
  MAX_TRANCHE_AHEAD_WINDOWS + 1 +
  ceil((EVIDENCE_DELAY_SECONDS + REORG_MARGIN_SECONDS) / 384) + 2
```

must be strictly less than `MAX_LIVE_WINDOWS=268`. The two extra windows cover boundary ordering and bounded maintenance. At four moves per window, the ring position reused 268 windows later is consequently releasable. Each generation maintains an exact unreleased-tranche counter, so release and reuse do not scan 512 leaves.

4.2 Authenticated snapshot

For aligned window W, let its start timestamp be `GENESIS_TIMESTAMP+384W`. The target is the greatest `BEACON_GENESIS_TIME+12k` not later than eight L1 epochs before that start; L2-genesis alignment is irrelevant. Define `sealDeadline(W)=windowStart-H_LOOK`. A seal is valid only when its carrier execution block has `F_FINAL` L1-block depth and `block.timestamp<sealDeadline(W)`. Beacon slots are not used as a surrogate for execution-block depth. At or after the deadline an unsealed window is permanently all-`VACANT`; a late transaction cannot change it. The contract performs this procedure:

1. The contract itself, not caller data, queries EIP-4788 for $j = 1, \ldots, 64$ at `q=targetTimestamp+12j`; missing timestamps revert and are skipped. The first successful query identifies the *carrier* execution block. EIP-4788 returns its parent beacon-block root, not the carrier root.
2. For the first successful query the caller supplies the carrier execution header. EIP-2935 authenticates its hash and block number; the header timestamp equals q and its `parent_beacon_block_root` equals the EIP-4788 result. Fork-specific SSZ branches in that parent authenticate its slot and one execution payload's `blockHash`, `stateRoot`, `prevRandao` and timestamp. The parent slot must be at or before target, its payload timestamp must equal its beacon-slot time, and its payload block hash must equal the carrier header's parent hash. No carrier observed by all 64 contract calls means all `VACANT`. If any call succeeds, a malformed header, a later parent, or a fork/gindex mismatch reverts without recording state; adversarial witness bytes can never manufacture a vacant seal.
3. Verify Merkle–Patricia storage proofs from that execution state root for the registry header and `registryRoot`. The caller supplies all 64 canonical serialized cells, including empty cells; recompute their six-level Keccak tree and require equality to `registryRoot`. Filter eligibility and order the entries, then store the byte-exact 64-leaf `entryRoot` and seed. Every *present* active cell supplies its tranche leaf and nine-sibling proof at index `Wmod512`. An absent registry cell has `trancheRoot=0`, supplies no tranche proof and is synthesized ineligible. For a present cell, canonical `EMPTY`, `FREE`, or a nonempty leaf for a different stored window is valid but ineligible; only exact `RESERVED(W)` with `amount=L_LEASE` is eligible. Proofs occur in ascending cell-index order, and extra/missing proofs reject. The fixed cell count proves completeness with one MPT path rather than 64 independent storage paths. The carrier satisfies `currentExecutionBlock-carrierBlockNumber>=F_FINAL`; the source payload is necessarily older.

`sealWindow` is permissionless. An optional maintenance distributor may pay its event in a later transaction. A missing seal fails closed to `VACANT`; it cannot give an address an unauthorized ticket. The snapshot age plus 64-slot carrier scan, finality and seal margin fits the eight-epoch geometry and remains far below EIP-4788's 8,191-slot history window. Consensus constants include the Deneb SSZ generalized indices; an L1 fork that changes them requires an explicitly registered verifier, never runtime guessing.

For Deneb, Electra and Fulu, the generalized index from `BeaconBlock` root to `slot` is 8 and to `body.execution_payload` is 201. Within the composed tree, payload `state_root`, `prev_randao`, `timestamp` and `block_hash` are 6,434, 6,437, 6,441 and 6,444 respectively. The seal input names the L1 fork digest; only these configured digests are accepted. The protocol-lifetime ScheduleOracle holds a one-shot append-only `forkDigest->(verifier,codeHash,configHash,gasLimit)` registry controlled only by a delayed ProtocolVersionManager manifest. Seal dispatch uses a bounded `STATICCALL`, exact return length and rechecked `EXTCODEHASH`; failure resolves VACANT and cannot mutate an old seal. Gloas or any fork that changes the container requires a reviewed verifier, constants, fixtures and delayed registration before its activation. That registration may change witness parsing only: it cannot change the protocol-lifetime seed, eligibility, allocation or placement rules.

4.3 Consensus byte encoding and allocation

All hashes use Ethereum legacy Keccak-256. Concatenation below is packed fixed width; strings are the literal ASCII bytes without a length prefix:

$$
\begin{aligned}
seed_W &= H(\texttt{"slot-chain-seed-v2"} \| u256(settlementChainId) \\
&\quad \| u256(W) \| bytes32(prevRandao)), \\
qTie_i &= H(\texttt{"slot-chain-quota-tie-v1"} \| seed_W \| u64(regIndex_i)), \\
sTie_{p,i} &= H(\texttt{"slot-chain-slot-order-v1"} \| seed_W \| u16(p) \| address20(i)).
\end{aligned}
$$

Starting from zero, capped D'Hondt awards each of 384 tickets to the unsaturated entry maximizing `bond/(allocation+1)`, with `Q_MAX=76`. Quotients are compared by cross multiplication; a `uint192` bond times `Q_MAX+1` cannot overflow `uint256`. Equal quotients use smallest `qTie`. Remaining tickets are `VACANT`; six addresses are required to fill a window.

Placement processes positions in order. A real address is feasible if it has a ticket remaining, differs from the preceding real address, and removing it preserves `remaining[a] <=otherRemaining+1` for every real address. Choose the feasible value with smallest `sTie`; `VACANT` may repeat. The executable model and its golden vectors are normative test fixtures.

The seed and allocation/placement algorithm are protocol-lifetime scheduling rules, not release-profile rules. In particular, neither a seal nor its seed contains a Settlement protocol version. A sealed future window therefore has identical meaning before and after a Settlement migration, matching the protocol-lifetime ScheduleOracle and retained reservations. An incompatible scheduling change cannot ride an ordinary profile migration: it requires a separately specified schedule-epoch transition that waits until every old-epoch seal and liability is outside the retention horizon. This document defines no such transition.

5 Blocks, promises and proof statement

5.1 Signed header

A builder signs exactly:

```
(settlementChainId, l2ChainId, protocolVersion, verifyingContract,
 slot, parentHash,
blockHash, stateRoot, bodyRoot, anchorNumber, anchorHash,
forceRoot, forceCutoff, messageStart, messageEnd, dataManifestRoot,
coinbase, tier, contextId, admissionVersion, admissionRoot,
 episode, recoveryRevision, recoveryId)
```

The normative EIP-712 type string is:

```
SlotChainBlock(
 uint256 settlementChainId,uint256 l2ChainId,uint256 protocolVersion,
 address verifyingContract,
 uint64 slot,bytes32 parentHash,bytes32 blockHash,bytes32 stateRoot,
 bytes32 bodyRoot,uint64 anchorNumber,bytes32 anchorHash,bytes32 forceRoot,
 uint64 forceCutoff,uint64 messageStart,uint64 messageEnd,
 bytes32 dataManifestRoot,address coinbase,uint8 tier,
 bytes32 contextId,uint64 admissionVersion,bytes32 admissionRoot,
 uint64 episode,
```

```
uint64 recoveryRevision,
bytes32 recoveryId)
```

The hashed ASCII has one space between each field type and field name and no other whitespace; displayed line breaks, indentation and spaces after commas are typography only. The exact single-line literal and a standalone Keccak implementation are in `commitment-model.py`. `TYPEHASH` is legacy Keccak-256 of those ASCII bytes and equals the concatenation of these two fragments:

```
0xee6a8c8e31e8245cd527869508f6e464
d6084893991203876f734d1855aed87c
```

The EIP-712 domain is `("SlotChain","2",settlementChainId,verifyingContract)`. The explicit `l2ChainId` is the EVM `CHAINID` used by raw L2 transactions. Signatures are exactly 65-byte `r||s||v`; `v` is 27 or 28, s is low, and zero or recovered-zero addresses reject. EIP-2098 and EIP-155 transaction-style `v` values are not accepted. Tier 1 requires

```
contextId = H("slot-chain-normal-context-v1" || baseCanonicalHash ||
              u64(admissionVersion) || admissionRoot ||
              u64(armBlockNumber) || armBlockHash)
```

and its execution anchor is exactly that arm block. `episode=recoveryRevision=0` and `recoveryId=0`; tier 2 requires the exact live round values and `contextId=recoveryId`. The unsigned tier 3 uses that same recovery context. Consequently semantically unused bytes cannot create false equivocation evidence. The execution `blockHash` excludes the off-chain builder signature, avoiding a circular hash definition. The signed `bodyRoot` covers only canonical discretionary transaction bytes; the proof derives the full Ethereum transactions root after adding the profile-defined system and valid forced transactions.

5.2 Three validity tiers

For every tier and every block, the circuit requires the EVM header `timestamp=GENESIS_TIMESTAMP+slot` with checked addition, and derives `blockHash` from that exact header. The signed relative slot can therefore never disagree with EVM time, forced-expiry classification, or the canonical tip clock. Every signed block also carries the exact frozen `admissionRoot`; the circuit checks it together with `admissionVersion`.

1. **Normal signed.** Every block is signed by its scheduled builder under the normal window's frozen `(admissionVersion,admissionRoot)`. Slots strictly increase, each slot is no later than `currentL2Slot+CLOCK_SKEW`, and the first slot strictly exceeds the canonical base slot. Every block is at or above the window's frozen `minAdmissibleSlot`; parent gaps are at most `G_MAX=3,600`, equal to the objective final-lag trigger and within the 12-window schedule-proof cap.
2. **Recovery signed.** The first block extends the L1-final canonical tuple, binds the active episode and current recovery revision, and may cross an unbounded time gap. Every block still requires its scheduled builder signature under the current recovery round's frozen admission pair. Every slot is at or after that round's `roundStartSlot`.
3. **Escape unsigned.** Exactly one deterministic block extends the L1-final canonical tuple during recovery. It has no builder signature, no beneficiary-controlled coinbase, no discretionary transaction and no blob body. It contains only the protocol anchor and the

deterministic maximum prefix of the current round's frozen forced queue. When the queue is empty an SLA-triggered episode may commit an empty anchor-only escape block. Its slot is `max(roundStartSlot+ESCAPE_OFFSET,canonical.tipSlot+1)` and its anchor, queue root/cutoff, block hash and state root are unique functions of the current recovery round and proved execution result.

Every candidate uses exactly one `batchAnchor`; every block repeats that anchor and frozen `forceRoot/forceCutoff`. For a normal candidate the supplied RLP execution header hashes to the EIP-2935 value, its timestamp is no later than the first L2 block timestamp, and MPT proofs under its state root authenticate the historical forced queue and any other contract state. For recovery, the RLP header authenticates the stored parent number/hash and derives its timestamp and state root; the queue pair is authenticated directly by the round fields and is not asserted against that header's state root. Requiring one anchor, not up to 4,096 independent anchors, bounds both circuit and L1 work. Cutoffs therefore do not change within a candidate; the first blocks drain the frozen FIFO prefix and later blocks see an empty prefix.

The circuit verifies parent linkage, strict slot monotonicity, schedules, signatures for tiers 1–2, version binding, anchor/header/MPT authentication, forced-prefix ordering, execution, exact body/data coverage and public outputs. It rejects more than 4,096 blocks, 12 schedule windows, 16 sealed sessions, 2,100 referenced blob records, 256 forced items, 4 MiB forced bytes or 80 million total accounted forced gas per candidate. These candidate-level caps are independent of the per-block caps.

5.3 Verifier statement ABI

The L1 verifier accepts `verify(proof,uint256[2]digestLimbs)`. Settlement constructs the following Solidity struct in exactly this order and computes `statementHash` as legacy Keccak of the ASCII domain `"slot-chain-statement-v2"` followed by `abi.encode(S)`:

```
S = (uint256 settlementChainId, uint256 l2ChainId, uint256 protocolVersion,
     bytes32 executionProfileHash, address verifyingContract,
     uint64 releaseProtocolVersion, bytes32 releaseManifestHash,
     uint8 tier, bytes32 baseCanonicalHash,
     bytes32 candidateCommitment, uint64 blockCount,
     uint64 firstSlot, bytes32 tipHash, uint64 tipSlot,
     uint64 endL2BlockNumber, bytes32 endStateRoot,
     bytes32 endTerminalRoot, uint64 endTerminalCount, uint64 endCursor,
     bytes32 winningDataCommitment, uint64 nextDueAt,
     uint256 nextBaseFee, uint64 nextExcessBlobGas,
     uint64 anchorNumber, bytes32 anchorHash,
     bytes32 anchorStateRoot, uint64 anchorTimestamp,
     bytes32 forceRoot, uint64 forceCutoff,
     bytes32 forcedDescriptorCommitment,
     uint64 startCursor,
     uint64 admissionVersion,
     bytes32 admissionRoot,
     uint64 episode, uint64 recoveryRevision, bytes32 recoveryId,
     bytes32 scheduleRootsCommitment, uint8 scheduleWindowCount,
     bytes32 sessionRefsCommitment, uint8 sessionCount,
     uint16 dataRecordCount, bytes32 executionOutputsCommitment,
```

```
    address proofBeneficiary)
```

`digestLimbs[0]` and `[1]` are respectively the high and low 128 bits of `statementHash`; neither is reduced modulo the proof field. The circuit recombines them and proves the exact preimage. Settlement independently checks the immutable `protocolVersion->executionProfileHash` mapping, the base canonical state, current/frozen versions and recovery round, recomputes the bounded schedule-root and sealed-session-reference commitments from calldata, recomputes the bounded forced-descriptor commitment from authoritative queue storage, recomputes `winningDataCommitment` from the candidate and sealed-session commitments, requires `endL2BlockNumber=stored.l2BlockNumber+blockCount`, and constructs the new core from the explicit `(endL2BlockNumber,tipHash,tipSlot,endStateRoot, endTerminalRoot,endTerminalCount,endCursor)` fields plus that recomputed value and proved `(nextBaseFee,nextExcessBlobGas)`. It recomputes `executionOutputsCommitment` including the same explicit terminal pair; the circuit proves that pair equals the accumulator's exact post-state slots, so neither calldata nor Settlement can substitute a root. It rejects zero/ill-typed fields, cursor regression, or disagreement with the last proved block, then stamps `canonicalizedAtBlock=block.number` outside the proof. `candidateId=statementHash`. `nextDueAt` is authenticated inside the frozen queue as the boundary leaf's deadline, or `UINT64_MAX` when `endCursor=forceCutoff`. It is only a proof-consistency output. Normal submission additionally verifies the current-root boundary proof described in §7. After every commit and on every activation check, the sole authoritative live head value is `ForcedQueue.dueAt(canonical.messageCursor);` no recovery-round or canonical-core scalar caches it. For the one proof-first launch or migration candidate, Settlement requires `releaseProtocolVersion=targetProtocolVersion` and `releaseManifestHash= targetManifestHash` from the authenticated launch or migration gate; every ordinary candidate requires both fields to be zero. The circuit uses the nonzero pair to require the first block's activation Anchor call and the exact manifest, and requires the exact release/domain seals and registered route to be active in every later block of that candidate. The L1 coordinator verifies this proof before the old Settlement or queue authority changes, then adopts its output and consumes the target pair in the same atomic cutover. No ACTIVE Settlement can be left waiting on a newly activated verifier to clear a pending release. `proofBeneficiary` is a nonzero proof input, never builder-signed block data or coinbase. It receives the consumed queue-fee interval and any separate proof reward; copying proof bytes cannot redirect either payment.

5.4 Preconfirmation semantics

A builder signature proves authorship and makes same-slot equivocation slashable. It does not prove that the body reached a prover or that the next builder saw it. A later scheduled builder may sign a profitable skip that the protocol cannot distinguish from non-receipt. Accordingly:

- before L1 candidate acceptance, a promise is *observed*;
- after acceptance and before close, it is *provisionally proved*;
- after normal close or recovery commit, it is *canonical*; and
- only the underlying L1 finality policy makes it *L1-final*.

Wallets and applications MUST NOTdisplay the first two states as final. An escape commit explicitly revokes every omitted observed or provisionally proved promise and returns omitted

transactions to the mempool.

6 Blob data authentication

Data is posted into publisher-owned, bonded sessions. A session ID is

```
H("slot-chain-session-v1" || u256(settlementChainId) || address20(contract) ||
  address20(owner) || u64(sessionSequence))
```

The contract stores one checked `uint64nextSessionSequence`, initially zero. A successful open uses `s=nextSessionSequence`, requires `s<UINT64_MAX`, derives and returns the ID above, then stores `s+1`. The caller supplies neither ID nor sequence, and a rejected or no-cell open does not consume one. Sequence `UINT64_MAX` is the explicit exhaustion sentinel. The sequence is global rather than per owner: a Sybil cannot create permanent nonce keys, while the owner and Settlement address in the preimage prevent cross-owner and cross-release aliases. A fresh migration target starts at zero; the retired source retains its historical sequence.

DataSession is a bounded storage region of the immutable Settlement, and the contract address in the ID above is that Settlement. Settlement accepts native ETH only through the session open/post surfaces and has no Queue, reward, Market, Bridge or other native liability; each of those custodians is a separate account. Its configuration binds the shared migration gate and immutable top-level `dataRent` sink. Every other payable selector, fallback and receive path rejects, and conformance tests exercise `msg.value>0` against every such selector. This exclusive custody makes the balance invariant and surplus calculation complete without a cross-contract caller-supplied "pinned" or "boundary closed" flag.

The profile/runtime geometry is exact: 1,024 physical cells, at most two LIVE cells per owner, 2,100 records per LIVE cell, at most six blobs per post, eight maintenance inspections and an 86,400-second maximum TTL. Each cell has one exact semantic tag `FREE=0`, `LIVE=1` or `REFUND=2`. Storage is exactly `cells[1024]`, an ID-to-cell-plus-one mapping with at most one key per non-FREE cell, a live-owner-count mapping containing only owners with at least one LIVE cell, checked `liveCount`, `refundCount`, `occupiedCount=liveCount+refundCount<=1024`, and `gcCursor`. A REFUND cell stores only `(sessionId,owner,bondWei,claimDeadline)`; its old MMR words are stale and ignored. Clearing a cell deletes its ID key, and removing an owner's last LIVE cell deletes its owner key. No refund or historical-owner mapping exists. The three counters and cursor are `uint16`; owner LIVE counts are also `uint16` and never exceed two. PREACTIVE and historical Settlements have zero local LIVE count. OPEN increments local LIVE and occupied by one; LIVE-to-REFUND decrements local LIVE and increments refund without changing occupied; REFUND-to-FREE decrements refund and occupied. No other transition changes those fields. An eligible owner claim directly from LIVE executes those two logical transitions atomically: its net effect decrements local LIVE and occupied, deletes the owner/ID entries, and reduces LIVE liability and balance by the same bond; refund count and liability have net zero change.

There is deliberately no Router/global LIVE counter and therefore no cross-contract write on OPEN, claim or maintenance. The Router authenticates exactly one current Settlement. Inductively, a PREACTIVE target begins with zero LIVE cells; only the current ACTIVE Settlement may open one; READY requires that source's local LIVE count zero; and cutover freezes that zero-LIVE source before making the zero-LIVE target current. Abort leaves the same source current and never activates the target. The generation-authenticated READY handshake and activation-time static reads below close this induction.

The direct sidecars authenticate that premise through one exact bounded Router read, never through a caller-supplied flag:

```
activeSettlementStateV1() returns
  (bytes4 magic,address activeSettlement,uint64 generation,
   uint64 activeProtocolVersion,uint64 targetProtocolVersion,
   bytes32 targetManifestHash,uint8 phase)
```

The magic is `0x41535231` (ASCII `ASR1`), phases are exactly ACTIVE=0, ARMED=1 and READY=2, and returndata is exactly 224 bytes with canonical ABI padding. OPEN, POST and SEAL read the immutable Router with an exact 50,000-gas static call. They reject revert/OOG/short/trailing or noncanonical data, and require the returned active address to equal this Settlement, the returned active version to equal its immutable protocol version and phase ACTIVE. ACTIVE requires the returned target version and manifest to be canonical zero; ARMED and READY require both nonzero. This check and the session mutation share one EVM revert domain.

Payable `openSession(uint16expectedEmptyCell,uint64expiry)` neither calls `sync` nor performs GC. It requires the current non-PREACTIVE component, the shared gate ACTIVE, `expectedEmptyCell<1024`, that exact tag FREE, owner/live/sequence capacity and the exact profile-priced value. It derives and returns the session ID and increments the global sequence only on success. Every failure reverts, returning value and preserving sequence, cursor, cells and liabilities. A cell-hint race therefore costs the winner the full rent and bond but gives no identity authority. Payable `postData` follows the same sidecar rule: it checks current target and ACTIVE directly and reverts on every failure; it does not run a canonical transition while holding caller value.

The native-value split is exact. The symbols below name these exact profile fields:

```
BOND     = refundableBondWei
BASE     = baseRentWei
BYTE     = rentPerPublishedByteWei
BLOB_BPS = blobBaseFeeMultiplierBps
```

The profile requires `BLOB_BPS<=10000`. Opening requires checked `msg.value=BOND+BASE`; only `BOND` increments the LIVE liability and `BASE` is immediately surplus. A post with n manifests requires $1 \le n \le \min(6, 2100 - count)$, nonzero `BLOBHASH(i)` for every $i < n$ and `BLOBHASH(n)=0`, so the manifest array is in one-to-one order-preserving correspondence with every blob in that transaction. Define

```
publishedBytes = sum(manifest[i].chunkByteLength)
blobGasUsed    = 131072 * n
blobWeight     = blobGasUsed * BLOB_BPS
blobSurcharge =
  mulDivUp(blobWeight, block.blobbasefee, 10000)
postRent       = publishedBytes * BYTE + blobSurcharge
```

over mathematical nonnegative integers; `mulDivUp(a,b,d)` is exactly $\lceil ab/d \rceil$ using a full 512-bit product and reverts if the result does not fit `uint256`. L1 checks the declared `chunkByteLength<=126972`, but cannot read or decode blob contents. The validity circuit later checks the blob's length prefix, zero padding, decoded bytes and computed chunk root

against that declaration. The implementation must produce the exact ceiling without an intermediate-width approximation and then require every result and `postRent` to fit `uint256`; otherwise it reverts. The post requires `msg.value==postRent`; underpayment and overpayment both revert. All post value is immediately surplus and never enters either bond liability. Zero-byte padding still consumes a whole blob and therefore its whole blob-gas surcharge. Any blob/KZG/MMR/value/arithmetic failure rolls back the complete call, including the value, record count, peaks and emitted events.

The externally callable surface is frozen below. `DataPostV1` is the static tuple

```
(bytes32 fullBodyRoot, uint16 blockOrdinal, uint16 chunkIndex,
 uint16 chunkCount, uint32 chunkByteLength, bytes32 chunkRoot, bytes32 y,
 bytes32 commitmentHi, bytes16 commitmentLo,
 bytes32 proofHi, bytes16 proofLo)
```

where concatenating each high/low pair yields exactly 48 bytes; `bytes16` ABI right-padding must be zero. The expanded selectors and returns are:

```
openSession(uint16,uint64) payable returns (bytes32 sessionId)
postData(bytes32,(bytes32,uint16,uint16,uint16,uint32,bytes32,bytes32,
  bytes32,bytes16,bytes32,bytes16)[]) payable
  returns (uint16 firstRecordIndex,uint16 newCount,bytes32 newRoot)
sealSession(bytes32) returns (uint16 count,bytes32 root,uint64 expiry)
maintainDataSessions() returns
  (uint8 status,uint8 inspected,uint8 changed,uint16 nextCursor)
claimSessionBond(bytes32,address) returns (uint256 amount)
sweepSessionSurplus() returns (uint256 amount)
```

Open, post and seal are direct session sidecars: none calls `sync` or moves the maintenance cursor, all recheck current Settlement and ACTIVE, and every invalid input or state reverts. POST and SEAL additionally require `block.timestamp<expiry`; equality is already expired and is eligible for ordinary conversion or a direct claim. Seal also requires a nonempty unsealed LIVE session owned by the caller. Maintenance status is exact: NOOP=0, SYNCED=1 and SCANNED=2. Current ACTIVE first invokes `sync`; if it changes state the call returns SYNCED without a separate scan, otherwise it performs an ordinary exact scan and returns SCANNED. Current ARMED returns the result of its sync-owned migration work as SYNCED, or NOOP while a boundary remains open. READY/FROZEN and historical components perform only the overdue-REFUND scan; PREACTIVE returns NOOP. SCANNED reports eight inspections, while the other statuses report zero explicit maintenance inspections even if ARMED `sync` performed its separately accounted migration scan. An invalid claim reverts; a zero-surplus sweep returns zero, while sink failure reverts only that sweep. PREACTIVE claim and sweep revert without a CALL, event or write, and PREACTIVE maintenance is the stated NOOP; forced ETH is the only way its balance can change before activation.

For manifest i, the contract derives `recordIndex=oldCount+i`, `versionedHash=BLOBHASH(i)`, `publisher=msg.sender` and `validUntil=session.expiry`; none is caller supplied. It checks `1<=chunkCount<=9`, `chunkIndex<chunkCount`, the declared length bound, field-canonical `y`, the versioned hash/commitment relation and the point-evaluation proof. It derives z and the Appendix leaf separately for each tuple, appends exactly n leaves in array/blob order, and emits exactly n leaf events. One failure at any position reverts all earlier appends, events and value.

The canonical events, with no more than three indexed arguments, are:

```
SessionOpened(bytes32 indexed sessionId,address indexed owner,
  uint16 cell,
  uint64 sequence,uint64 expiry,uint256 bondWei,uint256 baseRentWei)
DataRecordAppended(bytes32 indexed sessionId,uint16 indexed recordIndex,
  bytes32 indexed versionedHash,bytes32 leaf)
SessionSealed(bytes32 indexed sessionId,
  uint16 count,bytes32 root,uint64 expiry)
SessionLiveToRefund(bytes32 indexed sessionId,address indexed owner,
  uint16 cell,
  uint64 claimDeadline,uint64 migrationGeneration)
SessionBondClaimed(bytes32 indexed sessionId,address indexed owner,
  address indexed recipient,uint256 amount)
SessionRefundForfeited(bytes32 indexed sessionId,
  address indexed owner,
  uint16 cell,uint256 amount)
SessionSurplusSwept(address indexed sink,uint256 amount)
DataSessionsMaintained(uint8 mode,
  uint16 startCursor,uint16 nextCursor,
  uint8 inspected,uint8 changed)
```

An ordinary conversion emits generation zero; a migration conversion emits the armed generation. A direct LIVE claim emits only `SessionBondClaimed`; `SessionLiveToRefund` means the cell persistently ended the call as REFUND. Maintenance event mode is ORDINARY=1, MIGRATION=2 or REFUND_ONLY=3. Events are the permanent leaf/discovery oracle, not contract storage.

The exact views are

```
dataSessionCellV1(uint16 cell) returns
  (uint8 tag,bytes32 sessionId,address owner,uint64 sequence,uint64 expiry,
   uint16 count,bool sealed,bytes32 root,bytes32[12] peaks,uint256 bondWei,
   uint64 claimDeadline)
dataSessionByIdV1(bytes32 sessionId) returns
  (uint16 cellPlusOne,uint8 tag,address owner,uint64 sequence,uint64 expiry,
   uint16 count,bool sealed,bytes32 root,uint256 bondWei,uint64 claimDeadline)
dataSessionAccountingV1() returns
  (bytes4 magic,uint16 liveCount,uint16 refundCount,uint16 occupiedCount,
   uint16 gcCursor,uint64 nextSessionSequence,uint256 liveBondLiability,
   uint256 refundBondLiability,uint64 migrationRefundGeneration,
   uint64 migrationRefundClaimDeadline,bool guardEntered,
   bytes32 dataSessionConfigHash)
```

The accounting magic is `0x44535631` (ASCII `DSV1`); narrow words, booleans, addresses and the fixed array are canonically encoded, and returndata lengths are exactly 704, 320 and 384 bytes respectively. FREE cell/by-ID misses return canonical zero for every output. A REFUND view returns only `sessionId`, owner, bond and deadline nonzero; it masks sequence, expiry, count, sealed, root and peaks to canonical zero without clearing their stale physical words. Runtime

code hash binds selector, event, tuple and fee-algorithm semantics. The returned configuration hash is the exact immutable-value commitment

```
H("slot-chain-data-session-config-v1" || u32(340) ||
  u256(settlementChainId) || u64(protocolVersion) || address20(settlement) ||
  address20(activeSettlementRouter) || address20(protocolVersionManager) ||
  address20(dataRent) || executionProfileHash ||
  u256(BOND) || u256(BASE) || u256(BYTE) || u16(BLOB_BPS) ||
  u64(86400) || u64(86400) || u16(1024) || u16(2) || u16(2100) ||
  u8(8) || u8(6) || address20(0x0a) || u32(50000) || u256(BLS_MODULUS) ||
  u32(131072) || u32(126972) || u16(9))
```

Every address/hash is nonzero except that the point-evaluation address is the canonical numeric `0x0a`; the BPS/rate fields may be zero. The delayed target registration must carry this exact value, the target Settlement's wider configuration commitment must include it as a named field, and the Router must compare the accounting return against that registered value. The current target-registration/control-plane codec has not yet been extended to do so and remains the explicit release blocker below; this section does not falsely equate the DataSession sub-configuration hash with the wider `targetConfigurationHash`.

The union-cell transition writes are normative. FREE-to-LIVE writes the new ID, owner, sequence, expiry and bond and explicitly writes `count=0`, `sealed=false` and the wrapped empty-MMR root. It may leave physical peak words stale, because a peak is read only when its bit in the newly written count is one; the first append at count zero therefore overwrites height zero before using it. LIVE-to-REFUND writes the REFUND tag and claim deadline and preserves only ID, owner and bond semantically. REFUND-to-FREE deletes the ID key and writes the FREE tag while its union words may remain physically stale. FREE and REFUND getters apply the masks above, and no append/root operation may read a stale word hidden by the tag or count. Consequently reuse of a formerly sealed, nonempty cell has exactly the same first-append root as a fresh cell.

Nonpayable maintenance alone owns `gcCursor`. An eligible call includes FREE and unmodified cells in this exact scan and update:

```
cell[j] = (oldCursor + j) mod 1024, for j = 0..7
gcCursor = (oldCursor + 8) mod 1024
```

In ACTIVE, an ordinary LIVE cell becomes REFUND iff `expiry<=block.timestamp` and its ID is not referenced by the stored normal best; its deadline is fixed as saturating `uint64` addition of the claim window to the stored expiry, never to the keeper's scan time. A pre-existing REFUND becomes FREE only when `block.timestamp>claimDeadline`. One scan never both creates and forfeits the same REFUND. Migration and historical modes use the stricter rules below. No call scans all 1,024 cells.

The refundable bond is a LIVE liability and then a REFUND liability; conversion moves the same amount between the two checked scalars without transferring ETH. The owner calls `claimSessionBond(bytes32sessionId,addressrecipient)` through the bounded ID-to-cell lookup. It accepts either that exact REFUND before its deadline, or in current ACTIVE mode an expired LIVE cell not referenced by the normal best before `satAdd64(expiry,claimWindow)`. ARMED LIVE claims are suppressed until its normal/recovery boundary closes, then use that generation's frozen migration deadline. It always requires the exact owner, nonzero

recipient and `block.timestamp<=claimDeadline`; checks-effects-interactions clears the mapping/cell/counters/liability before transfer. A failed recipient transfer reverts the whole claim. A shared nonreentrancy guard rejects every nested session or surplus-sweep call without changing state; a recipient may catch that nested rejection and return normally, in which case the outer claim succeeds exactly once. Claims remain unpausable on ACTIVE, ARMED, READY, FROZEN and historical components. After the deadline, maintenance clears the REFUND and its ETH becomes surplus; no owner or sink is called.

The Settlement account enforces `balance>=liveBondLiability+refundBondLiability`. Nonrefundable rent, forfeited bonds and forced ETH are surplus and never inflate a refund. A separate permissionless, non-gating `sweepSessionSurplus()` sends exactly the balance above both liabilities to the immutable `dataRent` sink under one reentrancy guard. Sink failure reverts only that sweep and cannot block GC, claims, READY, cutover or the forced lane. The calibrated rent/bond combination prices the full possible occupancy through TTL and the 86,400-second refund window. A Sybil can still buy all optional DA cells for that bounded period; the forced/escape lane is independent and this is not a protocol-liveness claim.

Each occupied cell owns a fixed `bytes32peaks[12]` and `uint16count<=2100`; it stores no record array. A peak word at height h is meaningful exactly when bit h of count is one; zero-bit words may be stale. Append uses the old count as the leaf index, combines the stored peak on the left while successive count bits are one, writes the carry at the first zero bit and increments count. The wrapped MMR root is recomputed from at most 12 meaningful peaks. The exact bag order and proof grammar are frozen in Appendix A.

Only the owner may append, and a candidate may reference only a session made immutable by `sealSession`. Sealing fixes `(root,count,expiry)`; there is no unseal or append-after-seal operation. A candidate supplies at most 16 sorted unique `(sessionId,count,root)` references. At submission each must equal a live sealed session. Normal acceptance requires `expiry>=normalDeadline+REORG_MARGIN_SECONDS`; recovery requires `expiry>=block.timestamp+REORG_MARGIN_SECONDS`. Opening requires its expiry to cover proving, settlement and the reorg margin:

$$\begin{aligned} expiry &\geq now + P\_PROVE\_MAX + W\_SETTLE\_SECONDS \\ &\quad + REORG\_MARGIN\_SECONDS, \\ expiry &\leq now + DATA\_TTL. \end{aligned}$$

Here `P_PROVE_MAX=recovery.proofTimeMaxSeconds=900` and `W_SETTLE_SECONDS=recovery.settlementWindowSeconds=1200` in the exact execution profile; they are not independent implementation constants. Profile validation separately enforces `DATA_TTL>=P_PROVE_MAX+W_SETTLE_SECONDS+REORG_MARGIN_SECONDS`.

For every blob attached to `postData(session,manifests[])`, the contract:

1. obtains `versionedHash=BLOBHASH(i)` from the same transaction. The exact KZG relation is `versionedHash=bytes1(0x01)||SHA256(commitment48)[1:32]`; the point-evaluation precompile below checks this equality as part of acceptance, so no alternate version byte, digest function or truncation is legal;
2. computes legacy Keccak over this packed fixed-width sequence:

```
"slot-chain-data-fs-v2" || u256(settlementChainId) ||
u256(protocolVersion) ||
sessionId || versionedHash || fullBodyRoot || u16(blockOrdinal) ||
```

```
u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
chunkRoot || address20(publisher) || u64(validUntil)
```

It interprets the digest big-endian and reduces it modulo the BLS scalar-field modulus to obtain z;

3. makes a 50,000-gas `STATICCALL` to the EIP-4844 point-evaluation precompile at address `0x0a` with the exact 192 bytes `versionedHash32||z32||y32||commitment48||proof48`. It requires call success and exactly 64 return bytes equal to `u256(4096)||BLS_MODULUS`, where the exact modulus is

```
hex32("73eda753299d7d483339d80809a1d805" ||
      "53bda402fffe5bfeffffffff00000001")
```

short, long, reverting or wrong-constant returns reject; and

4. appends the exact leaf from §A to the session MMR.

The proof receives blob polynomials privately, recomputes $P(z) = y$, and decodes the canonical blob codec. A discretionary body byte string is `u32(txCount)||(u32(txLength)||signedRawTx)*`; its root is `H("slot-chain-body-v1"||u32(byteLength)||bodyBytes)`. Blob payload is `u32(chunkByteLength)||chunkBytes||zeroPadding`, split into 31-byte chunks; each blob field element is `0x00||chunk31`. Exactly 4,096 elements are used, unused payload bytes are zero, and noncanonical encodings reject. A manifest entry binds `(blockOrdinal,chunkIndex,chunkCount,chunkByteLength,fullBodyRoot,chunkRoot,sessionId,recordIndex)`. For each block entries are in strict chunk-index order, cover exactly `0..chunkCount-1`, have `chunkCount<=MAX_CHUNKS_PER_BODY=9`, and concatenate to at most `MAX_BODY_BYTES=1,142,748`. The circuit recomputes every chunk root, the full discretionary body root and then the full Ethereum transactions root. The signed `dataManifestRoot` is *per block*. Its leaves contain only that block's entries, use local positions `0..m-1`, repeat that block's candidate-wide `blockOrdinal`, and are sorted by chunk index. Candidate calldata concatenates these independently rooted lists in increasing block ordinal; no global manifest tree exists. Duplicate, extra, overlapping or uncovered data is invalid. `dataManifestRoot` is the exact per-block tree in Appendix A, not a free-form publisher value.

Data publication has no consensus-coupled reimbursement. A separate service may pay it, but empty payment state cannot revert canonical settlement.

6.1 Canonical reconstruction availability

Blob sessions make an in-flight candidate provable; their TTL and garbage collection are not permanent state archival. On canonical commit, full nodes must retain the canonical raw bodies, receipts, Ethereum trie nodes and every runtime-bytecode preimage needed to execute from `CanonicalCore.stateRoot`. Bytecode is mandatory because an Ethereum account leaf authenticates only `codeHash`, not the corresponding bytes. A reconstruction package is a content-addressed set of those objects plus the canonical block headers; each code object is keyed by `H(runtimeBytecode)` and must cover every account reachable during replay, including implementation code reached through proxy, beacon or delegation indirection. A consumer rejects any object whose transaction/body/header hash, trie path, bytecode hash or final state root disagrees with the L1-canonical commitments, so an archive can be anonymous and malicious without gaining safety authority.

Renewing a recovery round refreshes its L1 anchor, queue cutoff and proof target; it cannot

invert a state root or recreate destroyed bytes. Therefore the 2,164-second protocol bound begins only once a prover has the canonical reconstruction package and any required unexpired kind-0 bytes. A warm full node satisfies this condition; a cold-start prover is bounded by data download plus proof time while at least one archive or Ethereum's applicable DA history still serves the package. If no complete copy exists, safety remains intact but progress stops. No governance action may replace the state root, skip the missing prestate, or call that condition a builder failure.

7 Settlement and recovery state machine

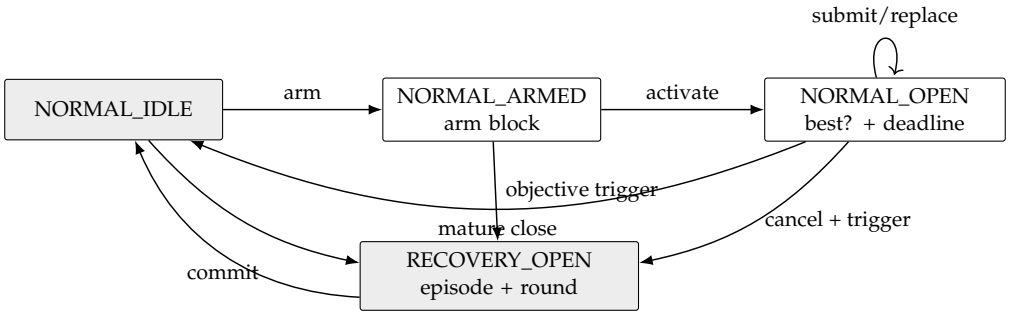


Figure 1: The complete mode machine. Payments are outside every arrow.

7.1 Normal context and candidates

Normal context creation is two-phase. Anyone calls `armNormalContext()` in `NORMAL_IDLE`; the call begins with `sync()`, stores only `armBlockNumber=block.number`, and emits `NormalContextArmed`. It accepts no caller-supplied anchor or hash. In a later L1 block anyone calls `activateNormalContext()`. It again begins with `sync()` and requires the arm's block-number age to be from 1 through `MAX_ARM_AGE_BLOCKS` (255), reads `armBlockHash=blockhash(armBlockNumber)`, and hashes a supplied RLP header to that value. It then freezes `baseCanonical`, `deadline=block.timestamp+W_SETTLE_SECONDS`, `minAdmissibleSlot=max(0,currentL2Slot-DELTA_TIP)`, the current stable `(admissionVersion,admissionRoot)`, the arm header fields, and `requiredThrough=deadline+T_INCLUDE_MAX_SECONDS`. It derives the sole normal `contextId` and emits `NormalContextActivated`. No candidate is accepted while armed. If age exceeds 255 blocks, any caller may invoke `armNormalContext()` again; after its leading `sync()` it atomically replaces only the stale arm number with the current block, emits `NormalContextRearmed`, and returns `REARMED`. A nonstale arm cannot be replaced. Thus an abandoned arm cannot suppress normal operation.

Builders collect and issue tier-1 signatures only after the activation event. The execution anchor for every candidate is the exact arm block, whose hash and parsed header are already frozen; replacements hash their supplied RLP header to that stored hash and never repeat an EIP-2935 recency lookup. A reorg that removes the arm block necessarily changes the context. If the arm block survives but the activation transaction is reorged, reactivation intentionally recreates the same context: the builder must retain and honor its prior same-context signature. Evidence independently authenticates the arm hash as canonical through EIP-2935 and is accepted only while that history entry is available. The enforced evidence horizon is strictly shorter than 8,191 L1 slots. Thus neither a shallow reorg nor a builder-selected historical anchor can create false or free equivocation.

Activation is permissionless, reserves no builder, and does not block submissions. An empty activated window merely cancels at deadline, so an opener gains no censorship right. Every candidate first proves its frozen-prefix boundary in the circuit. It also supplies a canonical single-leaf Merkle proof at `endCursor` under the queue's current root/count, or proves `endCursor==currentCount`. Settlement verifies the canonical depth-64 singleton proof, which contains exactly 64 sibling hashes, and requires the resulting `currentBoundaryDueAt` to exceed `requiredThrough`. This prevents an old but valid anchor from hiding a post-anchor append. Because due times are monotone, one current boundary leaf proves that every item due through the landing horizon is consumed. Later candidates use the same base, deadline, minimum, admission pair and horizon. Their total order is lexicographic:

$$(\text{block count descending},\ \text{tip slot descending},\ \text{tip hash ascending}).$$

Only a strictly better candidate replaces `normalBest`; the stored best contains `endCursor` and the minimum expiry of every referenced data session. Provisional replacement changes no canonical state and pays nobody. At or after the deadline, `sync()` reads `ForcedQueue.dueAt(best.endCursor)` and commits only if that live boundary is greater than `block.timestamp` and `block.timestamp+REORG_MARGIN_SECONDS<=best.minDataExpiry`. Thus a late close cancels rather than committing a block that omits an already-due head, even beyond the assumed inclusion bound, or whose authenticated data has passed its resubmission margin. Data-session GC cannot evict a session referenced by the stored best; its own leading `sync()` clears an expired best first. Because `FORCE_DELAY>=W_SETTLE_SECONDS+T_INCLUDE_MAX_SECONDS`, an append after open cannot violate the frozen acceptance horizon. Settlement then recomputes recovery causes against the new core.

7.2 Activation

While normal, `sync()` opens one persistent recovery episode if either:

- `currentL2Slot>canonical.tipSlot+DELTA_FINAL_LAG`; or
- the forced queue's stored `dueAt[canonical.messageCursor]` is no greater than `block.timestamp`.

A due forced head has precedence over an immature normal window, which is canceled. At a mature deadline, a best that consumes the required due prefix commits before causes are recomputed; an omitting best is canceled. No best is promoted across activation. The contract increments `episode`, records causes and the base-canonical hash, then runs the bounded seat-duty pass in §9: an objectively missed primary duty is allocated, failed over or marked breached as applicable, while a full duty ring fails open to economic vacancy. This path performs no Market call, bond movement or token transfer. Breach enforcement and bond terminalization are later permissionless noncanonical operations against the retained exact receipt; force-only recovery creates no aggregator penalty.

Every round freezes `roundStartSlot`, the preceding L1 block number/hash, current `forceRoot/forceCutoff`, current `(admissionVersion,admissionRoot)`, and `escapeSlot=max(roundStartSlot+ESCAPE_OFFSET,canonical.tipSlot+1)`. Its `expiresAt` is `GENESIS_TIMESTAMP+escapeSlot+DELTA_TIP`. The exact ID is:

```
recoveryId = H("slot-chain-recovery-v2",
               u256(settlementChainId), address20(contract),
```

```
u64(episode), u64(revision), bytes32(baseCanonicalHash),
u64(roundStartSlot), u64(anchorNumber), bytes32(anchorHash),
bytes32(forceRoot), u64(forceCutoff), u64(admissionVersion),
bytes32(admissionRoot), u64(escapeSlot), u8(causes))
```

At `block.timestamp>expiresAt`, and never before, `sync()` replaces the expired target with revision $r + 1$ and fresh round fields, then returns `SYNCED`. It does not change episode/base/cause, create another duty, promote, terminalize a bond or pay. The old proof is already inadmissible, so permissionless refresh cannot grief a still-live proof. New queue appends enter the refreshed cutoff. An envelope admitted during recovery receives `dueAt=max(enqueuedAt+FORCE_DELAY,lastDueAt,currentRound.expiresAt+1)`; therefore it cannot become due while a frozen round that omits it remains admissible.

7.3 Recovery acceptance

Recovery is first-valid, not heaviest. A tier-2 or tier-3 candidate MUST:

1. extend the exact current stored canonical state;
2. bind current `episode`, `revision`, `recoveryId`, frozen admission pair, anchor and queue target;
3. land at least `F_L1` L1 blocks after its anchor, have first slot at least `roundStartSlot`, tip no older than `DELTA_TIP`, and no slot later than wall clock plus skew;
4. consume the exact maximum forced-message prefix from the current cursor, bounded by the current round's frozen cutoff; and
5. satisfy its tier-specific signature/data restrictions and validity proof.

The first valid L1 transaction atomically commits the tuple, records the current L1 block *number only*, clears the recovery episode and returns to normal. Later proofs are stale. Candidate rejection cannot undo activation because of the early-return wrapper in §3.

8 Forced queue and unsigned escape

`ForcedEnvelope` is a tagged union. Kind 0 contains `(sender,nonce,l2ChainId,rawTxHash, byteLength,gasLimit,maxFee,validUntil,refundAddress,deposit)`. Admission canonically decodes the EIP-2718 transaction, requires its recovered sender and all duplicated fields to match, and requires `msg.sender==sender`. Launch exposes no relayed kind-0 entry; any future EIP-712 relay requires a new function, profile and codec. The sole launch entry is

```
enqueueForcedTransactionV2(bytes rawTransaction, uint64 validUntil,
                           address refundAddress)
  payable returns (uint8 result, uint64 queueIndex)
```

Its selector is the first four Keccak bytes of `"enqueueForcedTransactionV2(bytes,uint64, address)"`. Word 0 is the canonical bytes offset `0x60`, word 1 the canonically padded validity deadline and word 2 the canonically padded nonzero refund address. The tail is one length word, the nonempty raw EIP-2718 bytes and zero right-padding; total calldata length is exactly `132+ceil32(rawTransaction.length)` bytes. The adapter re-encodes and byte-compares the complete call before state. It recovers the sender and derives nonce, chain ID, raw transaction hash, raw byte length, intrinsic gas, gas limit and maximum fee from those exact bytes; `msg.sender` must equal the recovered sender, the transaction chain must be the configured L2, and `msg.value` must equal the checked shared-schedule deposit. No duplicated caller field or caller hash is authoritative. Kind 1 admission receives one `BridgeAdmissionEnvelope` containing one `uint64liquidityFee` and the complete raw eleven-field `IBridge.Message`:

```
(uint64 id, uint64 fee, uint32 gasLimit, address from,
 uint64 srcChainId, address srcOwner, uint64 destChainId,
 address destOwner, address to, uint256 value, bytes data)
```

Its Solidity name and the envelope field order are exact:

```
struct MessageV1 {
  uint64 id; uint64 fee; uint32 gasLimit; address from;
  uint64 srcChainId; address srcOwner; uint64 destChainId;
  address destOwner; address to; uint256 value; bytes data;
}
struct BridgeAdmissionEnvelopeV2 {
  MessageV1 message; uint64 liquidityFee;
}
sendMessageV2(MessageV1 message, uint64 liquidityFee)
  payable returns (bytes32 msgHash, MessageV1 normalizedMessage)
enqueueBridgeCreditV2(MessageV1 message, uint64 liquidityFee)
  payable returns (uint8 result, uint64 queueIndex)
```

Their selectors are the first four Keccak bytes of respectively `"sendMessageV2((uint64, uint64,uint32,address,uint64,address,uint64,address,address,uint256,bytes) ,uint64)"` and the same expanded arguments under `enqueueBridgeCreditV2`. After either selector, word 0 is the Message offset `0x40` and word 1 is the canonically padded liquidity fee. The Message begins at argument byte `0x40`, has eleven head words in the shown order, and its data offset word is exactly `0x160`; that tail is one length word, exact data and zero

right-padding. Thus canonical calldata length is `452+ceil32(data.length)` bytes including selector. Each entry re-encodes the decoded arguments and requires byte-for-byte equality and exact length, so aliases, gaps, overlap, trailing bytes, high integer/address padding or nonzero right-padding reject before state.

The SourceBridge first applies the normalized overwrites, then freezes the legacy V1 hash codec exactly:

```
abi.encode("TAIKO_MESSAGE", normalizedMessage) =
  [0x40, 0x80] || [u256(13), "TAIKO_MESSAGE", zero-pad-to-32] ||
  [the eleven normalized Message ABI head words, dataOffset=0x160] ||
  [u256(data.length), data, zero-pad-to-32]
msgHash = keccak256(that complete 512+ceil32(data.length)-byte preimage)
```

The return from `sendMessageV2` has `msgHash` in word 0, Message offset `0x40` in word 1 and the same canonical Message tuple/tail. Both adapter returns are exactly 64 bytes: result QUEUED=1 carries the new index, SYNCED=2 carries `UINT64_MAX`, and ALREADY_QUEUED=3 is permitted only for a zero-value kind-1 query and carries the stored index. Kind 0 never returns ALREADY_QUEUED. Unknown result, bad padding, or any other sentinel/index combination rejects at its caller. The SourceBridge overwrites `id`, `from` and `srcChainId` with its next ID, the actual caller and immutable chain context before deriving `keccak256(abi.encode("TAIKO_MESSAGE",message))`, data length and `calldataHash=keccak256(message.data)` over exactly the unpadded dynamic bytes. Empty data therefore uses Ethereum's canonical `keccak256("")=c5d2460186f7233c927e7db2dcc703c0e500b653ca82273b7bfad8045d85a470`. For kind 1, `byteLength=uint32(message.data.length)` exactly after canonical ABI decoding; for kind 0 it is the exact EIP-2718 raw-transaction byte length. No caller-supplied length, ABI-padded length, descriptor length or event length is accepted. It requires a nonzero `liquidityFee` and a checked `message.value+message.fee>0`; a zero settlement amount has no ticket, reservation or DONE encoding and is rejected before custody or ID mutation. No hash, length or funding ticket supplied by the caller is authoritative. Only the configured `BridgeInboxAdapter` may enqueue the resulting fixed V11 projection and it has no queue expiry. The source Bridge retains message value plus execution and liquidity fees while recording per-credit pending, user-pull and LP-pull liabilities; every outflow must leave `address(this).balance>=pendingEscrow+userPullLiability+lpPullLiability`. The adapter separately temporarily retains its caller-funded processing fee only across the sync decision; a successful append forwards it entirely to ForcedQueue. Its only forced effect is an idempotent inbox/signal credit keyed by the immutable identifier

```
sourceDomainId = H("slot-chain-source-domain-v4" || u64(srcChainId) ||
  bytes32(sourceGenesisHash) || address20(bridgeCreditRegistry) ||
  address20(terminalSignalVerifier) || address20(srcBridge) ||
  bridgeExecutionHash || bytes32(registryNamespace))
destinationDomainId = H("slot-chain-destination-domain-v7" ||
  u64(destChainId) || bytes32(destinationGenesisHash) ||
  address20(bridgeInboxAdapter) || address20(activeSettlementRouter) ||
  address20(terminalSignalVerifier) || address20(inboxApplyRouterV2) ||
  address20(inboxCreditStoreV2) || address20(protocolReleaseAuthorityV2) ||
  address20(terminalDomainRegistrarV2) ||
  address20(terminalAccumulatorV2) || address20(nativeLiquidityPoolV2) ||
  address20(destinationBridge) ||
```

```
  bytes32(destinationBridgeExecutionHash) ||
  bytes32(destinationInfrastructureHash) ||
  bytes32(destinationNamespace))
creditId = H("slot-chain-bridge-credit-id-v6" || u64(srcChainId) ||
  sourceDomainId || u64(srcEpoch) || address20(srcBridge) ||
  destinationDomainId || msgHash || u64(liquidityFee))
escrowId = H("slot-chain-bridge-escrow-v2" || creditId)
```

Both domain descriptors are append-only and source-approved; no message sender supplies an arbitrary destination trust root. The destination descriptor binds the full stable recovery path: L2 genesis, non-proxy `BridgeInboxAdapter`, immutable `ActiveSettlementRouter`, `TerminalSignalVerifier`, `InboxApplyRouterV2`, `InboxCreditStoreV2`, protocol-lifetime `ProtocolReleaseAuthorityV2`, immutable `TerminalDomainRegistrarV2`, protocol-lifetime `TerminalAccumulatorV2`, immutable `NativeLiquidityPoolV2` and frozen destination Bridge facade. The destination infrastructure hash is the Appendix-defined commitment to the deployed runtimes and constructor immutables of every one of those components; address-only binding is insufficient. The source and destination Bridge addresses are fresh immutable non-proxy lifetime constants for that domain; neither Resolver rotation nor a replacement proxy may change their callback, custody, status or terminal identity. An incompatible implementation requires a separately named endpoint and new domain, while every old endpoint remains callable for its old credits. Arbitrary destination invocation and RETRIABLE processing remain later ordinary bridge operations. The two forced-envelope variants have separate byte-exact leaves.

Launch accepts kind 1 only for the L1-to-slot-chain direction, where:

```
srcChainId = settlementChainId = block.chainid
```

Consequently the adapter omits an EIP-2935/MPT proof of its own L1. It performs a bounded `staticcall` directly to the descriptor's non-proxy `BridgeCreditRegistry`, checks its runtime hash and exact fixed-size return, and compares the immutable `CreditRecord`. A reorg that removes the source record necessarily removes every later enqueue transaction; anyone may retry from the still-durable record on the surviving chain. Cross-L1 sources require a separately reviewed protocol version and are not silently routed through this path.

The kind-0 v2 and kind-1 v11 leaf, descriptor and disposition decoders are protocol-lifetime queue ABIs. Every future execution profile must retain their exact semantics for all entries below `queueCount`; a migration may add a new tagged kind but cannot reinterpret an existing byte. The permanent queue keeps every fixed-width descriptor and kind-1 credit record. For kind 0, a candidate supplies raw EIP-2718 bytes matching `rawTxHash` while unexpired; if no party retains those bytes, the head becomes permissionlessly `EXPIRED_NO_TX` after the bounded seven-day validity horizon, using only the stored descriptor. Thus migration neither assumes an expiring blob for a durable bridge credit nor turns missing user bytes into permanent head-of-line blocking.

The deployed V1 `sendMessage(Message)` selector, counter, event, signal, status and recall semantics remain byte-for-byte V1. No Resolver update or interface probe redirects a V1 caller. Launch V2 is a separate, fresh immutable non-proxy SourceBridge with one additive `sendMessageV2(Message,uint64)` selector, where the second argument is the liquidity fee. It normalizes caller calldata by overwriting `id`, `from` and `srcChainId` with its checked next ID, the actual external caller and the immutable chain context before hashing. It

resolves the final destination Bridge and rejects that address as the effective caller. It requires exact `msg.value=message.value+message.fee+liquidityFee`, derives the complete Message hash and exact `keccak256(message.data)` hash/raw length itself, sets checked `enqueueBy=block.timestamp+MAX_BRIDGE_ENQUEUE_DELAY`, and creates only a DIRECT NEW credit. There is no V2 Vault selector, DRAFT state, capsule, restorable token, reverse asset outflow or `getOrDeploy` path. Before incrementing the ID or writing escrow/Registry state it also preserves the V1 safety invariants exactly: `srcOwner`, `destOwner` and `to` are each nonzero; `destChainId` differs from the immutable source chain and equals the selected destination-domain record; the checked sum `message.value+message.fee` is nonzero. If `gasLimit=0`, `fee` must be zero. Otherwise the checked `uint256(gasLimit)-V2_POST_CALL_GAS_RESERVE` must be nonzero; this is the exact destination invocation budget rule. Registry insertion, adapter join, the durable V11 decoder and the validity circuit repeat all those predicates. The corpus sets each owner/target address to zero separately, exercises both gas/fee branches, rejects value and fee both zero, exercises one-below/exactly-at/one-above the reserve, and requires byte-identical nonce, balance, liability, Registry and Queue rollback.

The immutable `BridgeCreditRegistryV2` recomputes `creditId`, accepts writes only from its exact configured SourceBridge capability, and rejects duplicates. Authorization creation, Bridge custody and liability creation are one non-reentrant rollback domain. The Registry stores only this fixed-size authorization and never stores or returns raw Message bytes:

```
(bytes32 sourceDomainId, uint64 srcEpoch, address srcBridge,
 bytes32 bridgeExecutionHash, uint64 emittedAtBlock, bytes32 msgHash,
 bytes32 destinationDomainId, uint64 destChainId,
 uint64 enqueueBy, address sender, address srcOwner, address destOwner,
 uint256 value, uint64 fee, uint64 liquidityFee,
 bytes32 calldataHash, uint32 calldataLength,
 bytes32 escrowId)
```

The SourceBridge alone owns mutable custody:

```
SourceLiability(value, executionFee, liquidityFee, status, queueIndex,
                pullClass, pullBeneficiary, pullAmount)
```

The adapter's source read boundary is exactly two calls:

```
creditAuthorizationV2(bytes32 creditId) view returns
 (bytes32 sourceDomainId, uint64 srcEpoch, address srcBridge,
  bytes32 bridgeExecutionHash, uint64 emittedAtBlock, bytes32 msgHash,
  bytes32 destinationDomainId, uint64 destChainId, uint64 enqueueBy,
  address sender, address srcOwner, address destOwner, uint256 value,
  uint64 fee, uint64 liquidityFee, bytes32 calldataHash,
  uint32 calldataLength, bytes32 escrowId)

creditLiabilityV2(bytes32 creditId) view returns
 (uint256 value, uint64 executionFee, uint64 liquidityFee, uint8 status,
  uint64 queueIndex, uint8 pullClass, address pullBeneficiary,
  uint256 pullAmount, uint256 totalLiveLiability)
```

Their selectors are respectively `0x05ecb6c2` and `0xc978978a`; calldata is exactly 36 bytes, gas is capped at 200,000 for each call, and return length is exactly 576 and 288 bytes. All narrow

words have canonical zero high padding. Before either read the adapter checks the descriptor-pinned runtime and exact `componentConfigHashV2()` result. It joins every duplicated amount/identity, requires `status=NEW=1`, `queueIndex=UINT64_MAX`, `pullClass=0`, zero beneficiary and zero pull amount, and checks `totalLiveLiability>=value+fee+liquidityFee` and `BALANCE(sourceBridge)>=totalLiveLiability`. Short/trailing return, malformed padding, substituted field, insolvency or call fault rejects before Queue, Registry, Bridge or adapter mutation. Only the liability moves `NEW->QUEUED->DELIVERED/RECALLED` or `NEW->CANCELLED`. Those two bounded read-only views let the adapter cross-check every immutable field, exact NEW status and aggregate solvency. Queue append, `markQueuedV2`, Registry state, Bridge liability/balance and adapter record are one transaction. Liability remains exactly value plus fee plus liquidity fee through QUEUED. Strict post-`enqueueBy` cancellation and proved FAILED recall reclassify the full amount from pending escrow to the source owner's USER_PULL. Proved DONE authenticates the consumed L2 funding ticket and reclassifies the full amount to its immutable L1 recipient's LP_PULL; it never drops liability or creates unowned SourceBridge surplus. Liability decreases only when the corresponding pull succeeds. Withdrawal is unpausable and CEI-safe, derives the claimant from the transaction frame, accepts only a nonzero claimant-selected recipient, and restores the claim if the recipient call fails or reenters.

The L1 `BridgeDomainRegistry` retains append-only source-domain, destination-domain and frozen-endpoint-support descriptors. Only the delayed `ProtocolVersionManager` may stage a descriptor, atomically with the active immutable release manifest. A staged tuple is unusable until permissionless `confirmDestinationRegistration(protocolVersion, canonicalSequence,proof)` atomically reads the exact sequence-tagged full canonical tuple through `ActiveSettlementRouter`, requires its L1 canonicalization block to be at least `F_L1` deep, and authenticates the exact domain-registration record under that tuple's `stateRoot`. It never accepts a root or tuple from the caller. The active profile pins one immutable, acyclic `RegistrationMptVerifierDescriptorV2`; no default verifier exists. Its exact statement is

```
struct RegistrationStorageStatementV2 {
  uint256 settlementChainId; address activeSettlementRouter;
  address bridgeDomainRegistry; bytes32 routeKey;
  uint256 destinationChainId; uint64 protocolVersion;
  uint64 canonicalSequence; bytes32 stateRoot;
  address terminalDomainRegistrar; bytes32 registrarCodeHash;
  bytes32 storageTrieKey; bytes32 expectedValue;
}
```

`routeKey=keccak256(abi.encode(ROUTE_KEY_TYPEHASH,sourceDomainId,bridgeExecutionHash, destinationDomainId))`. The exact type literal is

```
"RegistrationStorageStatementV2(uint256 settlementChainId,"
"address activeSettlementRouter,address bridgeDomainRegistry,bytes32 routeKey,"
"uint256 destinationChainId,uint64 protocolVersion,uint64 canonicalSequence,"
"bytes32 stateRoot,address terminalDomainRegistrar,bytes32 registrarCodeHash,"
"bytes32 storageTrieKey,bytes32 expectedValue)"
```

with the displayed fragments concatenated without whitespace or newline. `REGISTRATION_STATEMENT_TYPEHASH` and `publicInputSchemaHash` are both Keccak of those bytes, and `statementHash=keccak256(abi.encode(TYPEHASH,fields...))`. Registry derives all twelve

fields from its staged route, immutable chain/contract identities and the Router-returned canonical history; proof calldata supplies none of them.

The packed proof schema is committed by

```
REGISTRATION_MPT_PROOF_SCHEMA_LITERAL = concat(
  "MptProofV1=be16(accountNodeCount)||be16(storageNodeCount)||",
  "(be16(nodeLength)||canonicalRlpNode)*accountNodeCount||",
  "(be16(nodeLength)||canonicalRlpNode)*storageNodeCount;",
  "rootToLeaf;EthereumKeccak;canonicalHexPrefix;canonicalRlp;",
  "absenceRejected;valueRequired")
proofSchemaHash = keccak256(REGISTRATION_MPT_PROOF_SCHEMA_LITERAL)
```

The two hash layers are

```
REGISTRATION_MPT_VERIFIER_CONFIG_TYPE_LITERAL = concat(
  "RegistrationMptVerifierConfigV2(bytes32 publicInputSchemaHash,",
  "bytes32 proofSchemaHash,bytes4 selector,uint16 maximumNodesPerPath,",
  "uint16 maximumTotalNodes,uint16 maximumNodeBytes,",
  "uint32 maximumProofBytes,uint64 verificationGasLimit)")
registrationMptVerifierConfigurationHash = keccak256(abi.encode(
  keccak256(REGISTRATION_MPT_VERIFIER_CONFIG_TYPE_LITERAL),
  publicInputSchemaHash, proofSchemaHash, selector, 66, 132, 600, 80000, 8000000))

REGISTRATION_MPT_VERIFIER_DESCRIPTOR_TYPE_LITERAL = concat(
  "RegistrationMptVerifierDescriptorV2(address verifier,bytes32 runtimeHash,",
  "bytes32 configurationHash,bytes32 publicInputSchemaHash,",
  "bytes32 proofSchemaHash,bytes4 selector,uint16 maximumNodesPerPath,",
  "uint16 maximumTotalNodes,uint16 maximumNodeBytes,",
  "uint32 maximumProofBytes,uint64 verificationGasLimit)")
registrationMptVerifierDescriptorHash = keccak256(abi.encode(
  keccak256(REGISTRATION_MPT_VERIFIER_DESCRIPTOR_TYPE_LITERAL),
  verifier, runtimeHash, registrationMptVerifierConfigurationHash,
  publicInputSchemaHash, proofSchemaHash, selector, 66, 132, 600, 80000, 8000000))
```

The configuration layer deliberately excludes verifier, runtime hash and itself; the descriptor repeats every configuration field and rejects unless its `configurationHash` equals the independently recomputed first layer. The immutable verifier exposes only

```
registrationMptVerifierConfigHashV2() external view returns (bytes32)
REGISTRATION_MPT_VERIFIER_CONFIG_READ_GAS = 50000
```

Registry first checks `EXTCODEHASH`, then makes a zero-value exact-four-byte `STATICCALL` with that gas stipend and accepts exactly one 32-byte word equal to the recomputed configuration hash. Empty, short, trailing, faulting or mismatched returns reject before proof dispatch.

The verifier interface is exactly

```
verifyRegistration(
  RegistrationStorageStatementV2 calldata statement, bytes calldata proof)
    external view returns (bytes32 statementHash)
```

with selector from "verifyRegistration((uint256,address,address,bytes32,uint256,uint64,uint64,bytes32,address,bytes32,bytes32,bytes32),bytes)". Its ABI head is the twelve statement words followed by the proof offset word 0x1a0; the tail is canonical bytes length/data/zero-padding with no trailing calldata. Its total calldata length is exactly 452+ceil32(proof.length) bytes; the 80,000-byte cap measures the packed inner proof.length, not the outer ABI envelope. BridgeDomainRegistry makes one zero-value STATICCALL forwarding exactly 8,000,000 gas and accepts exactly 32 return bytes equal to its locally recomputed statement hash. Revert, OOG, malformed return, substituted statement or a Boolean result rejects before support state changes.

proof is MptProofV1: big-endian u16(accountNodeCount)||u16(storageNodeCount) followed by each root-to-leaf canonical-RLP node as u16(nodeLength)||nodeBytes. Each path has at most 66 nodes, each node at most 600 bytes, the whole proof at most 80,000 bytes, and any noncanonical RLP, wrong inline/hash reference, surplus node/byte or gas use above 8,000,000 rejects. The account proof fixes the registrar address and code hash; the storage proof fixes one nonzero registrationCommitment[protocolVersion] word. No Solidity struct or adjacent slot is inferred from that one path. The registrar exposes registrationCommitmentV2(uint64)viewreturns(bytes32) with exact 32-byte returndata, and stores only:

```
registrationCommitment =
  H("slot-chain-destination-registration-v1" ||
    u64(protocolVersion) || releaseManifestHash ||
    u256(destinationChainId) || destinationNamespace ||
    destinationDomainId || address20(destinationBridge) ||
    destinationInfrastructureHash || executionProfileHash)

baseSlot = H("slot-chain-terminal-domain-registrar.registrationCommitment.v1")
elementSlot = H(bytes32(uint256(protocolVersion)) || baseSlot)
storageTrieKey = H(elementSlot)
```

The account trie key is keccak256(address20(terminalDomainRegistrar)). The root reference is exactly stateRoot; a child reference shorter than 32 bytes must equal the complete inline child RLP and a 32-byte reference must equal Keccak of the next complete node. Other reference lengths reject. Canonical RLP uses the shortest length form, forbids leading length zeros and uses long form only for payloads of at least 56 bytes. A branch has exactly 17 items; an extension/leaf has exactly two. Hex-prefix flags are only 0–3, the even form has a zero low nibble, an extension path is nonempty, and a leaf is accepted only at exact key exhaustion. The account leaf is exactly [nonce,balance,storageRoot,codeHash]; nonce and balance are minimal unsigned integers of at most 32 bytes and both roots are exact 32-byte strings. The storage proof starts at that authenticated storageRoot, consumes the exact storageTrieKey, and returns the canonical minimal-RLP unsigned value equal to expectedValue. Missing, surplus or out-of-order nodes, trailing container bytes and an alternative yet value-equivalent RLP all reject. The storage-trie leaf value is canonical RLP of the minimal big-endian unsigned integer represented by the nonzero 32-byte commitment (and therefore preserves leading-zero semantics through the expected value). L1 recomputes the full commitment from its staged release manifest and accepts only exact equality; substitution of any field, mapping key, code hash, account or root rejects. The confirmation block, not manifest staging, starts the 214-block support delay. Each release adds at most 64 entries; there is no global cap, deletion, redirect or reinterpretation. A conflicting duplicate or inactive manifest rejects. Every credit creation performs

an O(1) check that its exact `(sourceDomainId,bridgeExecutionHash,destinationDomainId)` support entry has been active for 214 blocks; adding D2 never makes it usable with an old source endpoint before D2 itself is final.

V1 and V2 never share a Bridge account. The existing V1 proxy, Resolver, balance, nonce, status, pause/lock, QuotaManager and every Vault flow remain unchanged. V2-aware applications explicitly accept a fresh Bridge address and ContextV2; applications hard-coding V1 remain V1-only.

The SourceBridgeV2 bundle is deployed permissionlessly through a manifest-pinned immutable CREATE2 factory. Its descriptor binds factory address/runtime/config, bundle salt/init-code hash and the derived bundle-deployer runtime, the legacy V1 Bridge address that must not be reused, resulting SourceBridge address/runtime/config/layout, the Bridge-unique Registry and V2-only QuotaManager, support Registry/registrar/ epoch, immutable terminal verifier and Router binding, pauser and SignalService. The bundle-deployer address must equal the final 20 bytes of

```
keccak256(0xff || factory || salt || keccak256(init_code)).
```

The bundle init-code hash commits every acyclic child code/configuration primitive and excludes only the three addresses that its constructor derives from its own address. The exact Appendix formulas derive the bundle via CREATE2 and the Bridge, Registry and QuotaManager via its constructor's CREATE nonces 1–3 in one atomic transaction. A front-run is accepted only when the complete current bundle, code and configuration are exact; otherwise activation fails closed. No child address is embedded in the bundle CREATE2 preimage, so the graph is acyclic; the outer descriptor separately binds every derived address and current runtime/configuration.

Router retains one protocol-lifetime BridgeDomainRegistry, immutable factories, source bundles by descriptor ID and an append-only version-to-descriptor map. Current pointers are aliases only. Every incompatible successor has a fresh domain and fresh SourceBridge/Registry/Quota accounts; historical adapters keep their original bundle for terminal release and refunds. Each bundle owns its private nonce, status, pull and aggregate-liability mappings. It has no owner, upgrade, reinitializer, delegatecall or public mutation helper. The source generation is a fixed nonzero registration number and every credit binds its exact Bridge execution descriptor. At launch all source components share settlement Ethereum; cross-L1 support requires a new reviewed kind/domain. Admission also requires `0<srcChainId<=UINT64_MAX`, nonzero `sourceDomainId`, `0<srcEpoch<=UINT64_MAX`, `0<emittedAtBlock<=UINT64_MAX`, nonzero `msgHash`, `destChainId=l2ChainId<=UINT64_MAX`, a registered nonzero `destinationDomainId` and a nonzero `srcBridge`; no narrowing cast or destination-chain inference is permitted.

`BridgeInboxAdapter` is a non-proxy immutable contract. Its profile-bound runtime and constructor values fix the credit registry, source Bridge, immutable settlement router and ProtocolVersionManager. Its sole mutable configuration is a zero-to-nonzero `destinationDomainId` slot sealed once by that manager during manifest activation; the setter deletes its authority bit and cannot run twice. Enqueue rejects while the slot is zero. It has no upgrade, delegatecall, mutable target, arbitrary-call or caller-supplied authenticated value path. A different destination uses a distinct adapter/domain; an old adapter remains executable for its existing records.

The adapter is the exactly-once machine `NONE->QUEUED(index)`, keyed by `creditId`. Any caller supplies the separate queue processing fee and the full message preimage. The

payable adapter initially retains `msg.value`, derives both domains and the ID, direct-reads the immutable record and live source liability, and validates all fixed bounds including `block.timestamp<=enqueueBy`. It first calls the router's nonpayable

```
syncIngressV2() returns
  (uint8 result, uint64 activeProtocolVersion, uint64 routerGeneration)
```

as a registered immutable adapter. Its selector is the first four Keccak bytes of `"syncIngressV2()"`; calldata is exactly four bytes and returndata is exactly 96 bytes. STAMP=1 requires both narrow words nonzero; SYNCED=2 requires both zero. Unknown result, noncanonical high padding, short/trailing return or fault rejects and rolls the adapter back. If that call changes mode, the adapter records no credit or queue leaf, adds the still-local full value to `claimable[msg.sender]`, and returns `SYNCED`; a CEI pull withdrawal is callable independently of protocol mode and the caller retries in a later transaction. If unchanged, the same adapter immediately calls the router's payable

```
appendFromAdapterV2(uint64 activeProtocolVersion, uint64 routerGeneration,
                    uint8 kind, bytes descriptorBytes)
  payable returns (uint8 result, uint64 queueIndex)
```

with the exact STAMP words, its immutable kind and canonical fixed descriptor. Its selector is the first four Keccak bytes of `"appendFromAdapterV2(uint64,uint64,uint8,bytes)"`. The four-word head is version, generation, canonically padded kind and offset `0x80`; the tail is the exact length, 220-byte kind-0 or 541-byte kind-1 descriptor, and zero right-padding. Total calldata is respectively 388 or 708 bytes, with no other length valid. The exact 64-byte return is QUEUED=1 plus the new non-sentinel index; every other result, padding, length or index rejects. This second entry point does not call `sync()` again: it requires the registered caller, ACTIVE phase, and exact current version and generation stamp, then forwards all value with the exact descriptor to the queue. The immutable adapter has no entry point that can split, defer or redirect this pair and makes the second call before any other external call in the same top-level transaction. EVM call execution supplies no interleaving between them. A stale stamp after any migration transition rejects and returns the value by call-level rollback; it cannot preserve a second state transition inside a reverting payable call. A successful router retains zero wei. The queue checks `queueCount<UINT64_MAX`, appends at the current count and timestamps the deadline; then the source Bridge changes NEW to QUEUED at that exact index and the adapter records the returned index, all in the same revert domain. No reservation, source proof, PENDING state or finalization call exists. A repeated zero-value query returns the stored index; a duplicate with funds reverts. The source event alone never means queued.

Kind 0 requires a nonexpired `validUntil`, `validUntil<=enqueuedAt+MAX_FORCE_VALIDITY_SECONDS`, and `accountedGas=max(gasLimit,intrinsicGas,MIN_FORCE_ACCOUNTED_GAS)`. The initial validity horizon is seven days. Kind 1 uses the permanent conservative bound `accountedGas=MAX_FORCE_MESSAGE_GAS=5,000,000` for its encoded credit and routed pin. Kind 0 requires `0<accountedGas<=5,000,000` and `0<byteLength<=131,072`; kind 1 permits an empty Message payload and requires `byteLength<=131,072`. The active profile fixes one shared five-word fee schedule

```
(fixedIngressWei, executionWeiPerAccountedGas,
 proofWeiPerAccountedGas, permanentWeiPerByte,
 maximumAcceptedFeeWei).
```

```
requiredDeposit = fixedIngressWei
  + accountedGas * (executionWeiPerAccountedGas
                    + proofWeiPerAccountedGas)
  + byteLength * permanentWeiPerByte.
```

All additions and multiplications are checked `uint256`; every coefficient and the cap is nonzero, `requiredDeposit<=maximumAcceptedFeeWei`, and the release is invalid unless the maximum gas/byte geometry and `UINT64_MAX*maximumAcceptedFeeWei` are representable. Each payable adapter requires `msg.value=descriptor.deposit=requiredDeposit` exactly; overpayment and underpayment both revert before retained state. `fixedIngressWei` is the release-calibrated upper bound for all admission costs independent of `accountedGas` and `byteLength`, including the fixed descriptor, event, cold-account and fixed storage-write geometry. `executionWeiPerAccountedGas`, `proofWeiPerAccountedGas` and `permanentWeiPerByte` are respectively upper bounds, with the release's explicit safety margin, for execution gas, proving cost and permanent publication/storage of one raw input byte. The profile records the L1 base-fee, blob-fee and L2/prover price caps and margin used to derive all four amounts; a release whose benchmark corpus exceeds any bound rejects. The five words are identical in every authorization of one profile and are componentwise nondecreasing across successor profiles, so an old exact adapter never receives a cheaper active schedule after migration. This deposit covers execution, proof credit and permanent calldata/state costs. It is distinct from the SourceBridge's already escrowed `value+fee+liquidityFee`. The adapter verifies kind-1 source escrow and all static bounds before calling the router; failure leaves neither deposit, queue leaf nor partial credit. The mandatory codec corpus pins kind-0 raw-transaction lengths and kind-1 `Message.data.length` at 0, 31, 32, 33 and 131,072 bytes (with kind 0 rejecting zero), and rejects a one-byte mismatch between the normalized Message, descriptor, fee and per-block byte accounting.

Only `ActiveSettlementRouter` may append, and it accepts these two entry points only from an exact typed `authorizedIngress[caller]` entry. The target release/profile precommits the complete authorization root and deployed adapter objects. Cutover atomically binds Kind 0 and stages the exact Bridge support route. After the canonical destination-registration proof and the 214-block support delay, anyone may one-shot bind that release's exact historical Bridge adapter; no post-cutover governance action is required. Address, object, domain or kind-role conflicts reject, while old exact adapters remain usable for status, cancellation and refund of their historical records. Only the exact adapter deployment named by the active authorization may append; after cutover an old adapter cannot enqueue a still-NEW predecessor credit. That source credit remains permissionlessly cancelable after its strict `enqueueBy`. Every adapter fixes this router and queue in constructor/configuration. The router calls the active Settlement's `sync()` only in nonpayable `syncIngress` under a non-reentrant guard and reports `SYNCED` without accepting value if any transition occurred. Its adapter caller preserves that transition, creates no admission record and keeps the entire fee locally claimable by the original caller. Otherwise it computes `enqueuedAt=block.timestamp` and `dueAt=max(enqueuedAt+FORCE_DELAY,lastDueAt,recoveryExpiry+1)` (the last term is zero outside recovery), and appends atomically. Neither a sync-result Boolean, enqueue timestamp nor due time is accepted in router calldata; payable append rechecks the shared gate is ACTIVE and the exact stamp before forwarding value, without a second sync decision. Router, queue and adapter are non-reentrant, and failure of any later append/Bridge/adapter write reverts the value transfer and all state. No public caller can invoke the forwarding path. `ForcedQueue` stores one authoritative `QueueDescriptor` per index. It is the complete fixed-width kind-0 or kind-1 leaf preim-

age in Appendix A, excluding only kind-0 `rawTx` bytes themselves but including their hash. Thus sender, nonce, chain IDs, byte/gas fields, fee, validity, refund address, enqueue/due times, deposit and every bridge-credit field remain durable. Kind 1 requires DIRECT and zero Vault/capsule fields. The Merkle leaf is always recomputed from this stored descriptor; admission atomically writes the descriptor, deposit/escrow accounting, append frontier and root or writes nothing. The fee is deliberately nonrefundable after append: it buys FIFO storage, proving and processing even when an expired or invalid kind-0 item has no Ethereum transaction. The queue stores `depositPrefix[i+1]=depositPrefix[i]+deposit[i]` with checked addition. It is also the sole owner of `cursor`; neither Settlement, Protocol nor either ingress adapter has a second descriptor array, root, count, cursor or due-time cache. Settlement reads the queue's `dueAt` directly for the canonical head and a stored best's live boundary; there is no callback, cached deadline or duplicated scalar. Missing indices return `UINT64_MAX`. Thus `dueAt` is nondecreasing and force-due/coverage checks are one authoritative O(1) read. The queue exposes exactly two cursor-authority transitions:

```
advanceCursor(uint64 expectedStart, uint64 end,
              address proofBeneficiary)              // active Settlement only
setActiveSettlement(address expectedOld, address new) // router only
```

`advanceCursor` requires `cursor=expectedStart<=end<=count` and writes the cursor and, in O(1), moves `depositPrefix[end]-depositPrefix[expectedStart]` from unconsumed escrow to `claimable[proofBeneficiary]`. The beneficiary is the exact nonzero public input authenticated by the accepted proof; it is not chosen by the queue caller. The queue remains the ETH custodian and enforces the solvency lower bound `address(this).balance>=unconsumedEscrow+totalClaimable`; every claim-ledger mutation updates `totalClaimable` by the same checked amount, so no global sum or scan is performed. ETH forced into the contract without an append is surplus, never enters either liability and cannot make an append, advance or withdrawal revert. It exposes an unpausable CEI pull withdrawal. There is no per-item callback and no "unused" refund. `setActiveSettlement` requires the stored old address and is called only inside bootstrap or an atomic version switch. The ingress router never advances consumption, and a frozen Settlement cannot do so.

The queue is a protocol-lifetime, depth-64 append-only Merkle vector with fixed capacity `UINT64_MAX` items: the final leaf is deliberately unused. It stores root, `uint64` count/cursor, cumulative deposit prefix and a 64-node append frontier. Admission rejects only when `count=UINT64_MAX`; no version migration rotates or resizes the queue. A canonical DFS range multiproof authenticates the consumed leaves and, when present, the first excluded boundary leaf under the frozen `(forceRoot,forceCutoff)`. It supplies no fully covered subtree and no unused hash; at most 257 sibling hashes authenticate a 65-leaf block range. Skipping, reordering, changing the start index or claiming an early boundary changes the reconstructed root. Kind-0 raw bytes are enqueue calldata and their hash is in the leaf; blobs are forbidden on this availability path. The deliberately unused final tree leaf avoids a nonexistent height-64 frontier slot when the old index has all 64 bits set. At one billion appends per second the usable space lasts more than five centuries, so exhaustion is outside any credible Ethereum lifetime while remaining an explicit hard stop.

Each proved block consumes the unique maximum FIFO prefix satisfying all three bounds simultaneously. `forceGasBudget` is 13,000,000 exactly for the first release-activation block and 20,000,000 for every steady block:

$$count \le 64, \qquad \sum byteLength \le 1{,}048{,}576, \qquad \sum accountedGas \le forceGasBudget.$$

Every admitted item individually fits all three. The circuit exposes `(start,end,count, totalBytes,totalAccountedGas,forceRoot,forceCutoff,nextDueAt,forceGasBudget)` and verifies the range multiproof. Maximality means either `end=forceCutoff`, or the authenticated next leaf would exceed at least one remaining per-block cap; an early stop with a fitting leaf rejects. Candidate totals are independently capped as stated in §5. Launch enforces the separate activation and steady Anchor/force/margin inequalities in §12, plus the analogous witness-byte bounds. Because an item is at most 5,000,000 accounted gas, even the activation cap always admits the due FIFO head; a 20m backlog is split maximally under 13m in that one block and 20m thereafter rather than making activation invalid.

For the candidate's exact consumed interval, Settlement reads at most 256 descriptors plus the first excluded boundary descriptor when one exists under the frozen cutoff. Internal per-block boundaries are already the next consumed descriptor. It computes

```
H("slot-chain-force-descriptor-list-v2" || u64(start) ||
  u16(consumedCount) || u8(hasBoundary) ||
  (u64(index)||u8(kind)||u16(descriptorByteLength)||descriptorBytes)*)
```

as `forcedDescriptorCommitment`. The number of rows is the consumed count plus the zero-or-one boundary and is at most 257. `descriptorBytes` is the exact kind-specific packed sequence used by the leaf after its ASCII domain and index; its length is the unique constant for that kind. The circuit reconstructs every leaf, range proof, per-block maximum, candidate total and processing-fee transfer from these L1-authenticated descriptors. It additionally requires raw bytes for an unexpired kind 0 and the durable credit record for kind 1. An expired kind 0 needs neither historical calldata nor any unstored preimage.

Forced envelopes are auxiliary protocol inputs, not blindly copied into the Ethereum transaction list. Each block's transaction order is: (1) the exact profile-defined AnchorV4 transaction, (2) one deterministic `InboxApplyRouterV2` system transaction, (3) upfront-valid kind-0 raw transactions in FIFO order, then (4) discretionary transactions. Sequential pre-state classification puts expired, wrong-nonce, underfunded or underpriced kind-0 items only in `InboxApplyRouterV2`'s disposition vector; they have no Ethereum transaction, nonce or receipt. Upfront-valid kind 0 appears as an ordinary Ethereum transaction, whose success or revert has ordinary nonce/gas/receipt semantics. Kind 1 always produces its idempotent inbox credit inside `InboxApplyRouterV2` and never calls the eventual destination. The system calldata commits `(forceStart,forceEnd,(queueIndex,disposition,txIndex,resultHash)*)`; the circuit derives the transaction and receipt roots from this exact order. The full consumed fee interval is pull-credited to the proof beneficiary by the same L1 canonical commit. `refundAddress` remains part of the signed kind-0 descriptor for transaction-level compatibility but creates no queue-fee entitlement in this version; changing consumption or payment requires changing the descriptor and proof statement.

Disposition bytes are fixed: `0=EXPIRED_NO_TX`, `1=NONCE_NO_TX`, `2=FUNDS_NO_TX`, `3=FEE_NO_TX`, `4=INCLUDED_TX` and `5=BRIDGE_CREDIT`. Classification tests in that order; the first applicable no-transaction reason wins. Codes 0–3 require `txIndex=UINT32_MAX` and `resultHash=0`. Code 4 requires the exact Ethereum transaction index; `resultHash` is legacy Keccak-256 of the raw signed EIP-2718 transaction bytes. This ordinary Ethereum transaction hash is known before execution. It is deliberately not a receipt hash: putting a later receipt hash in the earlier InboxApply calldata would create a fixed point through cumulative gas. Code 5 requires `txIndex=UINT32_MAX` and this exact durable credit hash:

```
H("slot-chain-bridge-credit-result-v11" || u64(queueIndex) || creditId ||
  msgHash || u256(srcChainId) || sourceDomainId ||
  u64(srcEpoch) || address20(srcBridge) ||
  bridgeExecutionHash || u64(emittedAtBlock) || destinationDomainId ||
  u256(destChainId) || u64(enqueueBy) || address20(sender) ||
  address20(srcOwner) || address20(destOwner) || u256(value) ||
  u64(fee) || u64(liquidityFee) || calldataHash ||
  u8(refundMode) || address20(refundVault) ||
  refundCapsuleHash || escrowId)
```

InboxApplyRouterV2 recomputes it from the authenticated descriptor before routing the exact tuple and `resultHash` to `InboxCreditStoreV2`. That immutable store, neither Bridge nor legacy SignalService, owns the persistent destination pin. Resolver rotation and SignalService upgrades cannot retarget, fabricate or orphan an existing credit. Unknown codes or noncanonical auxiliary fields reject.

AnchorV4 and InboxApplyRouterV2 use a profile-defined, unsigned custom L2 system transaction type whose sender is injected by the post-cutover fork, not recovered from ECDSA. That type is invalid under every legacy fork, and the two reserved sender addresses are chosen with no known signing key. The circuit requires their exact profile transactions once each at indices 0 and 1 and rejects any ordinary, forced or discretionary transaction recovered from either sender. These reserved senders and the router/store pin ABI are permanent across every V2 profile; a later fork must still dispatch old routes through this router. InboxApplyRouterV2 stores the last applied L2 block number and rejects a second application in that block. An ordinary caller therefore cannot invoke the privileged path before or after cutover.

The launch fork deploys one protocol-lifetime non-proxy immutable `InboxApplyRouterV2` before every endpoint-specific immutable `InboxCreditStoreV2`. Their permanent operations are:

```
struct InboxCreditV2 {
  uint64 queueIndex; uint64 srcChainId; bytes32 sourceDomainId;
  uint64 srcEpoch; address srcBridge; bytes32 destinationDomainId;
  bytes32 msgHash; bytes32 resultHash; bytes32 sourceContextHash; uint256 value;
  uint64 executionFee; uint64 liquidityFee;
}
struct InboxRowV2 {
  uint64 queueIndex; uint8 disposition; uint32 txIndex;
  bytes32 resultHash; bytes kind1Descriptor;
}
InboxApplyRouterV2.apply(uint64 forceStart, InboxRowV2[] calldata rows)
markReceivedFromInboxBatch(InboxCreditV2[] calldata rows)
    returns (bytes4 magic)
routeConfigHashV2() view returns (bytes32)
verifyInboxCredit(uint64 srcChainId, bytes32 sourceDomainId,
                  uint64 srcEpoch, address srcBridge,
                  bytes32 destinationDomainId, bytes32 msgHash)
    view returns (bytes32 creditId, uint64 processBy, uint256 value,
                  uint64 executionFee, uint64 liquidityFee,
                  bytes32 sourceContextHash)
```

```
getInboxCreditSlot(bytes32 sourceDomainId, address srcBridge,
                   bytes32 destinationDomainId,
                   bytes32 creditId) pure returns (bytes32)
liquidityQuoteV2(bytes32 creditId) view returns
  (bytes32 resultHash, uint256 value, uint64 executionFee,
   uint64 liquidityFee, uint64 processBy)
DestinationBridge.liquidityReservationStateV2(bytes32 creditId) view returns
  (bytes32 destinationDomainId, address destBridge, address inboxCreditStore,
   uint8 status, uint64 terminalIndexPlusOne)
```

Every endpoint store constructor authorizes the same InboxApplyRouterV2; only that router may call the store batch operation, and only after that endpoint's one-shot activation. Router state includes checked `nextQueueIndex`, which the circuit requires to equal `CanonicalCore.messageCursor`. Every block calls the router once with its complete contiguous zero-to-64 forced-row interval, including kind-0 dispositions, in global queue order. It first validates the whole vector, exact result hashes, no duplicate `(domain,creditId)`, and each one-shot route `destinationDomainId->(store,Bridge,routeConfigHash,codeHash)`. Only the registrar may append a route; domain, store and Bridge are each unique and no route can be deleted or redirected. The router rechecks code/config/domain and Bridge on every use, partitions only pin side effects into maximal contiguous same-domain runs, then makes at most 64 fixed-selector, zero-value calls and requires the exact success magic. It advances `nextQueueIndex` only after all calls succeed. A missing route, conflict or later call failure reverts the cursor and every earlier store write. There is no sorting and no loop over historical routes. The batch selector is the canonical Solidity selector for the signature above. Precisely, it is the first four Keccak bytes of `"apply(uint64,(uint64,uint8,uint32,bytes32,bytes)[])"`. Calldata after the selector has `forceStart` in word 0 and the canonical rows offset `0x40` in word 1. At byte `0x40` is the row count $n \leq 64$, followed by n element offsets relative to the beginning of the element-head area (byte `0x60` of the arguments). Each element encoding is five head words in the displayed field order; its bytes offset is exactly `0xa0`, followed by the bytes length, data and zero right-padding. A kind-0 row requires zero descriptor length. A kind-1 row requires exactly the Appendix 541-byte durable descriptor body, padded to 544 bytes. Element offsets must be contiguous and minimal; aliases, gaps, overlap, trailing calldata, nonzero padding or a length inconsistent with the row kind rejects. Consequently the largest 64-row calldata is 49,252 bytes including selector. The bound statement's `inboxSystemCalldataHash` is Keccak of this complete byte string, while `inboxSystemTxHash` is Keccak of the complete type-`0x7f` envelope that contains it. The canonical Inbox rows in the L1 activation call are the data-availability preimage for both hashes; a proof-only hash is insufficient. Success returndata is exactly one 32-byte ABI word equal to `0x49425632` (ASCII "IBV2") followed by 28 zero bytes; empty, short, long, malformed or different returndata rejects even if the call itself succeeds. The getter likewise must return exactly 32 bytes, and its value is

```
H("slot-chain-inbox-route-config-v1" ||
  address20(inboxApplyRouterV2) || address20(destinationBridge) ||
  address20(terminalDomainRegistrarV2) ||
  destinationDomainId)
```

The router pins that hash and the store runtime hash at registration and rechecks both by bounded `STATICCALL/EXTCODEHASH` before any batch call. Runtime hash, not a caller assertion, pins the getter semantics. A successful no-op implementation cannot advance the cursor

because the same transaction's post-state circuit must find every fresh pin at the Appendix-defined slot; the conformance corpus includes a magic-returning no-op. That pin stores the exact authenticated result hash, source-context hash, value, execution fee, liquidity fee and `processBy`; none is inferred from a later Message. `liquidityQuoteV2` returns exactly five canonical ABI words and is callable only through a registrar-routed Store identity. The Pool checks the Store runtime/configuration and route before reading it; short/trailing/malformed return or a zero result hash rejects without reserving. The validity circuit proves those stored words equal the consumed V11 descriptor/result hash, so an under- or over-reservation cannot be created by substituting a quote. `liquidityReservationStateV2` likewise returns exactly five canonical ABI words under the manifest-pinned Bridge code/configuration. It preserves the existing IBridge enum encoding NEW=0, RETRIABLE=1, DONE=2, FAILED=3 and RECALLED=4. NEW and RETRIABLE are the only reservable statuses and require `terminalIndexPlusOne=0`; every terminal status has the checked terminal index plus one. Padding, short/trailing data or an inconsistent Store/domain/ Bridge tuple rejects. The canonical empty vector requires `forceStart=nextQueueIndex`, makes no store call and leaves the cursor unchanged, but still records the exact L2 block number; a second apply in that block rejects.

`kind1Descriptor` is empty for dispositions 0–4 and exactly the Appendix-defined 541-byte kind-1 descriptor for disposition 5; all other lengths reject. The router decodes that descriptor, recomputes `creditId` and the complete `bridge-credit-result-v11` hash, and derives the destination route. The circuit separately proves the same row/descriptor against the consumed L1 queue leaf. No absent owner, value, execution-hash, capsule or escrow field is treated as authoritative calldata.

Every registered store is nonproxy, unpausable, callback-free and permanently implements this exact all-or-revert pin ABI. Each run is completely validated before its first write. Activation is blocked unless the profile's system gas margin covers the maximum *reachable* cold path under the same count and accounted-gas caps proved by the circuit. In a steady block the route-call maximum is `min(64,floor(20,000,000/5,000,000))=4`, hence at most four route calls and sixteen fresh pin writes. In the first activation block the analogous 13,000,000-gas cap admits at most two kind-1 rows and eight fresh pin writes. The mandatory benchmark set includes all-cold four-kind-1 steady execution, the 64-row steady mix of 61 minimum-gas kind-0 rows plus three kind-1 rows, and the 64-row activation mix of 62 minimum-gas kind-0 rows plus two kind-1 rows. It also proves that the next non-fitting FIFO leaf is excluded. An unreachable 64-kind-1/256-write trace is not an activation requirement. Each kind-1 row conservatively carries `accountedGas=MAX_FORCE_MESSAGE_GAS=5,000,000`; opcode repricing may reduce throughput but cannot leave an old head underfunded. The 20,000,000 per-block forced-gas budget therefore admits at most four kind-1 rows per block. The store validates widths and nonzero fields and performs all writes atomically. The slot is exactly

```
H("slot-chain-inbox-credit-slot-v4" || sourceDomainId ||
  address20(srcBridge) || destinationDomainId || creditId)
```

Each pin occupies exactly four mapping values under the profile-pinned storage layout: `inboxResultHash[slot]`, `inboxSourceContextHash[slot]`, `inboxValue[slot]` and `inboxPackedTerms[slot]`. Packed-term bits 0–63 are `processBy`, 64–127 `executionFee`, 128–191 `liquidityFee`, and 192–255 are zero. A nonzero result hash is the received bit. An absent result stores those four words, with `processBy=block.timestamp+BRIDGE_PROCESS_TTL_SECONDS`. The initial TTL is 2,592,000 seconds. The contract exposes no proxy, upgrade,

delegatecall, mutable writer/reader/domain target, arbitrary storage primitive or pause gate. A repeated row succeeds without writes only when the stored result, context, amounts and deadline match and never extends `processBy`; any conflict, ordering error or excess row reverts the whole batch before a write. `verifyInboxCredit` recomputes `creditId` and the slot, requires the received bit and nonzero result, and requires `msg.sender` to be the constructor-fixed destination Bridge for that exact local domain before returning all six exact words. Runtime hash, constructor router/Bridge/gate values, the one-shot sealed local-domain slot and initial empty mappings are migration-proved; the noncircular construction descriptor deliberately excludes the derived domain value. This is a new bytes32-domain ABI; it never overloads legacy `getSignalSlot(uint64,address,bytes32)`.

BridgeV2 introduces these exact static context tuples:

```
struct SourceContextV2 {
  uint64 protocolVersion; uint8 kind; bytes32 creditId; bytes32 msgHash;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  address sourceBridge; bytes32 sourceBridgeExecutionHash;
  uint64 emittedAtBlock; uint64 queueIndex;
}
struct DestinationContextV2 {
  uint256 destinationChainId; bytes32 destinationDomainId;
  address destinationBridge; bytes32 releaseManifestHash;
  bytes32 executionProfileHash;
}
```

Here `protocolVersion`, manifest and profile are the immutable destination release registered for that domain; `kind=1`. The remaining words are recomputed from the durable pin and V11 descriptor. Address, `uint8` and `uint64` ABI padding is canonical. The Bridge rejects any caller-supplied word mismatch before status, reservation, value or external effects. The typed context commitment is

```
SOURCE_CONTEXT_TYPE_LITERAL =
  "SourceContextV2(uint64 protocolVersion,uint8 kind,bytes32 creditId,"
  "bytes32 msgHash,bytes32 sourceDomainId,uint64 sourceRegistrationEpoch,"
  "address sourceBridge,bytes32 sourceBridgeExecutionHash,"
  "uint64 emittedAtBlock,uint64 queueIndex)"
SOURCE_CONTEXT_TYPEHASH = keccak256(concat(the four quoted fragments))
sourceContextHash = keccak256(abi.encode(SOURCE_CONTEXT_TYPEHASH,
  all SourceContextV2 fields in the displayed order))
DESTINATION_CONTEXT_TYPE_LITERAL =
  "DestinationContextV2(uint256 destinationChainId,bytes32 destinationDomainId,"
  "address destinationBridge,bytes32 releaseManifestHash,"
  "bytes32 executionProfileHash)"
DESTINATION_CONTEXT_TYPEHASH = keccak256(concat(the three quoted fragments))
destinationContextHash = keccak256(abi.encode(DESTINATION_CONTEXT_TYPEHASH,
  all DestinationContextV2 fields in the displayed order))
```

Each literal is the byte concatenation of its displayed quoted fragments with no inserted whitespace or newline. InboxApplyRouterV2 derives it only from the V11 descriptor, queue index and registrar-bound release version, and the Store pins it with the credit. `verifyInboxCredit`

returns exactly six canonical ABI words including that hash. Destination Bridge recomputes it from the supplied tuple and requires equality; queue index, emission block and source execution hash are therefore not caller assertions hidden behind `creditId`. The existing send ABI remains V1; the complete additive V2/recovery ABI is:

```
struct LiquiditySettlementV1 {
  bytes32 ticketId; address l1Recipient; uint256 settlementAmount;
}
sendMessageV2(Message, uint64 liquidityFee) -> (bytes32 msgHash, Message)
messageContextV2(bytes32 msgHash)
  -> (bytes32 creditId, SourceContextV2, DestinationContextV2)
markQueuedV2(bytes32 creditId, uint64 queueIndex)
cancelUnqueuedMessageV2(bytes32 creditId)
finalizeDoneV2(bytes32 creditId, uint64 protocolVersion,
               uint64 canonicalSequence, uint64 leafIndex,
               LiquiditySettlementV1 settlement,
               bytes32[64] siblings)
processMessageV2(Message, SourceContextV2, DestinationContextV2)
  -> (Status, StatusReason)
retryMessageV2(Message, SourceContextV2, DestinationContextV2,
               bool isLastAttempt)
failMessageV2(Message, SourceContextV2, DestinationContextV2)
expireMessageV2(bytes32 creditId)
recallMessageV2(bytes32 creditId, uint64 protocolVersion,
                uint64 canonicalSequence, uint64 leafIndex,
                bytes32[64] siblings)
withdrawSourceCreditV2(address payable recipient)
withdrawLiquidityClaimV2(address payable recipient)
messageStatusV2(bytes32 creditId) -> Status
invocationContextV2() view returns (SourceContextV2, DestinationContextV2)
ActiveSettlementRouter.canonicalAt(uint64 protocolVersion,
                                   uint64 canonicalSequence)
  -> CanonicalHistoryV2
TerminalSignalVerifier.verifyTerminalSignal(
  bytes32 destinationDomainId, address destBridge,
  bytes32 creditId, uint8 terminal, bytes32 liquiditySettlementHash,
  uint64 protocolVersion,
  uint64 canonicalSequence, uint64 leafIndex,
bytes32[64] siblings) -> bytes32
```

Immediately around an allowed recipient CALL, Bridge installs those complete tuples in transient storage. `invocationContextV2` succeeds only while that exact outer call frame is live; it reverts before installation, after the finally-style clear, and during any failed/reverted attempt. The shared non-reentrant guard forbids a nested Bridge attempt from replacing the context. An implementation that lacks EIP-1153 support for the launch fork is invalid; persistent scratch storage is not an alternative.

Recipient invocation is a frozen release policy, not an application allowlist. The profile publishes a list of at most 64 addresses, sorted by unsigned 160-bit value and free of duplicates, denied addresses and

```
INVOCATION_POLICY_TYPEHASH = keccak256(
  "InvocationPolicyV2(uint16 count,bytes32 addressesHash,bytes4 hookSelector)")
addressesHash = keccak256(address20(denied[0]) || ... || address20(denied[n-1]))
invocationPolicyHash = keccak256(abi.encode(
  INVOCATION_POLICY_TYPEHASH, uint16(n), addressesHash,
  IMessageInvocable.onMessageInvocation.selector))
```

The destination Bridge configuration, source support record and release profile all bind that hash and exact list; historical lists cannot be changed. The list must contain the zero address, destination Bridge, both SignalServices, native QuotaManager, manifest pauser, every V1 Vault named by the release, any DelegateController, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, NativeLiquidityPoolV2, TerminalAccumulatorV2, TerminalSignalVerifier, every context/helper endpoint and every other system or Bridge-trusting endpoint enumerated by the profile. SourceBridge rejects a denied `message.to` before any ID, escrow or Registry write. Destination Bridge repeats the check; a historical or corrupted omission transitions to FAILED with zero target, quota, fee and Pool consumption, enabling the ordinary source refund proof. If an exact reservation is already live, that same terminal journal releases it, restores the ticket's unreserved balance and decrements `reservedCount(domain)`; otherwise the Pool is unchanged. A release fault rolls back FAILED, its terminal leaf and every Pool write. The corpus includes a target denied only at destination after source authorization and a pre-existing reservation, proving no LP capital or retirement counter is stranded. ProtocolVersionManager materializes the source-side list once at finalized domain-support confirmation. BridgeDomainRegistry exposes only

```
invocationPolicyV2(bytes32 destinationDomainId, address target) view returns
  (bytes32 invocationPolicyHash, bool denied, bytes4 magic)
```

with exactly 96 bytes, canonical Boolean/padding and magic `0x49505632` (`IPV2`). SourceBridge pins Registry code/configuration and a gas limit, requires the returned hash to equal the supported domain and rejects on any call/length/value fault. Destination Bridge uses its deployment-proved local mapping with the same hash/list; neither side accepts a Merkle proof, caller Boolean or mutable application registry.

All other addresses are permissionless and may acquire code after send through CREATE or CREATE2; code presence is decided only at execution. No-code accounts are treated as EOAs. Empty data with zero value succeeds without CALL. Empty data with positive value and data of length one to three use an ordinary CALL and therefore the recipient fallback/receive path. For data length at least four, a contract CALL is permitted only when the first four bytes equal `IMessageInvocable.onMessageInvocation(bytes).selector` and the complete calldata is the canonical ABI encoding of that one bytes argument; malformed or another selector is invocation-prohibited. Invocation-prohibited processing makes no target CALL: it creates the destination owner's value pull and the successful processor's execution-fee pull, consumes quota/reservation and appends DONE atomically. EOA delivery uses ordinary CALL semantics and never interprets a selector. The profile corpus covers zero/1/3/4-byte boundaries, unknown and post-CREATE2 accounts, each denyset member, malformed hook ABI and a target attempting context/reentrancy substitution. This permission does not let a later release turn a previously unprivileged counterfactual address into an old-Bridge-trusting system endpoint. Every future privileged endpoint that accepts a Bridge/context call must constructor-pin exactly one fresh `destinationDomainId` and its fresh DestinationBridge, and must revalidate the complete transient `DestinationContextV2` inside its hook. It must reject

every historical Bridge and any registry-wide "recognized Bridge" predicate. Each successor profile, deployment commitment and verifier enforce that rule for every newly privileged endpoint before publishing its address. The conformance corpus pins the counterfactual sequence send-to-X, deploy-new-privileged-X, call-from-old- Bridge and requires X to reject without effect. An application that chooses to trust historical Bridges is ordinary application risk and cannot be named a protocol/Vault/system endpoint by a release. `CanonicalHistoryV2` is the fixed tuple

```
(uint64 protocolVersion, bytes32 executionProfileHash,
 uint64 canonicalSequence, uint48 l2BlockNumber, bytes32 blockHash,
 uint64 tipSlot, bytes32 stateRoot, uint64 messageCursor,
 bytes32 winningDataCommitment, uint256 nextBaseFee,
 uint64 nextExcessBlobGas, bytes32 terminalRoot,
 uint64 terminalCount, uint64 canonicalizedAtBlock)
```

and the only caller-supplied proof payload is

```
(uint64 protocolVersion, uint64 canonicalSequence,
 uint64 leafIndex, bytes32[64] siblings)
```

The verifier reads the exact sequence-tagged ring entry from Router, requires `leafIndex<terminalCount`, derives the exact domain-separated leaf from index, domain, Bridge, credit ID, terminal and liquidity-settlement hash, and folds exactly 64 siblings to the stored root. There is no caller-supplied Canonical tuple, dynamic array, RLP node, current EVM state proof or trailing element. `terminal` is exactly 1 for DONE or 2 for FAILED; all other values reject. FAILED requires `liquiditySettlementHash=bytes32(0)`. DONE requires

```
liquiditySettlementHash = H("slot-chain-liquidity-settlement-v1" ||
  ticketId || address20(l1Recipient) || u256(settlementAmount))
settlementAmount = message.value + message.fee.
```

For DONE, `ticketId`, `l1Recipient` and the resulting `liquiditySettlementHash` are each nonzero; checked addition rejects overflow. A Pool deposit whose derived ticket ID is zero reverts before incrementing its sequence or accepting value. SourceBridge recomputes that hash from the caller-supplied fixed settlement tuple and the immutable credit, so a proof copier cannot redirect the LP pull. The verifier constructor fixes only `ActiveSettlementRouter` and depth 64; it is generic across every append-only destination endpoint. Its explicit domain and Bridge arguments are low-level leaf inputs, never source authority. Each historical source Bridge terminal transition loads `destinationDomainId` from its immutable `CreditAuthorization`, bounded-static-reads the corresponding permanent `destBridge` from `BridgeDomainRegistry`, and passes those values. It rejects if the descriptor is absent or differs from the proof; no caller or proof chooses either value. A D1 source endpoint can therefore finalize D1 and a later registered D2 without changing its verifier. Here `Message` is the existing IBridge tuple and `msgHash=hashMessage(Message)`. Destination processing first requires the context's destination Bridge to equal its own address and the context's domain to equal its one-shot sealed local destination domain. It then recomputes that domain from the pinned chain ID, genesis, BridgeInboxAdapter, active settlement router, terminal verifier, InboxApplyRouterV2, InboxCreditStoreV2, protocol-release authority, terminal-domain registrar, terminal accumulator, Bridge address/frozen execution hash, infrastructure hash and namespace. Only

then do they recompute `creditId` from the Message hash and context before any status, value or external effect. No caller-supplied foreign domain may query a pin through a different funded Bridge. Source terminal functions load the immutable authorization and current permanent liability by credit ID, while destination expiration loads the permanent inbox pin; neither needs the Message preimage. Queue marking is authorized only to the exact adapter in the recorded destination domain. The read-only status view accepts the derived key. The SourceBridge records both contexts. Its profile-pinned V2 event carries the credit ID, message hash, both contexts and the normalized Message; destination changes emit the credit ID and new status.

Native delivery uses a protocol-lifetime immutable `NativeLiquidityPoolV2`, not a finite protocol reserve and not native minting. Any LP may fund a ticket and bind the L1 address that will receive the source-side reimbursement:

```
depositLiquidity(address l1Recipient) payable returns (bytes32 ticketId)
reserveLiquidity(bytes32 ticketId, bytes32 destinationDomainId,
                 address destBridge, bytes32 creditId)
withdrawUnreserved(bytes32 ticketId, address payable recipient,
                   uint256 amount)
reservationV2(bytes32 creditId) view returns
  (bytes32 ticketId, address l1Recipient, uint256 amount, uint8 state)
consumeReservationV2(bytes32 creditId) returns
  (bytes32 ticketId, address l1Recipient, uint256 amount)
releaseReservationV2(bytes32 creditId)
```

`ticketId=H("slot-chain-liquidity-ticket-v1"||u256(chainId)||address20(pool)||address20(depositor)||address20(l1Recipient)||u64(ticketSequence))`. The pool keeps checked per-ticket unreserved/reserved amounts, immutable nonzero depositor/L1 recipient, aggregate totals and one reservation per credit. Reservation and unreserved withdrawal require exact `msg.sender=ticket.depositor`; approval, operator and bearer transfer do not exist. Withdrawal requires a nonzero recipient, debits before the CALL and atomically restores on recipient failure or reentrancy. A different ticket, caller or recipient substitution cannot move accounted funds. Reservation is depositor-only and, through the registrar's immutable route, bounded-static-reads both the exact Store quote and exact Bridge reservation state. It requires the existing inbox pin, exact domain/Bridge/Store/credit, status NEW or RETRIABLE, zero terminal index, `block.timestamp<=processBy`, and `amount=message.value+message.fee>0`; it records the pin's nonzero `liquidityFee` but never trusts a caller amount. First exact reservation wins. An LP chooses whether that fee compensates its capital and cross-chain cost; the protocol supplies no oracle or coercive fixed price. `NativeLiquidityPoolV2` is the sole writer of checked `uint64reservedCount[domain]`. A successful ABSENT-to-RESERVED transition increments it exactly once as the final write in the same journal as the ticket and aggregate reservation debits. A failed read, validation, duplicate reservation, foreign-Bridge call or reverted transfer leaves it unchanged. A RESERVED-to-CONSUMED or RESERVED-to-RELEASED transition decrements it exactly once in the same journal as the funds, aggregate and terminal transition; underflow reverts the complete journal. Deposit requires nonzero value and recipient, a nonexhausted ticket sequence and at least one registrar-sealed destination domain. Before the first tx0 domain activation it always reverts, so an attacker cannot front-run the launch's proved zero-ticket/zero-reservation prestate; forced ETH remains unaccounted surplus. Later releases preserve arbitrary live tickets exactly and cannot reset that gate or sequence. The Pool pins

both runtime/configuration hashes and requires exact five-word returndata from each view before changing a ticket. A terminal transition ordered first makes a later reservation reject; a reservation ordered first is necessarily consumed or released by that transition in the same rollback domain. A DONE/FAILED/expired credit can therefore never acquire a stranded reservation or inflate `reservedCount(domain)`.

Only the registrar-bound Bridge for that domain may invoke the pool's internal methods:

```
consumeReservationV2(bytes32 creditId)
releaseReservationV2(bytes32 creditId)
```

ReservationState is exactly ABSENT=0, RESERVED=1, CONSUMED=2 and RELEASED=3. Consume performs CEI, marks the reservation CONSUMED, decrements ticket/aggregate/domain reserved counts, transfers its exact amount to the calling Bridge, and returns the same immutable ticket/recipient/amount as exactly three canonical ABI words (96 bytes). DestinationBridge derives the DONE settlement hash only from that return; a caller tuple, pre-read cache, short/trailing/malformed return or substituted field rejects the nested frame. Release has no return data, marks it RELEASED and returns the exact amount to the same ticket's unreserved balance. Either is one-shot; nonempty release returndata or any reentrant/transfer fault reverts the entire Pool/Bridge journal. The hard pool invariant after every entry is

```
address(pool).balance >=
  totalUnreservedWei + totalReservedWei
sum(ticket.unreservedWei) = totalUnreservedWei
sum(live reservation.amount) = totalReservedWei
for every domain d:
  sum(reservation.state == RESERVED
      and reservation.domain == d)
    = reservedCount[d].
```

The final equality is a model invariant, not a production scan. Every reservation stores its immutable domain and registrar-bound Bridge so neither the counter nor its reclaim authority can be caller-substituted. Forced ETH is surplus and creates no ticket. Deposit and unreserved withdrawal are CEI-safe and non-reentrant; a recipient failure restores the ticket.

Destination success runs in a nested all-or-revert execution frame. It first consumes the reservation, then uses those exact funds for Message value and the execution-fee pull, executes the target, consumes quota and appends DONE with the ticket/recipient/amount settlement hash. If target, quota, pull, terminal or post-call reserve handling fails, that nested frame reverts the Pool transfer, reservation and every target/Bridge effect before the public wrapper returns NEW or RETRIABLE as appropriate. No failed attempt consumes LP capital. Manual failure or strict expiry releases a still-live reservation and appends FAILED with zero settlement hash in the same journal; a release fault reverts the terminal transition. An unreserved credit remains NEW and reports UNFUNDED, so absence of an LP cannot corrupt status or block unrelated credits.

That rollback boundary is one exact external self-call, never an ordinary internal call and never a second public reentrancy frame:

```
executeAttemptV2(
  MessageV1 message, SourceContextV2 sourceContext,
```

```
  DestinationContextV2 destinationContext, address processor,
  uint8 expectedEntryStatus, bool isLastAttempt)
  external returns (uint8 resultingStatus, uint8 statusReason)
finalizeFailedAttemptV2(bytes32 creditId)
  external returns (uint8 resultingStatus, uint8 statusReason)
error TargetCallFailedV2(bytes32 attemptDigest)
```

Its selector is the first four Keccak bytes of the exact expanded signature

```
executeAttemptV2((uint64,uint64,uint32,address,uint64,address,uint64,address,
address,uint256,bytes),(uint64,uint8,bytes32,bytes32,bytes32,uint64,address,
bytes32,uint64,uint64),(uint256,bytes32,address,bytes32,bytes32),address,uint8,bool)
```

The failure-finalizer selector is the first four Keccak bytes of `"finalizeFailedAttemptV2(bytes32)"`; the custom-error selector is the first four Keccak bytes of `"TargetCallFailedV2(bytes32)"`. The immutable implementation records `SELF=address(this)` at construction; both external entries require `msg.sender==SELF` and `address(this)==SELF`, so a direct caller, proxy or delegatecall context rejects before a read or write. This entry does not acquire the shared non-reentrant guard: the public `processMessageV2/retryMessageV2` wrapper already holds it, authenticates the Message/pin/contexts, derives `processor=msg.sender`, checks the entry status/owner/last-attempt policy, and rechecks that the supplied reservation is exact before making the self-call with zero value. The inner entry defensively recomputes credit, pin, contexts, policy and entry status, then owns the complete consume–target–quota–pull–terminal journal. The failure finalizer rechecks RETRIABLE, the same pin, absent terminal and exact live reservation, then owns release–FAILED–terminal in one journal. No other entry may invoke one of those steps.

The wrapper ABI-encodes only those derived values, reserves the profile-pinned `V2_OUTER_CATCH_GAS_RESERVE`, and makes a bounded external call to `SELF`; its success return must be exactly two canonical ABI words (64 bytes). A failed low-level target CALL never returns normally and never bubbles or hashes target returndata. Instead the Bridge computes

```
attemptDigest = H("slot-chain-destination-attempt-v1" || creditId ||
  sourceContextHash || destinationContextHash || address20(processor) ||
  u8(expectedEntryStatus) || u8(isLastAttempt ? 1 : 0))
```

and reverts the complete inner frame with the exact 36-byte canonical custom error `TargetCallFailedV2(attemptDigest)`. This restores the consumed reservation, Pool transfer, target effects and Bridge state to their entry values. The outer wrapper locally recomputes the digest and recognizes only that selector, exact length and exact word. Target returndata is discarded before the Bridge creates this error, so a target-crafted identical error cannot spoof the catch.

After an authenticated target-failure catch, non-owner initial failure and non-last retry return the unchanged NEW or RETRIABLE state without writes; an owner initial failure writes RETRIABLE in the outer frame; and an owner-last retry makes the zero-value `finalizeFailedAttemptV2(creditId)` self-call. That second entry releases the still-RESERVED ticket, decrements the domain count, writes FAILED and appends its zero-settlement terminal leaf atomically. If it reverts or returns anything but the exact canonical 64-byte FAILED result, the outer call reverts, leaving the entry status and reservation unchanged. Any first-frame revert other than the authenticated target-failure error, and any

short/long return, noncanonical enum or unexpected status, discards revert bytes and maps only to `(expectedEntryStatus,ATTEMPT_ROLLED_BACK)` without an outer write. The release profile freezes both selectors, tuple grammars, custom-error selector/domain, valid status/reason matrix, self-call gas caps and catch reserve. Direct-inner, delegatecall, callback reentry, nested process/retry, target-crafted error, target failure with reservation then retry, owner-last failure with release/leaf, malformed return, and post-target quota/pull/terminal fault vectors prove the separation and byte-identical rollback.

The destination quota is exact: its only V2 key is

```
nativeQuotaKey = H("slot-chain-native-quota-v2" ||
                   destinationDomainId || "NATIVE_ETH")
destinationQuotaDebit = checked(message.value + message.fee).
```

Both a successful real CALL and invocation-prohibited DONE debit that amount in the same nested terminal journal. Target failure, UNFUNDED, NEW, RETRIABLE, manual failure and expiry debit zero. A missing key, overflow, insufficient quota or quota-call fault rolls back the target call, Pool transfer/reservation, Bridge status/pulls and terminal append together. Refund, source recall, withdrawal and terminal proof paths never read or debit quota. The fixed corpus covers value zero/fee positive, value positive/fee zero, both-zero rejection, addition overflow, quota exactly equal/one wei short and a post-target quota miss with byte-identical rollback.

On a proved DONE leaf, SourceBridge reclassifies exactly `value+fee+liquidityFee` from pending escrow to the authenticated `l1Recipient`'s LP pull. On FAILED or pre-enqueue cancellation, it reclassifies the same total to the source owner's user pull. Its exact conservation invariant is

```
totalLiveLiability = pendingEscrow + userPullLiability + lpPullLiability
address(sourceBridge).balance >= totalLiveLiability.
```

DONE never turns locked L1 ETH into ownerless surplus. Source liability falls only after the matching pull transfer succeeds. This removes deterministic lifetime-reserve exhaustion while leaving existing V1 native exits untouched. It does not make liquidity free: delivery liveness is explicitly economic and requires at least one LP willing to reserve pre-existing L2 ETH.

The destination domain's fresh immutable non-proxy Bridge owns V2 status, retry state, ETH custody and terminal index for the domain's entire lifetime. Its address, runtime and storage never rotate and it never delegates execution. This prevents another implementation or address from seeing an empty status and processing a pinned credit twice, and keeps RETRIABLE liquidity in the same account. The sole V2 admission is `verifyInboxCredit`; there is no caller-supplied nonempty source-proof path on L2. `processMessageV2` is NEW-only. A non-owner invocation failure remains NEW; only a destination-owner failed initial invocation becomes RETRIABLE. `retryMessageV2` is RETRIABLE-only. Invocation-prohibited handling precedes owner checks and permissionlessly creates the destination owner's value pull. For a real invocation, zero gas or `isLastAttempt=true` is owner-only; a non-last failure is side-effect free, while an owner last failure becomes FAILED. Manual failure is owner-only and pausable. Strictly after `processBy`, `expireMessageV2(creditId)` is raw-preimage-free, permissionless and unpausable from NEW or RETRIABLE. DONE, FAILED and RECALLED are terminal.

The V2 fee is a success-only bounty. No NEW, RETRIABLE, FAILED, manual-fail or expiry path pays it. A non-owner successful real invocation receives all fee; owner self-processing

also receives all fee. For a real CALL, insufficient raw reserved Pool liquidity returns UNFUNDED before the attempt. The nested attempt then consumes the exact reservation; a transfer or raw-balance failure reverts that frame rather than authorizing from a free-balance view. A successful target is followed by exact quota consumption, execution-fee pull-credit creation, terminal append and the hard `balance>=aggregateV2Liability` assertion in the same journal. Any quota, liability-floor, reservation or terminal failure rolls back Pool, target and Bridge, including an owner-last success. Deterministic no-CALL success paths may precheck their exact capacity. Every entry and withdrawal shares one top-level non-reentrant frame. Send/process/retry/manual-fail may be paused; cancellation, expiry, terminal append, DONE finalization, FAILED recall and pull withdrawal are outside every pause and ordinary quota gate. On transition to DONE or FAILED, the immutable destination Bridge makes exactly one call to the protocol-lifetime `TerminalAccumulatorV2`. Every process, retry, fail and expire entry shares one non-reentrant guard, and arbitrary recipient execution finishes before a terminal status is installed. The Bridge then writes DONE or FAILED and `terminalIndex=UINT64_MAX`, calls only `appendTerminalV2(creditId)`, and replaces the sentinel with the returned index before returning. The accumulator accepts no terminal or domain argument. The permanent ABI is:

```
TerminalAccumulatorV2.appendTerminalV2(bytes32 creditId)
    returns (uint64 terminalIndex)
DestinationBridge.terminalCommitmentV2(bytes32 creditId)
    view returns (bytes32 destinationDomainId, address destBridge,
                  uint8 terminal, uint64 terminalIndex,
                  bytes32 liquiditySettlementHash)
TerminalAccumulatorV2.terminalStateV2()
    view returns (uint64 count, bytes32 root)
event TerminalAppended(uint64 indexed index,
  bytes32 indexed destinationDomainId, address indexed destBridge,
  bytes32 creditId, uint8 terminal, bytes32 liquiditySettlementHash,
  bytes32 leaf,
  uint64 count, bytes32 root)
```

All selectors are canonical Solidity selectors of those signatures. Append returndata is exactly 32 bytes with a zero-padded `uint64`; commitment returndata is exactly five ABI words (160 bytes), with canonical high padding, `terminal=1` for DONE or 2 for FAILED and `terminalIndex=UINT64_MAX`. State returndata is exactly two ABI words. Short, trailing, noncanonical, overflow, revert or out-of-gas returndata rejects atomically. The accumulator uses exactly 50,000 gas for the commitment `STATICCALL`; the release gas benchmark must show 30% margin or revise this normative constant and the profile before launch. With that fixed gas stipend and exact return length it `STATICCALL`s the registered caller's `terminalCommitmentV2(creditId)`, requires the sealed domain, caller address, DONE/FAILED status, sentinel and the terminal-specific settlement-hash rule, then appends:

```
leaf = H("slot-chain-terminal-leaf-v2" || u64(terminalIndex) ||
         destinationDomainId || address20(destBridge) ||
         creditId || u8(terminal) || liquiditySettlementHash)
```

Any call or return failure reverts status, sentinel, leaf and count together. During a V1 recipient callback there is no concurrent sentinel, so a message targeting the accumulator cannot forge

a leaf even though `msg.sender` is the Bridge. V2 additionally rejects the complete manifest denyset, including the accumulator, before any CALL.

The accumulator stores no per-credit "already appended" set. The exact immutable registered Bridge is the sole writer for its domain, and its permanent terminal status/index plus non-reentrant APPENDING sentinel make a second or concurrent append for one credit unreachable. The append and Bridge transition share one EVM revert domain, so a failed suffix restores frontier/count/root and the Bridge sentinel/status/index together.

The accumulator stores exactly a checked `uint64count`, wrapped root and 64-word frontier. At old index i it starts with the new leaf, combines `frontier[h]` on the left while bit h of i is one, writes the carry only to the first zero-bit frontier slot, increments count, then folds all 64 heights with canonical empty nodes to obtain the tree root and wrapped count/root. Stale frontier values at zero bits are ignored. Append rejects old `count=UINT64_MAX`; the final leaf is unused.

The complete leaf preimage, leaf, new count and root are emitted in `TerminalAppended`. Proofs are reconstructed from canonical L2 receipts by an open-source deterministic prover and at least two independently administered, replaceable log archives. Anyone verifies a supplied 64-sibling proof on L1; archives have no signing key or correctness authority. Loss of every leaf log copy can stop future proof generation and therefore terminal release, just as loss of every canonical state preimage stops recovery. Archive-independent proof generation is not claimed; it would require the rejected unbounded completed-subtree store. Production must benchmark carry depths 0–63 and the complete cold post-call suffix before fixing `V2_POST_CALL_GAS_RESERVE`. The same harness must measure and freeze nonzero profile fields `V2_ATTEMPT_PRE_CALL_GAS_BOUND` and `V2_OUTER_CATCH_GAS_RESERVE`. For positive Message gas, the wrapper requests exactly `V2_ATTEMPT_PRE_CALL_GAS_BOUND+message.gasLimit` for the self-call and first applies the exact EIP-150 sufficiency relation while retaining the catch reserve; the inner target receives exactly `message.gasLimit-V2_POST_CALL_GAS_RESERVE`. For owner-only zero-gas execution, it forwards all gas above the outer reserve and the inner retains the post-call reserve. Production vectors cover first/cold Pool and quota slots, target OOG, custom-error construction/decoding, the second failure-finalizer self-call and outer guard cleanup. Until all three measured values and a 30-percent margin are published, the execution profile and document remain non-implementation-ready.

Registration is append-only. Two protocol-lifetime non-proxy contracts are:

```
ProtocolReleaseAuthorityV2
TerminalDomainRegistrarV2
```

Only the registrar may register an accumulator writer. Release authority is bound to the fork-reserved system sender and manifest namespace, not a particular Anchor version or endpoint. Activation passes the value, never merely its hash:

```
struct ComponentDescriptorV2 {
  address component; bytes32 runtimeHash; bytes32 configHash;
}
struct DestinationBridgeDescriptorV2 {
  address bridge; bytes32 runtimeHash; bytes32 configurationHash;
  bytes32 storageLayoutHash;
  bytes32 bridgeKernelProfileHash; address inboxCreditStore;
```

```
  address terminalAccumulator; address terminalDomainRegistrar;
  address quotaManager; address nativeLiquidityPool;
}
struct ReleaseManifestV2 {
  uint64 protocolVersion;
  uint256 settlementChainId; uint256 destinationChainId;
  bytes32 destinationGenesisHash; bytes32 executionProfileHash;
  bytes32 manifestNamespace; bytes32 destinationNamespace;
  address anchorV4; bytes32 anchorRuntimeHash;
  bytes32 destinationDomainId; address destinationBridge;
  bytes32 destinationBridgeExecutionHash;
  DestinationBridgeDescriptorV2 destinationBridgeDescriptor;
  bytes32 destinationInfrastructureHash;
  bytes32 migrationVerifierDescriptorHash;
  bytes32 ingressAuthorizationRoot;
  address nativeLiquidityPool; bytes32 poolRuntimeHash;
  bytes32 poolConfigurationHash;
  ComponentDescriptorV2[10] components;
}
AnchorV4.activateRelease(ReleaseManifestV2 calldata manifest,
                         uint64 retirementQueueCount)
ProtocolReleaseAuthorityV2.activateRelease(
    ReleaseManifestV2 calldata manifest, uint64 retirementQueueCount)
TerminalDomainRegistrarV2.activateDomain(
    ReleaseManifestV2 calldata manifest, uint64 retirementQueueCount)
```

The release manifest is a fully static Solidity tuple: no string, bytes or dynamic array occurs in it. Every address/integer has canonical ABI padding and the ten components have fixed order. Its ABI tuple contains exactly 58 words (1,856 bytes). In the hash preimage, word 0 is `RELEASE_MANIFEST_TYPEHASH`, words 1–12 are the outer fields through `destinationBridgeExecutionHash`, words 13–22 are the ten destination descriptor fields, words 23–28 are the remaining outer fields through `poolConfigurationHash`, and words 29–58 are the ten three-word components. The typehash is Keccak of this exact single ASCII string (shown wrapped only for typesetting):

```
ReleaseManifestV2(uint64 protocolVersion,uint256 settlementChainId,
uint256 destinationChainId,bytes32 destinationGenesisHash,
bytes32 executionProfileHash,bytes32 manifestNamespace,
bytes32 destinationNamespace,address anchorV4,bytes32 anchorRuntimeHash,
bytes32 destinationDomainId,address destinationBridge,
bytes32 destinationBridgeExecutionHash,
DestinationBridgeDescriptorV2 destinationBridgeDescriptor,
bytes32 destinationInfrastructureHash,bytes32 migrationVerifierDescriptorHash,
bytes32 ingressAuthorizationRoot,address nativeLiquidityPool,
bytes32 poolRuntimeHash,bytes32 poolConfigurationHash,
ComponentDescriptorV2[10] components)
ComponentDescriptorV2(address component,bytes32 runtimeHash,bytes32 configHash)
DestinationBridgeDescriptorV2(address bridge,bytes32 runtimeHash,
bytes32 configurationHash,bytes32 storageLayoutHash,
```

```
bytes32 bridgeKernelProfileHash,address inboxCreditStore,
address terminalAccumulator,address terminalDomainRegistrar,
address quotaManager,address nativeLiquidityPool)
```

The byte string contains no whitespace or newline between the three displayed type declarations. Dependencies are in ascending type-name order. Its commitment is the type-hashed `keccak256(abi.encode(...))` below. The function selector is Keccak's first four bytes over the canonical expanded ABI signature

```
activateRelease((uint64,uint256,uint256,bytes32,bytes32,bytes32,bytes32,
address,bytes32,bytes32,address,bytes32,
(address,bytes32,bytes32,bytes32,bytes32,address,address,address,address,address),
bytes32,bytes32,bytes32,address,bytes32,bytes32,
(address,bytes32,bytes32)[10]),uint64)
```

again with the displayed newlines removed. Tx0 calldata is that four-byte selector followed by the 58 manifest words and one canonical padded `uint64retirementQueueCount` word: exactly 1,892 bytes. Appendix golden vectors pin every boundary word and the final selector/hash. The dependency order is profile and primitive descriptors, Bridge bundle init/config, addresses, release manifest, then registration commitment; neither Bridge init/config nor the profile may depend on the manifest or derived domain. It is

```
releaseManifestHash = H(abi.encode(RELEASE_MANIFEST_TYPEHASH,
                                   canonicalReleaseManifestV2))
```

Zero values, duplicate component addresses, a Bridge different from component 10, a component count/order mismatch, or any named/hash field inconsistent with the Appendix grammars rejects.

The first AnchorV4 custom system transaction calls exactly `activateRelease(manifest, retirementQueueCount)`. Its canonical uint64 word must equal the migration statement's authenticated L1 ForcedQueue count; there is no optional/default watermark or Inbox-cursor substitute. The raw tx0 bytes contain the fixed manifest words immediately followed by that count and contain no MPT proof, statement hash, candidate digest or tx0 hash. This avoids a hash cycle. The validity circuit authenticates the already published L1 release manifest and exact L1 origin as separate statement inputs, requires this tx0 at index zero, and constrains its raw EIP-2718 hash and decoded values. Anchor recomputes the manifest commitment, stores the active release once, calls the release authority and registrar with the identical value/count, and rolls all three back on any failure.

The authority recomputes the commitment, requires `msg.sender` and its `EXTCODEHASH` to equal the manifest Anchor, requires `tx.origin` to be the reserved Anchor-system sender, and requires Anchor's one-shot active manifest/count to match. It permanently records the release and exact retirement watermark. A direct call by that reserved sender, an EOA, Bridge or an Anchor from another manifest rejects. Before either side hashes or stores the manifest, L1 requires `settlementChainId=block.chainid<=UINT64_MAX`; L2 independently requires `destinationChainId=block.chainid<=UINT64_MAX`. The registrar constructor fixes only the protocol-lifetime release authority, InboxApplyRouterV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2. Its non-reentrant activation recomputes the same manifest hash, requires the authority's exact stored version/watermark, and compares every protocol-lifetime

row plus the fresh release-owned Store, QuotaManager and DestinationBridge against the manifest. Components on L1 were already checked by ProtocolVersionManager; the L2 registrar never pretends to inspect cross-chain code.

Every destination release is fresh-only. Tx0 authenticates exact non-proxy runtime/configuration/layout for Store, QuotaManager and Bridge; arbitrary pre-activation Bridge surplus; zero nonce, liability and private status/pull/ terminal mappings; full initial nonzero V2-only quota; and `v2Active=false`. It authenticates the lifetime Pool code/configuration and its unchanged aggregate solvency, then one-shot registers the fresh domain/Bridge/Store as an append-only reservation authority, installs the inbox route and accumulator writer, and seals Store plus `v2Active=true` in one rollback domain. Tx0 moves no native value, creates no LP ticket/reservation and mints nothing. There is no activation-gate account, legacy proxy branch, exact-reuse branch, owner, implementation slot or delegate target. A historical domain or Bridge can never be reactivated even if code is identical. Any partial state, duplicate identity, Pool insolvency, conflicting route or late failure reverts tx0 entirely.

InboxApplyRouterV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2 are protocol-lifetime objects repeated byte-exactly by every successor manifest. Their cursor, old routes/releases/registrations, Pool tickets/reservations and accumulator frontier/root/count remain intact; only the absent release, route, Pool domain authority and one-to-one writer are appended. The registrar stores the single Appendix-defined `registrationCommitment[protocolVersion]` only after the full seal. The accumulator has no public registrar or root setter and calls only the registered Bridge's bounded terminal getter during append. Status, sentinel, frontier/root/count update and terminal index are atomic. The application-level leaf sequence, not a versioned EVM storage proof, remains the source release statement across later Settlement profiles.

Release rotation and old-endpoint reclamation are bounded and explicit. Successful tx0 replaces the registrar's one `activeDestinationBridge` with the fresh Bridge and records for the predecessor `retirementQueueWatermark[oldBridge]=retirementQueueCount`; neither may be rewritten. The count is the migration statement's exact L1 Queue count. Thus the retired adapter cannot append after the watermark, while `InboxApplyRouterV2.nextQueueIndex>=watermark` proves that every earlier queued predecessor descriptor has been applied.

The same L2 tx0 journal appends a local successor receipt; an L2 Bridge never reads the L1 Router receipt. Its ID and permanent views are:

```
destinationReceiptId = H("slot-chain-destination-activation-receipt-v2" ||
  u256(destinationChainId) || address20(terminalDomainRegistrarV2) ||
  u64(successorIndex) || u64(oldProtocolVersion) ||
  u64(newProtocolVersion) || oldManifestHash || newManifestHash ||
  oldDestinationDomainId || newDestinationDomainId ||
  address20(oldDestinationBridge) || address20(newDestinationBridge) ||
  u64(retirementQueueCount) || u64(activatedAtBlock))
destinationActivationReceiptV2(bytes32 destinationReceiptId) view returns
  (bytes4 magic, bytes32 returnedReceiptId, uint64 successorIndex,
   uint64 oldProtocolVersion, uint64 newProtocolVersion,
   bytes32 oldManifestHash, bytes32 newManifestHash,
   bytes32 oldDestinationDomainId, bytes32 newDestinationDomainId,
   address oldDestinationBridge, address newDestinationBridge,
   uint64 retirementQueueCount, uint64 activatedAtBlock, uint8 sealed)
```

```
destinationSuccessorReceiptV2(bytes32 oldDestinationDomainId,
                              address oldDestinationBridge) view returns
  (bytes32 destinationReceiptId, uint64 successorIndex, bytes4 magic)
```

The full return is exactly 448 bytes with magic `0x44525632` (`DRV2`), `sealed=1` and canonical padding; the index return is 96 bytes with magic `0x44535632` (`DSV2`). Registrar derives all fields from its current active release and the proof-bound identical manifest/count, increments a checked `uint64successorIndex`, and writes the immutable receipt, predecessor index, new active tuple and all route/Pool/writer seals in one rollback domain. Genesis uses canonical-zero predecessor fields and writes no predecessor index. A successor requires nonzero old fields, distinct new domain/Bridge, exact stored old manifest, absent predecessor index and exact watermark; partial or duplicate sealing rejects.

Each release-owned `InboxCreditStoreV2` maintains a checked `uint64pinnedCount`, incremented only when its authorized InboxApply route writes an absent credit pin. The accumulator maintains a checked per-domain `uint64terminalizedPinnedCount`. It increments only in the same journal as a terminal append when the caller is that registrar-bound domain Bridge, the route and Store identities match, the pin exists, status is DONE or FAILED, and the `(domainId,creditId)` terminal guard was previously absent. A duplicate, unpinned credit, foreign Store/Bridge or failed append changes no counter, leaf, guard or status.

Anyone may call the old Bridge's bounded `reclaimSurplus()` only after a distinct successor has the Registrar-local sealed destination activation receipt above, InboxApply has reached the immutable watermark, `terminalizedPinnedCount[oldDomain]=oldStore.pinnedCount`, and the old Bridge's aggregate pull liability is zero. The call reauthenticates the old Bridge's aggregate execution-fee pull liability is zero and `NativeLiquidityPoolV2.reservedCount(oldDomain)=0`. Unconsumed old-domain reservations must first be consumed by success or released by a terminal failure; unrelated unreserved ticket funds remain withdrawable by their LP. The call bounded-static-reads both Registrar views, checks exact code/config/ length/magic and old/new version/manifest/domain/Bridge/watermark/seal, then reauthenticates the old manifest and complete lifetime Router/authority/registrar/accumulator/Pool graph, requires the old route and domain writer unchanged, and transfers the entire remaining raw Bridge surplus to the profile's typed NATIVE_ETH `bridgeSurplus` sink; it never enumerates a mapping or consumes a Pool ticket. Its result is

```
enum ReclaimResult { REJECTED, RECLAIMED_ZERO, RECLAIMED_VALUE }
returns (ReclaimResult result, uint256 amount)
```

so sink rejection cannot masquerade as a valid zero-balance reclaim. Only after an accepted full transfer does it set the one-shot `retired` bit; partial transfer, callback/reentrancy, later processing, double reclaim and any state mismatch revert or return REJECTED without mutation. Historical status, terminal index/events and proof verification remain readable forever. Historical SourceBridges remain callable until every USER_PULL and LP_PULL is withdrawn; a DONE reimbursement is never reclaimable as surplus. Repeated release rotation changes no ticket owner, reservation, source liability or terminal proof.

Every execution proof additionally authenticates the accumulator account and proves that the candidate's `terminalRoot` and `terminalCount` equal its exact post-state root/count slots. These fields are part of `CanonicalCore` and `executionOutputsCommitment`; tier-2 and tier-3 use the same public-output constraint. A caller cannot supply an unconstrained root; no checkpoint publisher exists in this revision. The same proof authenticates the protocol-lifetime

InboxApplyRouterV2 account and, for each block, requires its pre-state `nextQueueIndex` to equal that block's `messageStart` and its post-state value to equal `messageEnd`. Before proof acceptance, the L1 base invariant is `baseCanonical.messageCursor=ForcedQueue.cursor`, while the proof's first L2 pre-state cursor equals the same start. The final proved L2 post-state cursor and new `CanonicalCore.messageCursor` both equal the candidate end. After proof verification and before its history write, the active Settlement calls `ForcedQueue.advanceCursor(start,end,proofBeneficiary)`. That call moves the exact cumulative processing-fee interval to the beneficiary's pull claim. A failed call reverts the entire canonical commit. L1 never writes the L2 router; the end value is authenticated L2 poststate adopted by the proof. Launch bootstrap proves all three base values equal in the imported L2 state. A later migration occurs only at a canonical boundary, checks the two L1 values directly, and inherits L2 equality inductively from the last accepted proof; the target profile must preserve that same cursor slot and transition. No activation transaction may repair, overwrite or select among divergent cursors.

The vector root is

```
H("slot-chain-terminal-root-v2" || u64(terminalCount) || treeRoot)
```

where the Appendix fixes empty nodes, node hashing and bit order.

Every versioned Settlement is a permanent non-proxy contract with no `SELFDESTRUCT`, delegate target, history pause or code upgrade. Its internally written history cell contains the exact sequence plus the full `CanonicalCore` and `canonicalizedAtBlock`; a read requires exact tag equality. The protocol-lifetime non-proxy `ActiveSettlementRouter` keeps an append-only mapping from protocol version to exact Settlement address, `EXTCODEHASH` and execution-profile hash. Historical entries cannot be deleted, redirected or reactivated. Its only switch is:

```
struct CanonicalCoreV2 {
  uint48 l2BlockNumber; bytes32 tipHash; uint64 tipSlot; bytes32 stateRoot;
  uint64 messageCursor; bytes32 winningDataCommitment; uint256 nextBaseFee;
  uint64 nextExcessBlobGas; bytes32 terminalRoot; uint64 terminalCount;
}
struct MigrationActivationFixedV2 {
  uint8 transitionKind; uint64 migrationGeneration;
  uint64 sourceProtocolVersion; uint64 targetProtocolVersion;
  uint64 sourceCanonicalSequence; bytes32 candidateDigest;
  CanonicalCoreV2 outputCore; address proofBeneficiary;
  uint256 anchorNumber; bytes32 anchorHash; uint64 forceCutoff;
  uint64 preInboxLastAppliedPlusOne;
}
activateVersionWithMigration(
  MigrationActivationFixedV2 calldata fixedInput,
  ReleaseManifestV2 calldata manifest,
  InboxRowV2[] calldata rows,
  bytes calldata importedHeaderRlp,
  bytes calldata proof)
```

The selector is the first four Keccak bytes of this exact canonical expanded signature, with no whitespace:

```
activateVersionWithMigration((uint8,uint64,uint64,uint64,uint64,bytes32,
(uint48,bytes32,uint64,bytes32,uint64,bytes32,uint256,uint64,bytes32,uint64),
address,uint256,bytes32,uint64,uint64),
(uint64,uint256,uint256,bytes32,bytes32,bytes32,bytes32,address,bytes32,
bytes32,address,bytes32,
(address,bytes32,bytes32,bytes32,bytes32,address,address,address,address,address),
bytes32,bytes32,bytes32,address,bytes32,bytes32,
(address,bytes32,bytes32)[10]),
(uint64,uint8,uint32,bytes32,bytes)[],bytes,bytes)
```

The first two tuples are static and occupy 21 and 58 words. Therefore the top-level ABI head is exactly 82 words: words 0–20 are `fixedInput`, 21–78 are `manifest`, and words 79–81 are the rows, imported-header and proof offsets. The rows offset is exactly `0x0a40`. The other offsets are the immediately following canonical tails: rows encode as one length word, n element offsets and n contiguous five-word element heads plus their bytes tails; then the header and proof each encode one length word, bytes and zero right-padding. Aliases, gaps, overlap, nonminimal offsets, nonzero padding or trailing calldata reject. There are at most 64 rows; kind-0 descriptor bytes are empty and kind-1 bytes are exactly 541. GENESIS_IMPORT requires a nonempty canonical RLP header of at most 2,048 bytes; VERSION_MIGRATION requires zero header bytes. Proof length is nonzero, no greater than the descriptor's `maximumProofBytes`, and that bound is at most 131,072 bytes. It is permissionless only while the Router's exact gate is READY. Delayed ProtocolVersionManager governance may pre-register and arm or cancel a target; it has no activation-only callback, privileged landing path or post-READY step. The Router derives the target Settlement, manifest, execution profile and immutable verifier solely from that stored gate/registration and rejects a caller-supplied address or authority. The call carries only those bounded values, the registered manifest value, canonical Inbox rows, the launch header preimage when applicable and raw proof bytes. The Router reconstructs the complete typed statement and invokes the profile-pinned verifier directly before any write. Launch header/MPT/SignalService authentication is part of that same verifier and typed statement, not an opaque Boolean or a second adapter output. A later switch relies on induction from the last validity proof and checks on L1 that the old `CanonicalCore.messageCursor=ForcedQueue.cursor`; it does not pretend to read the live L2 Inbox cursor. Before any irreversible write the Router reads the old Settlement's exact full `Canonical`, sequence and `lastCanonicalCommitBlock`; checks the queue's root, count, frontier, cursor, accounted liabilities, actual-balance lower bound and `lastDueAt`; verifies the target proof's imported base, release pair, execution profile, queue range and poststate; and requires the new empty PREACTIVE Settlement to share the exact protocol-lifetime queue, registry, schedule and gate objects. The target proof executes the release Anchor and directly validates L2 component rows 4–10. A missing component, route collision, bad verifier or proving outage therefore leaves the old Settlement and queue authority unchanged.

"Empty PREACTIVE target" is an exact predicate, not a claimed constructor default. The target Settlement's internal DataSession region requires all 1,024 cell tags FREE; empty ID-to-cell and live-owner maps; `liveCount=refundCount=occupiedCount=0`; `nextSessionSequence=0`, `gcCursor=0`; `migrationRefundGeneration=0`, `migrationRefundClaimDeadline=0`, the shared session/sweep guard in its canonical NOT_ENTERED state; and zero live- and refund-bond liability. The actual target Settlement ETH balance is deliberately not required to be zero: CREATE2 prefunding or forced ETH is surplus and cannot veto activation. Deployment may occur while the source has live sessions, because the protocol-lifetime gate is shared; cutover sepa-

rately requires that gate to be READY after authenticating the source-local live-session count zero. Exact fresh creation code plus the absence of every PREACTIVE mutation surface proves the fixed cells and hidden maps empty. Logs are not storage and are not part of this predicate. Before cutover the Router uses the exact `dataSessionAccountingV1()` ABI to recheck every observable scalar, guard word and configuration hash; it does not pretend to enumerate a mapping. It also repeats the source-readiness handshake defined below. Both reads must report the same generation/version/manifest, closed boundary, zero LIVE count, NOT_ENTERED guard and registered DataSession configuration immediately before any authority, queue or canonical write.

Settlement constructor provenance is authenticated by the delayed registration's exact 232-byte `SettlementDeploymentDescriptorV1`:

```
(address20(factory), bytes32(factoryRuntimeHash),
 bytes32(factoryConfigurationHash), bytes32(salt), bytes32(initCodeHash),
 address20(targetSettlement), bytes32(targetRuntimeHash),
 bytes32(targetConfigurationHash))
```

Its hash is `H("slot-chain-settlement-deployment-v1"||u32(232)||packedDescriptor)`. Every field is nonzero and `targetSettlement=low20(keccak256(0xff||address20(factory)||salt||initCodeHash))`. Registration must publish the complete init-code artifact and immutably bind this descriptor inside `targetRegistrationHash`; arm, activation, cancellation and the activation receipt must bind that same hash. The Router checks factory/target `EXTCODEHASH`, the exact four-byte `componentConfigHashV2()` call with 50,000 gas and exactly 32 return bytes, and the derivation at arm and activation. Hidden mapping emptiness follows only from that authenticated constructor and the lack of PREACTIVE mutation entry points; it is never inferred from runtime/config getters alone. The same authenticated Settlement init code and its target configuration commitment bind the internal DataSession storage layout, shared gate, `dataRent` sink and the rule that every non-session payable selector, fallback and receive path rejects; there is no second custody contract or cross-contract session authority. The current authorization/receipt codecs do not yet carry the new `targetRegistrationHash`. Their exact extension, golden vectors and cross-language tests are therefore an explicit release blocker in §14; target emptiness is not claimed closed until that control-plane round lands.

Only after every proof and configuration check succeeds does the same revert domain freeze the old Settlement, transfer queue authority, advance the proved queue interval, adopt the proof-authenticated L2 Inbox cursor, initialize the new Settlement directly at the proved post-state and next sequence, write its first version-namespaced history cell, consume the target pair and change the router/gate to the new ACTIVE version. The imported base remains in the old version's permanent history; it is not duplicated as a target canonical history entry. Mature reward/refund/queue claims remain claim-only and do not gate cutover. No external target call occurs after authority transfer. Failure of any step reverts the proof adoption, queue payment, freeze, history write and switch together.

The router's stable `syncIngress` status/refund preflight remains callable by every historically registered immutable BridgeInboxAdapter. The payable `appendFromAdapter` path is narrower: it requires the caller to equal the current active authorization's exact adapter deployment and authorization ID. Its nonpayable half resolves the active version, calls that Settlement's bounded `sync()`, preserves a SYNCED transition without accepting value, and otherwise returns the exact version/generation stamp. Its payable half makes no second sync decision and forwards the exact value and descriptor only after that stamp still matches the single-

ton ForcedQueue's ACTIVE authority and the same active adapter/authorization binding; a predecessor adapter cannot append after its retirement watermark. Historical adapter status, cancellation and pull-refund functions remain callable without an append capability. No Settlement calls the ingress/version router while committing canonical state; it calls only the queue's non-callback `advanceCursor`. Router `canonicalAt(protocolVersion,canonicalSequence)` performs a bounded fixed-selector `STATICCALL` to the permanently registered Settlement, requires the exact return length, code hash, version/profile and cell tag, and returns no caller fields. This gives target replacement the same delayed trust root as the validity verifier without adding a canonical-path dependency.

The non-proxy, unpausable `TerminalSignalVerifier` accepts only the fixed 64-sibling proof against one exact routed tuple and the exact domain/Bridge/credit/terminal leaf. Source finalize/recall verifies and mutates the source credit in one transaction; there is no publication stage to front-run. Sparse 4,096-block L2 jumps cannot choose a cell, and at most one canonical write per L1 block gives a live version's 256 cells at least 3,072 seconds of retention. Frozen versions retain their last 256 cells forever. The Bridge stores `terminalIndex[creditId]`. Any archive reconstructs the siblings from the canonical complete `TerminalAppended` event sequence, and L1 verifies them against the selected history cell. No archive signature or privileged indexer is accepted. Source terminal functions never call legacy SignalService. No V2 failure, recall, delivery finalization or view falls back to `msgHash`. The release profile pins every selector, event, status, liability and accumulator slot plus positive and negative lifecycle vectors.

V2 launch contains no asset codec or Vault flow. ERC20, ERC721, ERC1155 and all official Vault calls remain V1, with their existing V1 status, quota and recall semantics. A future asset protocol must define bidirectional round-trip and delivered-backing accounting, immutable asset policy and canonical/deployment identity, mapping-rotation history, destination release-versus-mint supply conservation and an unpausable exit. It requires a new kind, domain, release, profile, codec and audit and cannot reinterpret DIRECT.

The fresh L2 Bridge's sealed `v2Active` state enables processing only after the proof-authenticated tx0 atomically seals its Store and Bridge. Source V2 send remains disabled until that canonical registration is proven to L1 and its 214-block support delay completes. Every old `sendMessage` call and all L2-to-L1 sends retain V1 process/retry/fail/recall. InboxApplyRouterV2 makes only the bounded manifest-routed store calls above, and neither router nor store invokes a message recipient. Legacy proof-based `proveSignalReceived` remains unchanged for V1 and is never a V2 terminal dependency. AnchorV4 and InboxApplyRouterV2 are profile-defined system transactions: their custom type, reserved senders, system nonces, zero beneficiary tip and fee treatment are consensus fork rules, not ordinary user accounts. Their gas and the batch loop still count against the block limit. Deployment and migration require vector-tested InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2 and TerminalAccumulatorV2 bytecode. Legacy SignalService remains V1-only and may be upgraded without entering a V2 trust path. Authorizing an unreviewed caller is invalid.

An unexpired kind-0 item must reveal raw bytes hashing to its stored `rawTxHash`; the circuit decodes them and matches every duplicated descriptor field before execution. Once `validUntil` is strictly before the L2 block timestamp, the circuit emits the unique `EXPIRED_NO_TX` disposition from the authenticated durable descriptor without needing those bytes. Consequently a disappeared sender or pruned history can delay its own item only until the bounded validity horizon, never wedge the FIFO permanently. A user that wants execution remains an available source for its own signed bytes. The 2,164-second recovery target is conditional

on unexpired head bytes being retrievable; the unconditional protocol bound adds at most `MAX_FORCE_VALIDITY_SECONDS`.

For a normal signed block, `forceRoot/forceCutoff` are authenticated by its L1 anchor. For recovery they equal the current round's directly stored pair. The circuit consumes only up to that cutoff, so a later append cannot invalidate an already signed block or in-flight proof. All blocks in one candidate use the same cutoff. A due current head cannot be bypassed: `sync()` cancels an immature window, or closes a mature best only when its live boundary is not due, before opening recovery.

The 1,500-second force delay applies to the queue head, not to an arbitrary position behind existing prepaid work. An admitted sender can compute its worst-case number of escape blocks from the gas ahead at admission. As with L1 itself, the protocol does not promise backlog-independent latency against an adversary willing to buy all available capacity; it promises that builders cannot selectively veto the FIFO head.

An escape block is deterministic under a governance-frozen `executionProfileHash`. That profile commits the L2 fork rules; complete header-field derivation (gas limit, base fee, extra data, withdrawals/requests roots and randomness); the deployed AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2 and TerminalAccumulatorV2 bytecode/constructor hashes; reserved senders, nonces and gas; AnchorV4 checkpoint calldata and `ancestorsHash` transition; and the transaction/receipt derivation above. There is no protocol signature or account nonce. The permanent unsigned typed system envelope is exactly

```
0x7f || rlp([chainId, systemKind, systemNonce, gasLimit, to, value, data])
```

with minimal RLP integers, a 20-byte `to`, zero value, no signature fields and `systemNonce=block.number-firstV2BlockNumber` independently for each kind. `firstV2BlockNumber` is the protocol-lifetime value `launchImportedCanonical.l2BlockNumber+1`; launch installs it in the cutover commitment and execution profile, every later profile must reproduce it exactly, and checked subtraction rejects a pre-cutover block. Kind 0 is Anchor at transaction index 0 and kind 1 is InboxApply at index 1; the fork injects their distinct reserved senders. Ordinary accounts cannot submit type `0x7f`. System gas counts against the block limit, but the envelope has zero tip and is exempt from account balance/base-fee charging. Its receipt is exactly `0x7f||rlp([status,cumulativeGasUsed,bloom,logs])`; both transactions contribute normally to transaction/receipt roots and cumulative gas.

The type's state transition is fully separate from EIP-1559 signing. Decoding requires `chainId=block.chainid=destinationChainId<=UINT64_MAX`; a foreign, zero or nonminimal chain ID rejects before execution. The profile maps each kind to exactly one reserved sender and target. Both `CALLER` and `ORIGIN` are that reserved sender. Sender and target start warm in addition to the fork's ordinary precompile warm set. No sender account nonce is read, compared or incremented; no sender balance is read, debited or credited; `GASPRICE` returns zero while `BASEFEE` retains the block's ordinary value. There is no access list, creation target, blob field or value transfer. Intrinsic gas is exactly `21000+4*zeroCalldataBytes+16*nonzeroCalldataBytes`; execution starts with `gasLimit-intrinsicGas`. A malformed envelope or intrinsic gas above the exact kind gas limit makes the block invalid.

The transaction hash is legacy Keccak-256 of the complete typed envelope above, and that same byte string is the transaction-trie value. After execution let `spentBeforeRefund=gasLimit-gasRemaining` (therefore including intrinsic gas). Under the launch fork's Lon-

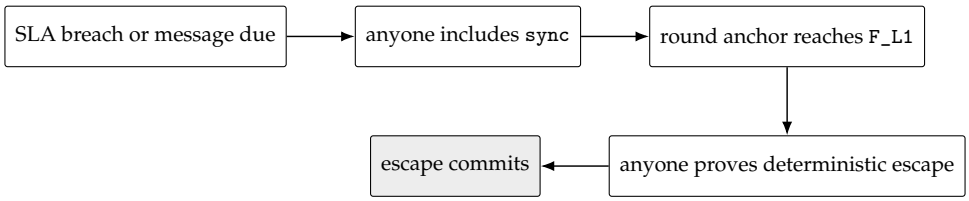
don refund quotient 5, `refundGas=min(refundCounter,floor(spentBeforeRefund/5))` and `gasUsed=spentBeforeRefund-refundGas`; unused or refunded gas creates no account credit. REVERT or exceptional halt produces status zero and reverts state/logs under ordinary EVM rules while retaining that gas accounting; success produces status one. The typed receipt bytes above are the receipt-trie value, and `cumulativeGasUsed` is the ordinary running sum in exact transaction order. The block rejects any mismatch in envelope bytes, injected environment, warm set, state change, transaction hash/root, receipt bytes/root, gas used or cumulative gas.

The steady kind-0 call has gas limit 1,000,000, selector `0x523e6854` and ABI tuple

```
(uint48(anchorNumber), anchorHash, anchorStateRoot)
```

from the candidate's one batch anchor. The first pending-release activation kind-0 call has gas limit 12,000,000 and the activation selector/calldata defined above. Its kind-1 forced budget is 13,000,000 and that block permits no discretionary gas. Kind 1 always targets the protocol-lifetime InboxApplyRouterV2, has gas limit 5,000,000, selector `0x6b326168` for `apply(uint64,(uint64,uint8,uint32,bytes32,bytes)[])`, and uses the exact ABI rows derived in §8; the empty vector is still a canonical call. Activation uses 12m Anchor + 13m forced + 5m margin, exactly the 30m block limit. Steady blocks use 1m Anchor + 20m forced + 5m margin, at most 30m. Its timestamp is `GENESIS_TIMESTAMP+escapeSlot`, coinbase is the protocol sink and there is no discretionary body. A repeated batch anchor still performs the exact per-block ancestors transition. Changing the execution profile requires a delayed protocol-version upgrade and new full-block golden vectors; runtime guessing is forbidden.

For a cartel with no usable builder block, the recovery sequence is:



With the initial bounds, activation inclusion (120 s), the frozen `ESCAPE_OFFSET=1,900s` (covering `T_DEPTH_MAX=900s`, proving at 900 s and 100 s margin), submission inclusion (120 s), and clock margin (24 s) total 2,164 seconds, below `DELTA_FINAL_LAG=3,600s`. Tip admissibility is separate: the escape slot is at the end of the offset, so only submission and skew—144 seconds—must fit `DELTA_TIP=1,200s` when proof generation meets its bound.

9 Perpetual aggregator seat market

The aggregator market is an optional service and payment layer around the permissionless proof path. It has four installed ranks—one primary and three standbys—and a continuous reverse auction with four shared waiting cells. It has no auction epoch or fixed close window. A healthy funded primary persists until competitive handover, voluntary exit, its one objective duty, funding expiry or migration. No seat, operator action, premium balance or Market call is a precondition for proof submission, canonical commit, forced recovery, recovery renewal or migration readiness.

9.1 Authority, identity and bounded call graph

Settlement alone owns the installed roster, immutable seat terms and service intervals, one staged-handover commitment, duties, successor selection, failover, breach receipts, migration state and retained duty history. Every canonical transition is bounded, scans at most four seat/duty cells and performs no external call. If optional seat state is absent, stale, economically unmaintained or locally full, Settlement closes the affected economics to vacancy and continues canonical recovery.

The protocol-lifetime non-proxy `AggregatorSeatMarket` alone owns pending offers, native-ETH SLA bonds, premium custody, reserves and pull credits. It has no canonical writer, migration coordinator, `armMigration` or `abortMigration` entry point. It stores append-only exact Settlement authorizations installed only through Release Manager. A Market call may read Settlement only through a separately specified bounded `STATICCALL`; it checks exact chain, version, address, runtime/configuration hash, selector, return length and magic. No caller-supplied authority, phase, generation, operator, close time or validity Boolean is accepted.

The sole offer-book state read is the no-argument selector `seatMarketTargetStateV1()` with canonical ABI return

```
(address target, uint256 settlementChainId, uint64 protocolVersion,
 bytes32 runtimeHash, bytes32 configurationHash, bytes4 magic,
 uint8 phase, uint64 seatGeneration)
```

where `target=address(this)`, `runtimeHash=address(this).codehash`, `magic=0x53454154` (ASCII `SEAT`), and phase is exactly ACTIVE=1, ARMED=2, READY=3 or FROZEN=4. Return length is exactly 256 bytes; address, `uint64`, `bytes4` and `uint8` padding must be canonical. Market checks the target's pinned `EXTCODEHASH` before a descriptor-gas-bounded `STATICCALL` and compares every word with the append-only authorization. Revert, OOG, short/trailing data or any mismatch fails closed. `AggregatorSeatMarket.syncSeatGeneration()` returns exactly `(uint8result,uint8purgedCount,uint64generation)` as three ABI words, where result is UNCHANGED=0 or SYNCED=1 and `purgedCount<=4`; it reads the current target, terminalizes every stale waiting tranche in one rollback domain and commits the generation cache last. It is the sole permissionless *offer-book* entry point that reads Settlement. Insertion, requote, pending exit/refund and pull withdrawal perform no target read and require the already initialized cached generation. Stage/apply authenticate the same generation only inside their enclosing atomic Settlement call. Premium, installed-release, breach and duty-reclamation paths may make only the two separately bounded historical reads below:

```
seatMarketRecordV1(bytes32 seatTermId) view returns
  (bytes4 magic, bytes32 authorizationId, uint64 seatGeneration,
   bytes32 returnedSeatTermId, bytes32 trancheId, address operator,
   uint64 responsibilityStart, uint64 premiumCap, uint64 serviceCloseAt,
   uint64 termRemovedAt, uint64 lastLiabilityAt, uint16 liveDutyCount,
   bytes32 latestDutyId, uint8 latestDutyDisposition,
   uint64 latestDutyDispositionAt, bytes32 breachReceiptId,
   uint64 breachRecordedAt)
seatDutyRecordV1(bytes32 dutyId) view returns
  (bytes4 magic, bytes32 authorizationId, uint64 seatGeneration,
   bytes32 returnedDutyId, bytes32 seatTermId, bytes32 trancheId,
   uint8 disposition, uint64 dispositionAt, uint64 lastLiabilityAt,
```

```
   bytes32 breachReceiptId, uint64 breachRecordedAt)
```

Both magic words are `0x53485231` (ASCII `SHR1`); exact return lengths are 544 and 352 bytes. Every narrow word and address has canonical padding. The append-only target authorization pins both selectors, separate gas limits, runtime/configuration, version and chain. Market checks code hash, makes one bounded STATICCALL, and compares authorization, generation and returned ID before using a word. Revert, OOG, short/trailing/noncanonical data, unknown record, a live-duty count above four or an invalid disposition rejects. Historical records are permanent after close and cannot be redirected by a successor. These are the only post-install Settlement reads; no caller-supplied close cap, breach receipt, duty binding, magic or Boolean is authoritative.

The non-proxy `ActiveSettlementRouter` is the protocol-lifetime root of the Settlement graph and immutably owns the shared migration gate, `ForcedQueue`, `L1HeaderOracle` and InboxApply deployment descriptor. Every live, PREACTIVE and historical Settlement must identity-alias those exact objects. The Router neither reads nor writes a live L2 object. Release Manager keeps append-only target authorizations, but Router is the sole *L1* activation-receipt and successor-index store. The separately domained L2 Registrar receipt exists only for destination-Bridge reclamation; its counter, ID and magic are never accepted on L1, and the L1 receipt is never accepted on L2. Historical targets expose only premium accrual/claim, reserve reconciliation, breach enforcement, installed release, bond-credit claim and duty-history reclamation; they cannot accept new offers, stages, sessions or ingress.

Every successful switch appends one exact sealed receipt. Its identity is

```
receiptId = H("TAIKO_ACTIVATION_RECEIPT_V1" ||
  u256(settlementChainId) || address20(router) || u64(routerGeneration) ||
  u64(successorIndex) || u8(transitionKind) ||
  u64(sourceProtocolVersion) || u64(targetProtocolVersion) ||
  sourceManifestHash || targetManifestHash ||
  sourceAuthorizationId || targetAuthorizationId ||
  address20(sourceSettlement) || address20(targetSettlement) ||
  oldDestinationDomainId || newDestinationDomainId ||
  address20(oldDestinationBridge) || address20(newDestinationBridge) ||
  u64(queueWatermark) || candidateDigest || outputCanonicalHash ||
  u64(outputCanonicalSequence) || u64(activatedAtBlock))
```

The Router exposes only these fixed views:

```
activationReceiptV1(bytes32 receiptId) view returns
  (bytes4 magic, bytes32 returnedReceiptId, uint256 settlementChainId,
   address router, uint64 routerGeneration, uint64 successorIndex,
   uint8 transitionKind, uint64 sourceProtocolVersion,
   uint64 targetProtocolVersion, bytes32 sourceManifestHash,
   bytes32 targetManifestHash, bytes32 sourceAuthorizationId,
   bytes32 targetAuthorizationId, address sourceSettlement,
   address targetSettlement, bytes32 oldDestinationDomainId,
   bytes32 newDestinationDomainId, address oldDestinationBridge,
   address newDestinationBridge, uint64 queueWatermark,
   bytes32 candidateDigest, bytes32 outputCanonicalHash,
```

```
   uint64 outputCanonicalSequence, uint64 activatedAtBlock, uint8 sealed)
seatSuccessorReceiptV1(bytes32 oldAuthorizationId) view returns
  (bytes32 receiptId, uint64 successorIndex, bytes4 magic)
bridgeSuccessorReceiptV1(bytes32 oldDestinationDomainId,
                         address oldDestinationBridge) view returns
  (bytes32 receiptId, uint64 successorIndex, bytes4 magic)
```

The receipt return is exactly 800 bytes with magic `0x41525631` (`ARV1`), `sealed=1` and canonical padding; each index return is exactly 96 bytes with magic `0x41535631` (`ASV1`). The Router checks and increments one global `uint64successorIndex`, derives every field from the verified activation/current registrations, and writes receipt plus both nonzero predecessor indexes in the same journal before ACTIVE. A GENESIS_IMPORT has canonical-zero predecessor authorization/domain/Bridge and writes no predecessor index. VERSION_MIGRATION requires every predecessor field nonzero, a distinct target, absent indexes, exact queue watermark and `outputCanonicalSequence`; no key can be consumed twice. The full receipt is immutable and a partial/unsealed write cannot survive rollback.

Market rotation accepts a successor only after bounded code/config/length checks on both Router views and exact old/new authorization, versions, manifests, Settlements, generation, candidate/output and seal. Old-Bridge reclamation does not use these L1 views; it uses the proof-bound Registrar-local destination receipt specified in the release lifecycle below. A caller-supplied receipt, index, successor address or Boolean is never authority on either layer.

9.2 Orthogonal lifecycle and conservation

Lifecycle dimensions are deliberately independent:

```
OfferLocation  = NONE | PENDING | STAGED
TrancheUsage   = OFFER | STAGED | INSTALLED | CLOSED_UNINSTALLED
BondDisposition = NONE | OWNER_CREDITED | PENALTY_CREDITED
ReserveLifecycle = ABSENT | UNSTARTED | OPEN | CLOSED_TAIL
```

Only PENDING occupies a waiting cell and only STAGED occupies the single stage. Installation sets location to NONE but preserves the immutable `SeatTerm.offerId`. The stage retains the capacity of its selected waiting cell, so at all times

```
pendingCount + stagedCount <= 4
stagedCount <= 1.
```

Staging changes the pair by $(-1, +1)$; ordinary or lineup expiry restores the same offer by $(+1, -1)$ and can never overwrite or refund a fifth entry.

Every cross-contract record uses one fixed-width legacy-Keccak identity; no ABI dynamic encoding, string enum or caller-selected ID is accepted:

```
authorizationId = H("TAIKO_SEAT_TARGET_AUTHORIZATION_V1" ||
  u256(marketChainId) || address20(market) || u256(settlementChainId) ||
  u64(protocolVersion) || address20(target) || runtimeHash ||
  configurationHash || bytes4(expectedMagic))
trancheId = H("TAIKO_SEAT_TRANCHE_V1" || authorizationId ||
  u64(seatGeneration) || address20(operator) || u256(bondWei) ||
```

```
  u256(creationSequence))
offerId = H("TAIKO_SEAT_OFFER_V1" || authorizationId ||
  u64(seatGeneration) || trancheId || address20(payout) ||
  u256(askWeiPerSecond) || u256(eligibleAtTimestamp) ||
  u256(eligibleAtBlock) || u256(quoteSequence))
stageId = H("TAIKO_SEAT_STAGE_V1" || authorizationId ||
  u64(seatGeneration) || lineupCommitment || offerId ||
  u256(selectedRank) || u256(handoverAt) || u256(expiresAt))
seatTermId = H("TAIKO_SEAT_TERM_V1" || authorizationId ||
  u64(seatGeneration) || offerId || trancheId ||
  u64(installedAt) || u64(installRevision))
lineupCommitment = H("TAIKO_SEAT_LINEUP_V1" || u64(lineupRevision) ||
  primaryTermId || standby0TermId || standby1TermId || standby2TermId)
dutyId = H("TAIKO_SEAT_DUTY_V1" || seatTermId || u64(dutySequence) ||
  u64(baseCanonicalSequence) || u64(baseTipSlot))
selectionId = H("TAIKO_SEAT_SELECTION_V1" || seatTermId || trancheId ||
  offerId || u64(selectedCanonicalSequence) || u64(selectedAt) ||
  u64(targetTip) || u8(selectionSource) || predecessorDutyIdOrZero)
breachReceiptId = H("TAIKO_SEAT_BREACH_V1" || dutyId || seatTermId ||
  trancheId || u64(recordedAt))
```

Absent lineup terms and predecessor duty are `bytes32(0)`; selection source is DUTY_FAILOVER=1 or HEALTHY_EXPIRY=2. All widths are checked before hashing, all counters are monotone, and a hash collision reverts rather than overwriting an existing record. Market and Settlement recompute the same ID at every handoff; the release bundle contains cross-language golden and one-field- substitution vectors for each domain.

Each tranche stores immutable operator, bond, creation sequence, offer and term bindings, one-shot release timestamp and terminal disposition. The exact terminal credit is

```
creditId = H("TAIKO_SEAT_BOND_CREDIT_V1" || u256(marketChainId) ||
             address20(market) || trancheId || u8(OWNER|PENALTY))
```

and stores immutable beneficiary and amount plus one claimed bit. A terminal transition requires `disposition=NONE`, moves the exact bond from escrow to exactly one owner or penalty credit, and writes the disposition atomically. Claim marks the exact credit before transfer and never erases history.

Event	**Offer location**	**Tranche usage**	**Bond disposition**
Accept	NONE → PENDING	create OFFER	NONE
Displace/purge	PENDING → NONE	OFFER → CLOSED_UNINSTALLED	OWNER_CREDITED
Request pending exit	PENDING → NONE	OFFER → CLOSED_UNINSTALLED	NONE until delay
Finalize pending exit	unchanged	unchanged	OWNER_CREDITED
Stage	PENDING → STAGED	OFFER → STAGED	unchanged

Ordinary/lineup expiry	STAGED → PENDING	STAGED → OFFER	unchanged
Migration cancellation	STAGED → NONE	STAGED → CLOSED_UNINSTALLED	OWNER_CREDITED
Install	STAGED → NONE	STAGED → INSTALLED	unchanged
Healthy release	unchanged	remains INSTALLED	OWNER_CREDITED
Breach enforcement	unchanged	remains INSTALLED	PENALTY_CREDITED

A pending exit immediately leaves ranking/capacity and sets `pendingRefundAt=exitRequestedAt+EXIT_DELAY_SECONDS`; anyone finalizes at or after equality. A staged offer cannot exit. Never-installed tranches never use installed release, and installed tranches never use pending refund. An earlier installed release request cannot defeat later valid breach enforcement.

Market accounting obeys

```
accountedMarketBalance = bondEscrow + outstandingOwnerBondCredits
  + outstandingPenaltyCredits + freePremium + reservedPremium
  + outstandingPremiumClaims
actualMarketBalance >= accountedMarketBalance
reservedPremium = sum(live PremiumReserve.reservedWei).
```

Forced ETH above the accounted amount is surplus and creates no claim.

Staging moves exactly `askWeiPerSecond*SEAT_RUNWAY_SECONDS` from `freePremium` into the exact stage reserve. Apply rekeys it to the exact seat term with no balance delta. An installed standby is UNSTARTED and earns nothing; after canonical promotion, the first noncanonical economic call derives the true start from immutable Settlement service state and may not trust a Market sentinel.

For a started service,

```
claimMaturedThrough = max(0, now - PREMIUM_CLAIM_DELAY_SECONDS)
accrueTo = max(lastAccruedAt,
  min(claimMaturedThrough, Settlement.previewPremiumCap(seatTermId),
      premiumFundedUntil))
earnedWei = askWeiPerSecond * (accrueTo - lastAccruedAt).
```

Accrual moves the checked earned value from that reserve to the immutable payout pull credit before withdrawal. The `Settlement.previewPremiumCap` notation means the authenticated `premiumCap` word returned by `seatMarketRecordV1`; there is no third selector. That word is the earliest applicable immutable close, satisfaction, objective failover, unfunded eligibility end, or ring-full recovery cap; omitted maintenance can never increase it.

For an atomic healthy close let $a = lastAccruedAt$, $f = premiumFundedUntil$, $c = \min(authenticatedCloseAt, f)$ and $m = \max(a, \min(c, now - PREMIUM\_CLAIM\_DELAY\_SECONDS))$, with saturating subtraction and $a \le c \le f$. It allocates

```
maturedWei  = ask * (m-a)
tailWei     = ask * (c-m)
unearnedWei = ask * (f-c).
```

The transaction credits matured value, returns only unearned value and retains the tail as CLOSED_TAIL until `c+PREMIUM_CLAIM_DELAY_SECONDS`. Canonical/asynchronous closes

make no Market call; later permissionless reconciliation authenticates the permanent Settlement cap, credits earned value and returns only proven unearned value. Bond terminalization and duty-cell reclamation are forbidden while a reserve remains.

9.3 Continuous reverse auction and service clocks

Pending offers sort by ascending `askWeiPerSecond`, `eligibleAtTimestamp`, `eligibleAtBlock`, `quoteSequence`, then operator address. A fifth offer must strictly beat the worst under this full order or revert without retaining funds; displacement creates the exact owner bond credit atomically. Stale target/generation offers cannot rank or displace.

Requote is pending-only and requires the immutable operator, exact live offer/tranche, no exit, exact active authorization/generation, nonzero payout and an ask within the immutable maximum. It may lower the ask, or retain the ask while changing payout; increases and no-ops revert. It keeps the tranche and bond but derives checked fresh quote sequence and offer ID and resets both maturity clocks. It performs no ETH movement or Settlement call.

Direct and promoted primary service share:

```
SEAT_RUNWAY_SECONDS >= MIN_PRIMARY_TENURE_SECONDS
  + HANDOVER_EXECUTION_BUFFER_SECONDS + SLA_TAIL_SECONDS
HANDOVER_EXECUTION_BUFFER_SECONDS >= HANDOVER_DELAY_SECONDS
  + STAGE_GRACE_SECONDS + T_INCLUDE_MAX_SECONDS
SLA_TAIL_SECONDS = DELTA_SLASH_LAG - DELTA_RECOVERY_LAG
minimumTenureUntil = responsibilityStart + MIN_PRIMARY_TENURE_SECONDS
premiumFundedUntil = responsibilityStart + SEAT_RUNWAY_SECONDS
serviceEligibleUntil = premiumFundedUntil - SLA_TAIL_SECONDS.
```

All operations are checked-width. Standby tenure remains separately bounded by `MIN_STANDBY_TENURE_SECONDS`. Primary tenure gates only competitive replacement and voluntary primary exit; duty completion, failover, funding expiry, ring-full vacancy and migration override it.

Anyone may call `stageBest()`. Settlement first performs its bounded leading sync. If sync changes canonical/recovery state it returns SYNCED with no Market call. Otherwise Market scans at most four offers in total order and selects the first mature, current and structurally feasible transition: direct primary into vacancy, strictly cheaper primary replacement preserving all standbys, or deterministic standby insertion when fewer than four terms exist. The first structurally feasible candidate must be fully funded from the same gross reserve snapshot; staging cannot anticipate release of the outgoing primary reserve. Candidate selection, reserve debit, PENDING-to-STAGED and Settlement stage recording are one revert domain.

Replacement handover is `max(active.minimumTenureUntil,now+HANDOVER_DELAY_SECONDS)`; vacancy fill or standby insertion uses `now+HANDOVER_DELAY_SECONDS`. Expiry is handover plus `STAGE_GRACE_SECONDS`. The stage binds target, generation, selected rank, outgoing primary or zero, offer, reserve and

```
lineupCommitment = H("TAIKO_SEAT_LINEUP_V1" || u64(lineupRevision) ||
  primarySeatTermId || standbySeatTermIds[0] ||
  standbySeatTermIds[1] || standbySeatTermIds[2]).
```

Absent ranks are zero. It also requires `stageExpiresAt+T_INCLUDE_MAX_SECONDS<=serviceEligibleUntil` for a live primary. No later quote replaces a stage or moves its deadlines.

Permissionless apply repeats leading sync and requires exact unchanged stage, lineup, authorization, generation, health, maturity and headroom. Opening recovery tombstones the stage before returning SYNCED. Apply derives the term only at its actual timestamp:

```
installRevision = checked(lineupRevision + 1)
seatTermId = H("TAIKO_SEAT_TERM_V1" || authorizationCommitment ||
  u64(seatGeneration) || offerId || trancheId ||
  u64(installedAt) || u64(installRevision)).
```

Market rekeys the reserve and binds the tranche in the same transaction in which Settlement changes the roster and revision.

Installed exit request is one-shot and operator-only. Primary exit matures at the exact stored `primaryExitAt=max(exitRequestedAt+EXIT_DELAY_SECONDS,minimumTenureUntil)`; standby exit matures at `standbyExitAt=max(exitRequestedAt+EXIT_DELAY_SECONDS, minimumStandbyTenureUntil)`. The immutable one-shot `exitRequestedAt` can be written only from zero, and a request requires `seatTermId!=selectedSuccessorTermId`; the latter is the exact term ID in the current immutable selection record, or zero when no successor is selected. Finalization is permissionless, begins with sync and removes the term only on a fresh retry if sync changed state. Until removal the bond and every attached duty remain live. Funding expiry may close service earlier; later sponsorship cannot extend an installed term or postpone its exit.

9.4 Duty, failover, release and reclamation

Service start allocates no duty. It stores one prospective canonical base and objective recovery/failover/slash thresholds. A qualifying ordinary commit at or before recovery refreshes that prospective base. After strict `now>recoveryAt`, Settlement's one four-cell pass allocates at most one duty; equality remains curable. If no duty cell is locally reusable, or the checked sequence is exhausted, the same sync closes optional seat economics to vacancy at objective `recoveryAt` and immediately opens SLA recovery. It creates no duty, calls no Market and never waits for `DELTA_FINAL_LAG`.

The pass applies objective failover/slash to existing duties before cure and then performs prospective/selection work. A commit after strict failover may still satisfy the failed-over predecessor but starts a selected successor only when its exact selection record names that duty. A commit after strict slash may advance canonical state but cannot cure BREACHED. Recovery candidates use one O(1) preflight followed by that same pass. Failed history adoption restores all state.

Selection records are immutable and bind unique selection ID, term/tranche/ offer, revision/time and source DUTY_FAILOVER or HEALTHY_EXPIRY. A healthy expiry allocates no fake duty. Every roster removal records immutable `termRemovedAt`. A selected-but-unstarted successor counts as seatless for the existing `DELTA_FINAL_LAG/G_MAX` recovery floor; a pointer can never suppress recovery or backdate liability. A usable later commit/recovery starts it with fresh thresholds, while an unusable revision vacates the lineup.

Installed release is permitted only for exact closed Settlement history, an INSTALLED tranche with no disposition, a refundable terminal duty and no live reserve. Anyone may create its one-shot request. Define

```
dispositionStableAt = dispositionAt + REORG_STABILITY_SECONDS (or 0)
```

```
evidenceSafeAt = lastLiabilityAt + EVIDENCE_DELAY_SECONDS
                 + REORG_STABILITY_SECONDS
finalizeReleaseAt = max(releaseRequestedAt + RELEASE_CHALLENGE_SECONDS,
                        dispositionStableAt, evidenceSafeAt)
reserveMatureAt = settlementCap + PREMIUM_CLAIM_DELAY_SECONDS (or 0)
ownerTerminalAt = max(finalizeReleaseAt, reserveMatureAt).
```

Permissionless finalization uses the exact authenticated `seatMarketRecordV1`, requires zero live duties and the matching term/ tranche/operator/times, reconciles and deletes the reserve, then credits only the immutable operator. For breach, `penaltyTerminalAt=max(recordedAt+ REORG_STABILITY_SECONDS,reserveMatureAt);` permissionless enforcement authenticates the exact receipt and duty binding from both historical getters, reconciles the reserve and credits only the immutable penalty sink. Neither path can overwrite a terminal disposition.

Market's monotone `isDutyHistorySafe(dutyId,seatTermId,trancheId)` requires the exact three-way binding, terminal bond disposition, no live offer/stage/roster/reserve, all timing horizons, and an identity that can never be reused. Immutable closed term and binding history remain queryable forever. Settlement's `reclaimDutyCell` begins with leading sync; if sync changes state it returns SYNCED with no Market call. Otherwise it authenticates the exact predicate and caches only a local reusable bit. Credit withdrawal is irrelevant, so a beneficiary cannot pin history.

9.5 Migration interaction

ProtocolVersionManager alone consumes delayed arm/cancel manifests. In one revert domain it writes Router ACTIVE-to-ARMED, then calls the old Settlement's manager-only completion callback. The callback rereads the complete router tuple, runs leading sync internally without a generic SYNCED early return and with READY transition explicitly disabled, recomputes its inputs, increments `seatGeneration`, closes services non-fault, excuses unresolved duties and selected successors, tombstones the stage and returns exact fixed-length `SEAT_ARMED_MAGIC` binding generation, versions, manifest and resulting seat generation. Any mismatch, short/trailing return or local fault reverts the global arm.

The two manager-only selectors are the canonical ABI selectors for

```
completeSeatMigrationArmV1(uint64,uint64,uint64,bytes32)
    returns (bytes response)
completeSeatMigrationAbortV1(uint64,uint64,uint64,bytes32)
    returns (bytes response)
```

Their four calldata words are, in order, Router generation, active version, target version and target manifest hash; every `uint64` word has zero high padding and trailing calldata rejects. The returned `bytes` payload is exactly 68 bytes:

```
bytes4 magic || u64(routerGeneration) || u64(activeProtocolVersion) ||
u64(targetProtocolVersion) || bytes32(targetManifestHash) ||
u64(seatGeneration)
```

with `magic=0x5341524d` (ASCII `SARM`) for arm and `0x53414254` (ASCII `SABT`) for abort. Full ABI returndata is exactly 160 bytes: offset `0x20`, length `0x44`, the payload and 28 zero pad

bytes. PVM validates the complete encoding and every field; raw 68-byte, short, long, nonzero-padding or substituted responses revert the Router word, Settlement callback and all local cleanup together.

Abort is symmetric: the manager consumes the matured exact cancel manifest, writes the old-version ACTIVE router word, calls the manager-only abort callback, runs post-abort sync, and requires `SEAT_ABORTED_MAGIC` bound to the canceled tuple and retained seat generation. Abort never resurrects a term, quote, duty, successor or stage. Market is not called by either canonical callback. Later Market rotation authenticates Router's immutable successor receipt, cancels the exact old stage reserve if present, purges at most four pending offers, disables the old authorization and advances one receipt per call. Historical economic claim/reconciliation surfaces remain available; only the exact ACTIVE tip can resynchronize generation and accept new offers.

10 Economics, slashing and payments

Event	Objective evidence	Effect
Same-context equivocation	Two distinct protocol-domain signatures for one key/slot/context	Slash that schedule window's tranche once; tombstone key; emit reporter entitlement.
Aggregator SLA breach	Exact duty passes its strict slash threshold uncured	Record BREACHED in Settlement; later permissionless enforcement reconciles the reserve and terminalizes that tranche once to the penalty credit.
Force-only activation	Due queue head	No aggregator penalty; open recovery.
Skip or absence	None that distinguishes fault from non-receipt	Not slashable; promise may be revoked.
Invalid proof/data	Verifier failure	Reject; no canonical or payment effect.

The release commits one canonical `taiko.slot-chain.economic-profile.v2` JSON artifact and its measurement commit. Units are not interchangeable: native amount is `wei`, native rate is `wei/second`, and builder amount is atomic units of the one profile-bound builder lease token. That token binds chain ID, nonzero address, `EXTCODEHASH` and decimals. Its fields are exactly `leasePerWindowAtomic`, `maximumBondAtomic` and `reporterRewardCapAtomic`; no generic untyped "bond" scalar is decoded. For one builder slash,

```
reporterRewardAtomic = min(reporterRewardCapAtomic, builderSlashAtomic)
builderPenaltyAtomic = builderSlashAtomic - reporterRewardAtomic.
```

Both operations are checked in that token's atomic units.

There is one top-level immutable sink table. Each entry is the typed pair `(assetId,address)`. `builderPenalty` must use BUILDER_LEASE; `dataRent`, `seatPenalty`, `forcedExpiry` and `bridgeSurplus` must use NATIVE_ETH. All sink addresses are nonzero and a nested or component-local sink is invalid; in particular DataSession cannot substitute a rent sink. Reporter reward goes only to the evidence submitter and the remainder only to the typed builder penalty sink. Seat SLA bond, premiums and ingress deposits are native ETH and never enter builder-token accounting.

A production profile must have `status=CALIBRATED`, contain every known key and no unknown key, and make every required identity/measurement nonnull. It rejects wrong unit strings, malformed decimal strings, asset-chain mismatch, zero identity, unchecked arithmetic or any failed relation, including:

```
dataSession geometry = (86400,86400,1024,2,2100,8,6)
1024 * refundableBondWei <= UINT256_MAX
refundableBondWei + baseRentWei <= UINT256_MAX
2100 * 126972 * rentPerPublishedByteWei <= UINT256_MAX
DATA_TTL >= P_PROVE_MAX + W_SETTLE_SECONDS + REORG_MARGIN_SECONDS
refundClaimWindowSeconds >= T_INCLUDE_MAX_SECONDS + REORG_MARGIN_SECONDS

PREMIUM_CLAIM_DELAY_SECONDS >= REORG_STABILITY_SECONDS

SEAT_RUNWAY_SECONDS >= MIN_PRIMARY_TENURE_SECONDS
  + HANDOVER_EXECUTION_BUFFER_SECONDS + SLA_TAIL_SECONDS

HANDOVER_EXECUTION_BUFFER_SECONDS >= HANDOVER_DELAY_SECONDS
  + STAGE_GRACE_SECONDS + T_INCLUDE_MAX_SECONDS

SLA_BOND_WEI >= MAX_ASK_WEI_PER_SECOND
                  * PREMIUM_CLAIM_DELAY_SECONDS

SLA_BOND_WEI >= MAX_ASK_WEI_PER_SECOND
                  * MIN_PRIMARY_TENURE_SECONDS
  + MAX_AVOIDED_SERVICE_COST_WEI + COLLUSION_SAFETY_MARGIN_WEI

maximumBondAtomic <= 2^192-1
maximumBondAtomic * (Q_MAX+1) < 2^256.
```

The first bond inequality prevents cheap closed-tail capital grief; the second bounds the direct subsidy when one actor fails a predecessor to promote its own maximum-ask standby. Neither claims to insure unbounded external application loss. Every other geometry, retention, proof-size, queue-solvency and fee relation in the parameter table is validated by the same checked profile decoder. The committed example remains UNCALIBRATED and therefore cannot be used to deploy or activate a release.

Equivocation evidence is idempotent per `(builder,window)`; separate windows slash separate tranches. `L_LEASE` is calibrated against at most 76 assigned slots but is not insurance for unbounded signed value. Tombstoning is atomic with the first valid slash; future snapshots exclude the key.

Evidence does not depend on retaining an obsolete `admissionRoot`. It contains two distinct low-*s* EIP-712 headers with the same recovered nonzero key, slot, tier, context ID, admission version, admission root, episode, revision and recovery ID. For tier 1 it recomputes the normal context from the same base, admission pair and arm-block anchor carried by both headers and authenticates that arm hash as canonical through EIP-2935; tier 2 recomputes the recovery context. The contract enforces at launch that the last tier-1 evidence/replay instant precedes expiry of that 8,191-block history entry. It supplies one proof under the signed admission root and one under the *current* admission root for the same address and `registrationIndex`; the latter

must be that still-retained active/liability generation. Its current stored tranche root separately authenticates the window leaf, which is `RESERVED(W)` or `LIABLE(W)`. The two complete struct hashes must differ. Signing two conflicting protocol headers in one such context is slashable even if the key was not ultimately selected at that schedule position; builders must not sign off-ticket headers. The evidence transaction derives W from the slot, requires the tranche leaf's stored window to equal W, and lands by `evidenceReplayDeadline(W)=evidenceDeadline(W) +REORG_MARGIN_SECONDS`. Evidence available by `evidenceDeadline(W)` therefore has the full margin for reorg detection and re-inclusion; first publication delayed into the replay tail has no replay guarantee. Address reuse is forbidden while that retained generation exists, so the current proof cannot select a newer incarnation. This rule keeps evidence verification fixed-depth while the liability-capacity invariant retains every slashable generation through the replay deadline.

Canonical settlement performs no reward calculation, token balance write or external call. It always emits a candidate-committed event containing the ID, beneficiary and reward class. Best-effort protocol claims use a fixed `MAX_REWARD_RECEIPTS=256` ring, selecting `uint8(uint256(candidateId))`: if that cell is empty or its prior receipt is both claimed/expired and past the reorg margin, Settlement records the three fields, commit block and `claimed=false`; otherwise it only emits the event. Ring occupancy can therefore forfeit a reward but can never revert or delay canonical state. A separate distributor verifies this stored receipt, sets `claimed=true` before transfer, and pays at most the funded class cap before `REWARD_CLAIM_WINDOW=86,400s`. Logs alone are explicitly not on-chain-verifiable entitlements. Burns are never recycled into the same episode's bounty. Normal service/data compensation likewise cannot affect candidate validity or canonical commit.

Seat premium and bond accounting follows the conservation and maturity rules in §9. Canonical progress never pushes ETH, waits for Market or infers a penalty from mere non-receipt. A breach requires the objective retained duty receipt. Each seat bond has one terminal disposition, while every premium withdrawal is backed by an exact reserve debit and delayed through `PREMIUM_CLAIM_DELAY_SECONDS`. The example economic profile remains UNCALIBRATED and is invalid for deployment.

11 Security and liveness analysis

Adversary/failure	Protocol result	Bound or residual
Invalid execution or forged state	Proof rejects.	Finalized-state safety reduces to proof-system and L1 safety.
Aggregator offline/byzantine	Objective duty recovery/failover proceeds without Market; any prover uses tier 2 or 3.	At most the configured recovery trigger plus 2,164 s under stated inclusion/proof and head-data bounds; economic enforcement is asynchronous.
No seat or no standby	State remains valid; escape has no seat dependency.	Same protocol bound; reward may be zero.
All builders collude/absent	Unsigned escape advances anchor and forced queue.	Meaningful discretionary throughput falls to forced envelopes; no builder censorship veto.

Builder equivocation	Competing signed branches; one may win normal order.	Direct per-window slash; pre-close promises remain revocable.
Builder skips/withholds body	Honest tail may be unprovable or omitted.	Not objectively slashable; eventual escape restores state progress but may revoke promises.
Data-session spam	Exact rent/bond buys at most 1,024 bounded LIVE/REFUND cells.	A funded Sybil may censor the optional session market through TTL plus refund grace; escape uses no blob session.
Missing/pruned kind-0 bytes	Unexpired head waits; after validity it is consumed as EXPIRED_NO_TX from its durable descriptor.	Adds at most seven days; a kind-1 descriptor is durable, while successful destination invocation still needs Message bytes.
V1 or V2 endpoint upgrade exploit	V1 remains untouched; every V2 source/destination Bridge is a fresh immutable non-proxy account.	A new endpoint/domain cannot redirect an old writer; every old endpoint retains its terminal/refund path.
Unauthorized source-support fill	Version manager rejects entries absent from the active delayed manifest.	At most 64 audited additions per release; no attacker-consumable lifetime cap.
Message bytes disappear	Enqueue expires to source cancellation; a pinned credit expires to destination FAILED.	The full DIRECT ETH value, execution fee and liquidity fee refund remains permissionless by credit ID; successful destination invocation is lost.
Source escrow drain	The fresh SourceBridge checks every payout against pending, user-pull and LP-pull aggregate liabilities.	DONE reclassifies the full escrow to the proof-bound LP recipient; cancel/FAILED reclassifies it to the user, and liability falls only on a successful pull.
No willing destination LP	The credit remains UNFUNDED without consuming Pool capital or blocking unrelated credits, then may expire to FAILED.	Delivery liveness is economic: permissionless tickets remove deterministic finite-reserve exhaustion but cannot compel capital at an unattractive `liquidityFee`.
Target or post-call tail fails	The only-self attempt frame reverts Pool consume, transfer, target and Bridge effects; target-failure policy runs only after the authenticated catch.	Retry keeps the same RESERVED ticket; owner-last failure releases it and appends FAILED in a second atomic journal.
Guardian pause or zero quota	Risky sends/process/retries stop; expiry, terminal verification and reserved refunds continue.	Frozen V2 proof/refund paths have no transitive pause or ordinary quota dependency.

Foreign destination replay	Local domain and `destBridge==address(this)` checks reject before status or value effects.	The immutable pin store and accumulator both bind one exact domain/Bridge pair.
Terminal proof-format change	Canonical state carries a stable application-level depth-64 vector root/count.	Source verification does not parse the destination EVM state format; later roots preserve old leaves.
Retired destination reclaim	Reclaim waits for successor receipt, Queue watermark, pin/terminal equality, zero pull liability and zero Pool reservations for the old domain.	Only unaccounted Bridge surplus reaches the pinned sink; tickets are never enumerated or swept.
Legacy call to installed V2 code	The fresh release-owned Bridge and Store remain inactive and the custom system transaction type is invalid.	Proved tx0 atomically seals Store and Bridge once; legacy V1 addresses are never redirected.
All canonical prestate copies lost	No party can construct the next EVM witness from a root alone; proof generation stops without changing canonical state.	Explicit information-theoretic availability limit; restore any root-verifiable archive copy.
Transparent-mempool front-run	First valid recovery transaction wins.	Proof statement binds beneficiary, episode, revision and outputs; copying cannot redirect payment or state.
Payment-vault exhaustion	Claim is zero or partial.	Never blocks safety or canonical liveness; economic liveness is conditional.
L1 reorg within policy	Contract state, burns and claims roll back together.	Clients replay canonical logs and proofs against new version/hash.
L1 censorship/outage	Neither activation nor proof can land.	Outside model after `T_INCLUDE_MAX_SECONDS`; no L2 protocol can force its L1 contract to execute.

The escape lane establishes strong transaction-entry permissionlessness even when the capital-ranked builder set is captured. It does not make the normal builder market egalitarian: top-64 ranking and per-window capital remain a meaningful entry barrier. The product must distinguish these two claims.

12 Implementation blueprint

12.1 L1 modules and bounded storage

- `BuilderRegistry`: protocol-lifetime non-proxy, with 64 active cells, 1,072 liability generations, 512-leaf tranche roots, tombstones, replacement counters and the versioned `admissionRoot`. Every transition is fixed-depth or constant-size.
- `ScheduleOracle`: protocol-lifetime non-proxy sealed window root/seed under the version-independent `slot-chain-seed-v2` and frozen allocation/ placement rules, plus the one-

shot delayed fork-verifier registry, in a ring covering `MAX_LIVE_WINDOWS=268`. `MAX_EARLY_SEAL_WINDOWS=8` covers the earliest snapshot geometry. In window units the capacity is at least:

```
MAX_EARLY_SEAL_WINDOWS + MAX_CANDIDATE_WINDOWS +
ceil((W_SETTLE_SECONDS + DELTA_FINAL_LAG + DELTA_TIP +
      REORG_MARGIN_SECONDS + EVIDENCE_DELAY_SECONDS) / 384) + 2
```

An entry is overwritten only after it is `EXPIRED`; the same transaction checks the release predicate and no stored normal best/round reference.

- Settlement's internal DataSession region: exclusive session ETH custody inside the immutable Settlement and a 1,024-cell FREE/LIVE/REFUND ring, at most two LIVE cells per owner, one global checked session sequence, exact live/refund/occupied counters and a live-owner-only mapping. Each LIVE cell has a derived owner/sequence ID, `bytes32peaks[12]`, checked count at most 2,100, sealed root/count, expiry and bond; REFUND reuses that cell for one bounded owner/bond/deadline claim. There is no on-chain leaf array, permanent per-owner nonce or claim mapping. Only nonpayable ordinary/migration/refund maintenance advances the cursor by exactly eight cells; payable open selects one caller-hinted FREE cell and never runs maintenance. Expiry and migration perform no callback, and surplus-sweep failure cannot gate any protocol transition.
- `ForcedQueue`: protocol-lifetime immutable non-proxy depth-64 append-only Merkle root/count/frontier, last due time, complete fixed-width per-index descriptors, deposit prefix, unconsumed escrow and pull claims. An expired kind-0 descriptor reconstructs its leaf and prefix accounting without historical calldata, while kind-1 credits remain durable. Only the router appends and only the active Settlement advances the cursor/payment interval.
- `BridgeDomainRegistry`: non-proxy append-only source, destination and frozen-endpoint-support mappings. Only bounded entries from the manifest being atomically activated by ProtocolVersionManager may be added, and destination support additionally requires a finalized canonical L2 proof of the exact registrar record; there is no delete, redirect or age-proportional iteration.
- `BridgeInboxAdapter`: exactly-once NONE/QUEUED records and caller-funded queue processing fees. It direct-reads same-L1 CreditRecord and Bridge liability state, rejects while PREACTIVE and, after router SYNCED, commits that transition, creates no record and pull-credits the still-local full caller fee before a nonreverting return. A successful append forwards the whole fee to ForcedQueue and atomically marks the permanent source record QUEUED. It is non-proxy and immutable; its runtime and constructor-fixed registry, Bridge, router and version manager plus its one-shot sealed destination-domain slot are profile/domain-bound, with no mutable target, delegatecall or arbitrary-call selector.
- `InboxApplyRouterV2` and `InboxCreditStoreV2`: the router is a protocol-lifetime non-proxy dispatcher with one global cursor and append-only registrar-authenticated domain routes; every endpoint store fixes that same router and its own Bridge/domain. Full-vector validation, at most 64 contiguous-run calls and EVM rollback make mixed-domain application atomic. Neither has an upgrade, resolver, callback or caller-selected target.
- `ProtocolReleaseAuthorityV2`: protocol-lifetime non-proxy release manifest authenticator for the manifest-derived Anchor caller, permanent reserved system-transaction origin and namespace. Its version records are one-shot and have no endpoint target.

- `TerminalDomainRegistrarV2`: non-proxy immutable version-activation coordinator bound only to the release authority and accumulator. It derives endpoint targets from the authenticated manifest and atomically seals the fresh Store, Bridge `v2Active` bit, inbox route and accumulator writer. Records are permanent, one-shot and one-to-one.
- `TerminalAccumulatorV2`: protocol-lifetime non-proxy depth-64 append-only vector with checked count/wrapped root, a 64-word frontier and the complete canonical leaf event. It has one-to-one domain/Bridge writer registration and no root setter. Its sentinel/static-read handshake derives the terminal from the Bridge; each transition stores the returned index atomically.
- Fresh immutable non-proxy SourceBridgeV2 and DestinationBridgeV2 bundles retain V2 custody, aggregate/per-credit liabilities, status, pull credits and ContextV2 identity. V1 accounts and Vaults remain untouched. Every successor uses fresh domain/Bridge/Registry/Quota accounts; none has an upgrade, delegatecall, owner or reinitializer.
- `Settlement`: permanent non-proxy state machine with `PREACTIVE`, `ACTIVE`, `MIGRATION_ARMED`, `MIGRATION_READY` and `FROZEN` states; canonical core, one normal best, recovery episode/round, seat pointer, 256-cell sequence-tagged full-core history and a 256-cell best-effort receipt ring. A delayed cutover stops opening new work and reaches MIGRATION_READY after the current transition settles, so an adversary cannot veto migration by continuously creating a best or recovery round. Rewards and burn disposal remain separate; no loop is proportional to chain age.
- `ActiveSettlementRouter`: non-proxy, no generic target setter. Its append-only version registry permanently binds Settlement address/code/profile; proof-first activation binds the exact old core, same ForcedQueue/registry/schedule objects and migration-ready state, then atomically installs the proved target poststate. Delayed exact-target abort restores the unchanged old authority before cutover. Its stable stamped two-call sync/append facade preserves old adapters, while `canonicalAt(version,sequence)` only reads exact historical cells.
- Terminal verifier: immutable, read-only and unpausable; it checks one fixed 64-sibling vector proof against an exact router-routed version and sequence cell. The leaf separately binds index, domain, Bridge, credit and terminal status.

The non-authoritative `CanonicalArchive` is independently mirrored off-chain canonical bodies, receipts, state-trie nodes and runtime bytecode indexed by canonical block, state root and code hash. Every object is commitment-verifiable; no archive address, signature or availability vote appears in Settlement or the verifier statement.

All setters enforce parameter inequalities atomically and are disabled after the launch freeze except through the existing delayed governance path. Every counter has an explicit width and checked increment; no sentinel shares a legal value.

12.2 Historical authentication

Native `blockhash` plus a supplied matching RLP header authenticates a fresh normal arm block; EIP-2935 authenticates later normal-context evidence and schedule carrier headers. A recovery round captures its immediately preceding block hash with `blockhash(block.number-1)` and later hashes a supplied RLP header to that stored value; it never tries to read an unavailable parent timestamp, queries an expired history entry or mutates the meaning of

a past header. Its queue pair is round storage, not an MPT claim. EIP-4788 plus SSZ branches authenticates the schedule's parent beacon/execution payload; its state root plus MPT proofs authenticates the registry. Deployments lacking either primitive require an equivalently verified oracle and separate review.

12.3 Executable execution profile

`executionProfileHash` is not a free-form governance label. Each `protocolVersion` maps immutably to one canonical CBOR profile whose bytes are published with the verifier key and hashed as `H("slot-chain-execution-profile-v1"||u32(byteLength)||profileBytes)`. The decoder rejects unknown keys, duplicate map keys, nonminimal integers, indefinite lengths and zero code hashes. The profile contains:

Every profile contains exactly one immutable `MigrationTransitionVerifierDescriptorV2`:

```
struct MigrationTransitionVerifierDescriptorV2 {
  address verifier; bytes32 runtimeHash; bytes32 configurationHash;
  bytes32 verifyingKeyHash; bytes32 proofSystemId;
  bytes32 publicInputSchemaHash; bytes4 selector;
  uint32 maximumProofBytes; uint64 verificationGasLimit;
}
```

All fields are nonzero, `selector` equals `bytes4(keccak256("verifyMigrationTransition(bytes, uint256[2])"))`, and the two hash layers are acyclic and byte exact. First, `configurationHash` is the canonical verifier-instance configuration:

```
MIGRATION_VERIFIER_CONFIG_TYPE_LITERAL =
  "MigrationTransitionVerifierConfigV2(bytes32 verifyingKeyHash,"
  "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
  "uint32 maximumProofBytes,uint64 verificationGasLimit)"
MIGRATION_VERIFIER_CONFIG_TYPEHASH = keccak256(
  concat(the three quoted fragments))
configurationHash = keccak256(abi.encode(
  MIGRATION_VERIFIER_CONFIG_TYPEHASH, verifyingKeyHash, proofSystemId,
  publicInputSchemaHash, selector, maximumProofBytes, verificationGasLimit))
```

It deliberately excludes `verifier`, `runtimeHash` and itself; the immutable verifier getter is exactly

```
migrationVerifierConfigHashV2() external view returns (bytes32)
MIGRATION_VERIFIER_CONFIG_READ_GAS = 50000
```

Its selector is the first four Keccak bytes of `"migrationVerifierConfigHashV2()"`. PVM and Router first require `EXTCODEHASH(verifier)==runtimeHash`, then issue a zero-value `STATICCALL` with exactly those four calldata bytes, no suffix and the fixed 50,000-gas stipend. Success returns exactly one 32-byte word equal to `configurationHash`; empty, short, trailing, faulting or mismatched returns reject before any state write. Second,

```
MIGRATION_VERIFIER_DESCRIPTOR_TYPE_LITERAL =
  "MigrationTransitionVerifierDescriptorV2(address verifier,"
```

```
  "bytes32 runtimeHash,bytes32 configurationHash,bytes32 verifyingKeyHash,"
  "bytes32 proofSystemId,bytes32 publicInputSchemaHash,bytes4 selector,"
  "uint32 maximumProofBytes,uint64 verificationGasLimit)"
MIGRATION_VERIFIER_DESCRIPTOR_TYPEHASH = keccak256(
  concat(the four quoted fragments))
migrationVerifierDescriptorHash = keccak256(abi.encode(
  MIGRATION_VERIFIER_DESCRIPTOR_TYPEHASH, verifier, runtimeHash,
  configurationHash, verifyingKeyHash, proofSystemId,
  publicInputSchemaHash, selector, maximumProofBytes,
  verificationGasLimit))
```

Each type literal is the byte concatenation of its displayed fragments with no inserted whitespace or newline. These are ordinary canonical ABI words: address occupies the low 20 bytes; `bytes4` occupies the high four bytes; unsigned integers occupy the low bytes; every unused byte is zero. The profile decoder recomputes both layers and rejects noncanonical padding or any mismatch. PVM independently recomputes them while validating the complete manifest and requires its `migrationVerifierDescriptorHash` to equal the second layer. The Router repeats those checks immediately before bounded verifier dispatch and uses only the descriptor recovered from the pinned profile. The profile hash, release manifest and append-only historical registration therefore commit one identical descriptor; no constructor default, caller-supplied descriptor or global verifier exists. The verifier's separate implementation configuration hash is not a synonym for the descriptor hash. A successor may rotate the descriptor only by activating a new profile and release. Golden/reject vectors replace each field in turn and cover zero values, short/long getter returns, caller-substituted configuration, nonzero narrow-field padding and the former self-referential fixed-point construction; all must reject except the one exact canonical vector. The same profile records nonzero `supportedL1BlockGasLimit` and `worstCaseActivationAdoptionGas`. The latter is the measured cold-state upper bound for the *complete* L1 Router execution after transaction intrinsic gas: canonical ABI decode/re-encode and allocation, every cold Router/Queue/history/profile/manifest read, statement and transaction hashing, `EXTCODEHASH`, the configuration getter and its call overhead, verifier dispatch consuming the full `verificationGasLimit` including EIP-150 and memory/call overhead, and every post-verifier callback/write in the atomic Router/Queue/old-Settlement/new-Settlement/seat/receipt suffix. It is not a post-verifier-only estimate or a governance number detached from the release bytecode. A benchmark that omits any prefix, call overhead or suffix cannot populate this field.

The public-input schema is the exact static Solidity tuple below. The two core hashes are the Appendix canonical hashes of the complete supplied base and output `CanonicalCore`; neither hides a caller-selected partial tuple.

```
struct MigrationTransitionStatementV2 {
  uint256 settlementChainId;
  address activeSettlementRouter;
  bytes32 routerRuntimeHash; bytes32 routerConfigurationHash;
  uint8 transitionKind; uint64 migrationGeneration; uint64 sourceProtocolVersion;
  uint64 targetProtocolVersion; uint64 sourceCanonicalSequence;
  bytes32 executionProfileHash; bytes32 targetManifestHash;
  bytes32 candidateDigest; bytes32 baseCanonicalHash;
  bytes32 outputCanonicalHash;
  address forcedQueue; bytes32 queueRuntimeHash;
```

```
  bytes32 queueConfigurationHash; bytes32 queueRoot;
  uint64 queueCount; uint64 startCursor; uint64 endCursor;
  bytes32 forcedDescriptorCommitment; address proofBeneficiary;
  uint256 anchorNumber; bytes32 anchorHash;
  bytes32 forceRoot; uint64 forceCutoff;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  bytes32 sourceBridgeExecutionHash;
  bytes32 releaseSystemCalldataHash; bytes32 inboxSystemCalldataHash;
  bytes32 releaseSystemTxHash; bytes32 inboxSystemTxHash;
  uint8 releaseSystemTxPosition; uint8 inboxSystemTxPosition;
  bytes32 importedHeaderHash; bytes32 importedStateRoot;
  bytes32 legacySignalCheckpointHash;
  bytes32 deploymentCommitment;
  uint64 preInboxLastAppliedPlusOne;
  uint64 postInboxLastAppliedPlusOne;
}
```

The typehash is Keccak of the exact canonical Solidity signature with the field names and order shown, and `statementHash=keccak256(abi.encode(TYPEHASH,fields...))`. `publicInputSchemaHash` is Keccak of that same signature. The plus-one cursor words use zero only for an uninitialized lifetime InboxApply router and otherwise encode the checked block number plus one. L1 derives every L1-authoritative field from current Router/Queue/history state or exact bounded calldata. `transitionKind` is exactly GENESIS_IMPORT=1 or VERSION_MIGRATION=2. For GENESIS_IMPORT, the imported header hash must equal the legacy Inbox's frozen finalized-block hash; its state root equals `importedStateRoot`. The checkpoint hash is exactly `H("slot-chain-legacy-checkpoint-v1"||u48(header.number)||importedHeaderHash||importedStateRoot)` and the circuit proves the legacy SignalService record at that height has those values. For VERSION_MIGRATION, `importedHeaderHash`, `importedStateRoot` and `legacySignalCheckpointHash` are all canonical zero; the base core and state root are derived exclusively from Router's exact sequence-tagged current Canonical. Mixing a legacy witness into a later migration, or zeroing one for launch, rejects. The sole deployment public input is derived as follows:

```
componentsHash = keccak256(abi.encode(manifest.components))
prestatePolicyHash = H("slot-chain-deployment-prestate-policy-v2" ||
                       u8(transitionKind))
poststatePolicyHash = H("slot-chain-deployment-poststate-policy-v2" ||
                        u8(transitionKind))
DEPLOYMENT_COMMITMENT_TYPEHASH = keccak256(
  concat(
   "DeploymentCommitmentV2(uint8 transitionKind,",
   "uint64 targetProtocolVersion,bytes32 targetManifestHash,",
   "bytes32 destinationInfrastructureHash,bytes32 componentsHash,",
   "bytes32 poolConfigurationHash,uint64 retirementQueueCount,",
   "bytes32 prestatePolicyHash,bytes32 poststatePolicyHash)"))
deploymentCommitment = keccak256(abi.encode(
  DEPLOYMENT_COMMITMENT_TYPEHASH, transitionKind, targetProtocolVersion,
  targetManifestHash, manifest.destinationInfrastructureHash,
  componentsHash, manifest.poolConfigurationHash, queueCount,
```

```
  prestatePolicyHash, poststatePolicyHash))
```

`componentsHash` encodes the ten three-word rows in manifest order with no dynamic offset. Router derives every field and supplies the same one hash as both statement input and verifier expectation; there is no caller-selected or "observed" hash. The type literal is the single concatenation shown, with no display newline or extra whitespace.

For GENESIS_IMPORT, the prestate-policy tag means exact lifetime code/config, empty Router routes/cursor, empty release/domain/writer records, accumulator empty root/count/frontier, Pool zero domain/ticket/reservation/accounted totals, fresh Store/Quota/Bridge private state and arbitrary forced ETH surplus. For VERSION_MIGRATION it means exact lifetime code/config and byte-for-byte preservation of every old route/release/domain/writer, Inbox cursor, accumulator frontier/root/count and Pool ticket/reservation/accounting word, with only the new release/domain/writer/route absent and the new Store/Quota/Bridge fresh. Both poststate tags mean the exact tx0 manifest/count is sealed, only those new append-only entries and endpoint activation are added, all old lifetime state is preserved, no Pool ticket/reservation is created and no ETH is moved or minted. The circuit proves each enumerated account/code/storage and balance relation directly under authenticated pre/post roots and proves the complete output core. It also proves the deployment policy encoded by the single `deploymentCommitment` holds, `queueCount=retirementQueueCount`, positions are zero and one, and the two transaction hashes are legacy Keccak of the exact type-`0x7f` RLP envelopes reconstructed from the separately bound calldata hashes, fixed targets, kind, nonce, gas, zero value and destination chain ID.

Verifier calldata is exactly

```
verifyMigrationTransition(bytes proof, uint256[2] digestLimbs)
    external view returns (bytes32 statementHash)
```

with ABI head `[0x60,high128(statementHash),low128(statementHash)]`, then the canonical bytes length/data/padding tail. Proof length is at most the descriptor bound. ActiveSettlementRouter makes one descriptor-gas-bounded `STATICCALL`, after checking `EXTCODEHASH` and configuration, and accepts exactly 32 return bytes equal to its reconstructed `statementHash`. The circuit recombines both unreduced 128-bit limbs and proves this exact typed preimage. Revert, OOG, short/trailing return, wrong hash, substituted code/key/schema or a Boolean-shaped result rejects before any state write. The target Settlement never receives a verifier address, result capability or callback; its authority begins only after Router validates and atomically adopts the exact result.

The commitment dependency graph is acyclic and normative:

```
kernel/component configs/runtime descriptors
  -> Bridge execution/infrastructure/domain
  -> canonical profile bytes -> executionProfileHash
  -> ReleaseManifestV2 -> releaseManifestHash
  -> destination registration commitment
```

The profile may contain the manifest schema version, manager/authority/registrar addresses, namespaces, ABI and activation/validation rules. It MUST NOT contain the release-manifest hash, registration commitment, destination-registration commitment, or any field derived from them; otherwise decoding rejects rather than attempting a hash fixed point. In particular

it contains one nonzero `poolNamespace`, distinct from the router, manifest and destination namespaces. That primitive profile value is an input to component-9 `poolConfigurationHash`; it is not derived from the Pool address, manifest or destination domain. The Pool address/configuration then participates in the infrastructure and domain hashes in the direction shown above, so either side can independently reconstruct the graph without a fixed point.

- settlement/L2 chain IDs, both genesis hashes, fork digest, EIP-2935 address and protocol-lifetime `firstV2BlockNumber`; BuilderRegistry, ScheduleOracle, its protocol-lifetime schedule-rules hash and every fork-verifier address/code/config/gas tuple, domain-registry and version-manager addresses, runtime hashes and exact immutable/configuration commitments; every append-only source, destination and support descriptor and manifest schema/validation binding; each nonzero domain namespace, fresh immutable SourceBridge/Registry/Quota and DestinationBridge/Store/Quota bundle runtime/config/layout, CREATE2 factory/salt/init-code derivation, and the ban on every V2 proxy, owner, upgrade and delegate target, terminal verifier, immutable settlement router, its append-only ingress authorization rules, immutable ForcedQueue runtime and exact configuration hash, the kind-0 ingress adapter and Bridge inbox adapter, InboxApply router, InboxCredit Store, release authority, terminal registrar and accumulator runtime plus every constructor immutable, infrastructure hash, ABI, CreditRecord/liability/status/index layouts and direct-call gas limit;
- gas limit, EIP-1559 elasticity/change denominator, blob target/update fraction, empty ommers/withdrawals/requests constants and exact derivation of base fee, blob fields, logs bloom, randomness, parent beacon root, requests hash and every post-fork header field;
- the Anchor, InboxApplyRouterV2, InboxCreditStoreV2, the release authority, terminal registrar and accumulator, and both legacy SignalService contracts, fresh source/destination Bridge bundles, Bridge-unique quota managers and NativeLiquidityPoolV2 address/runtime/configurati every DIRECT-only V2 selector and ABI tuple; the unsigned custom system transaction type and unforgeable reserved senders; system nonce rules, fee exception, chain-ID equality, caller/origin injection, warm set, intrinsic gas, refund rule, zero GASPRICE, account non-transition, exact typed trie values, one-shot Store/Bridge seal and gas ceilings;
- the exact AnchorV4 checkpoint/ancestors transition, forced disposition rules, InboxApplyRouterV2 global cursor, route and batch transition, immutable store inbox slot, result and deadline storage, Bridge V2 aggregate and per-credit liability, both endpoint statuses, terminal indices, vector count, root, 64-word terminal frontier and full leaf event, pull credits, cancellation, expiry, delivery and recall transitions and the sole inbox-pin path; and
- at least six complete vectors, each covering parent, prestate, input, transaction, receipt, header and poststate: empty escape, valid kind 0, expired kind 0 without bytes, kind 1 under two separately registered immutable source generations, source and destination domain replacement, direct-record absence and mismatch, enqueue and cancel ordering, router-SYNCED refund and retry, capacity race, delayed append, PREACTIVE kind 0 and kind 1 rejection, legacy attempts to invoke inbound V2, first post-cutover release activation, pin survival across arbitrary legacy SignalService upgrades, endpoint-support manifest authorization/finality and finalized L2 registration proof, squatting/unauthorized/out-of-order registration, mixed D1/D2/D1 routing, old-domain enqueue after new-domain activation, missing-route atomic rollback, multi-credit batch ordering/conflict, Message loss before enqueue and after pin, deadline expiration, DONE/FAILED terminal replay and a cap-limited multi-block prefix, plus extra-reserved-sender, unauthorized

clone, mismatched/local-foreign domains, pooled-balance drain attempts, zero ordinary quota during V2 refund, legacy SignalService pause during V2 terminal proof, forbidden upgrade/delegatecall, exact legacy-send compatibility, DIRECT-only V2 and rejected Vault callers, old-domain accumulator append after new-domain activation, canonical-sequence cell collisions, accumulator-target caller confusion, sentinel/wrong-credit/duplicate-append, terminal leaf/index/root substitution, historic-count proof reconstruction, unauthorized/fake/full-core-discontinuous router switches, same-block history writes, migration-ready anti-griefing, conflicting result/terminal hash and every V1/V2 boundary.

For escape, parent hash/height/state and next fee/blob scalars come from the stored canonical core; timestamp/slot, L1 origin and forced inputs come from the round. Every remaining header field and both system transactions are a pure profile function of those values and the execution result. The circuit loads the profile selected by the public protocol version and proves its hash equals Settlement's immutable mapping. Launch tooling rejects a version lacking the CBOR artifact, verifier-key binding or complete vectors. Thus a profile change is a new protocol version and review, not runtime interpretation.

12.4 Reorg rule

The canonical L1 log is the journal. A reorg truncates and replays, in order: canonical tuple, mode/version, normal arm/activation/best/deadline/frozen admission, episode/round, seat duties/services/selections, retained breach receipts, Market credits/reserves, forced cursor, data-session root/count, and reward eligibility. Off-chain workers index by `(settlementChainId, contract,blockHash,logIndex)` and discard orphaned jobs. Withdrawal/bridge execution waits for the configured L1 finality policy.

12.5 Genesis and migration

Deployment occurs at least `D_SNAP+F_FINAL+sealMargin` before activation, so first live sources are post-deployment and sealable. New Settlement begins `PREACTIVE`; registry and scheduling preparation plus off-chain data preparation may run, but target session open/post/seal, every candidate and every kind-0/kind-1 forced-ingress or bridge enqueue call rejects before creating target-local state, a deposit, adapter record or queue leaf.

The legacy Inbox does not share this design's queue or expose `freezeAndExport`. Migration therefore never pretends to adopt its blob-backed forced-inclusion records. A staged legacy upgrade first disables new legacy forced admission, then drains every admitted legacy record only through the legacy protocol's native consume/expiry/void rules. Existing records do not identify a general refund owner, so this design promises neither a fabricated refund nor conversion. Any separately governed claim mechanism must complete before migration and is outside this adapter. Activation is blocked until the audited native legacy predicates report an empty live queue and zero unsettled forced escrow. Only then is the new empty `ForcedQueue` placed behind `ActiveSettlementRouter`; its initial root/count/cursor and `dueAt(0)=UINT64_MAX` are golden migration values. Existing Taiko bridge messages remain on the legacy SignalService/Bridge retry path and are not silently converted into kind 1. Only new, explicitly escrowed inbox-credit messages use `BridgeInboxAdapter`; its direct-read descriptor binds the existing message hash, both domains, source epoch, permanent Bridge/execution identity, source emission block, enqueue deadline, sender, fee, ownership/value and derived calldata

hash/length fields, while destination invocation remains a later bridge operation.

After that drain, migration has three phases. First, before quiescence, the legacy L2 governance path deploys the profile-exact protocol-lifetime non-proxy InboxApplyRouterV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2 and NativeLiquidityPoolV2, then one fresh endpoint InboxCreditStoreV2, DestinationBridgeV2 and Bridge-unique QuotaManager. The global router's constructor fixes only the permanent system sender, registrar and namespace. The store fixes that router, fresh Bridge and registrar, but its local-domain slot remains zero; Bridge `v2Active`, private mappings, liability and nonce remain zero and its quota begins full. All component addresses are precomputed from the deployment coordinator's fixed CREATE nonce sequence recorded in the manifest, so constructor references do not require an address/init-code fixed point. It also installs any profile-required AnchorV4 runtime. Legacy SignalService and the entire V1 Bridge/Vault graph remain untouched. The L1 side permissionlessly deploys the manifest-pinned fresh SourceBridgeV2/BridgeCreditRegistryV2/ QuotaManager bundle through the exact CREATE2 factory, plus TerminalSignalVerifier and BridgeInboxAdapter. ProtocolVersionManager directly checks the deployed L1 components 1–3 and their constructor configuration, records those exact descriptor rows in the finalized manifest, computes the acyclic combined infrastructure/domain descriptor, seals the L1 adapter's domain once and burns that setter. The L2 release proof carries those same rows; its registrar directly checks local components 4–10 and recomputes the identical combined hash. Every V2 account is immutable and non-proxy. InboxApplyRouterV2, the Store and every L2 inbound-V2 process/retry/fail/expire path require the fresh Store/Bridge one-shot seal; only proved tx0 may set it. No separate activation gate exists. This is an ordinary proved L2 state transition; no L1 transaction is claimed to mutate an imported L2 state root. Its transaction bytes, receipts and resulting account/storage values are golden profile vectors.

Second, anyone advances the old Inbox until its finalized transition is quiescent; the gate requires the next proposal ID to equal the last finalized proposal ID plus one, the old core's finalized block-hash field, an empty native forced queue and no unsettled escrow. The caller supplies the canonical RLP of that final L2 header and fixed-depth MPT account/storage proof witness under its state root for the exact AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2, NativeLiquidityPoolV2 and the fresh Store/QuotaManager/ DestinationBridge runtime code hashes; exact non-proxy configuration/layout and the absence of any reachable upgrade/delegate target; InboxApplyRouterV2 permanent sender/registrar/namespace, `nextQueueIndex=0` and empty routes; inbox-store router/Bridge/registrar immutables, zero domain and empty mappings; empty release-authority/registrar records; accumulator empty domain writer, count/root/frontier; Pool zero tickets/reservations/accounted balances and no registered domain, with any forced ETH classified only as surplus; fresh Bridge zero nonce/liability/private mappings and `v2Active=false`; full quota; and arbitrary authenticated Bridge surplus. The proof follows the exact immutable account and constructor-bound reference topology encoded in the profile and rejects an unlisted hop, delegate target or mutable upgrade path. Every runtime and authenticated value must equal the immutable execution profile. These values are witness to the one immutable migration-transition verifier, not the output of a separate adapter or a stored Boolean. That verifier requires the header hash to equal the frozen legacy finalized hash, `number<UINT48_MAX`, and the legacy SignalService checkpoint at that exact number to equal its block hash and state root. Its typed public statement binds the header/state/checkpoint and the single deployment commitment. No

second verifier, behavioral getter or store-and-trust stage is accepted.

Third, the delayed launch manifest arms a reversible QUIESCENT phase. It stops new legacy proposal/forced ingress only after the old boundary above is exact, while retaining the old Inbox's authority. From the authenticated header and frozen execution profile the immutable migration circuit constructs the target proof's base fields exactly:

```
l2BlockNumber       = header.number
tipHash             = H(RLP(header))
tipSlot             = header.timestamp - GENESIS_TIMESTAMP
stateRoot           = header.stateRoot
messageCursor       = 0
winningDataCommitment = H("slot-chain-migration-data-v2" ||
                          u256(settlementChainId) || u256(l2ChainId) ||
                          tipHash || stateRoot || terminalRoot ||
                          u64(terminalCount))
nextBaseFee         = profile.nextBaseFee(header)
nextExcessBlobGas   = profile.nextExcessBlobGas(header)
terminalRoot        = authenticatedAccumulator.root
terminalCount       = authenticatedAccumulator.count
firstV2BlockNumber    = header.number + 1
```

It rejects a pre-genesis timestamp, checked-addition overflow, missing fork-required header fields or any profile mismatch. Before cutover, an untrusted prover supplies the exact migration-transition-verifier proof for the first V2 block from this base. The block number is `firstV2BlockNumber`, and the block must begin with the custom AnchorV4 activation transaction for the staged `(protocolVersion,releaseManifestHash)`. The circuit authenticates the staged manifest commitment under an already published L1 anchor, the exact imported parent, execution-profile hash, empty depth-64 queue and exact L2 component rows 4–10. Its top-level sender is the reserved Anchor-system sender. AnchorV4 first calls `ProtocolReleaseAuthorityV2.activateRelease`; the authority verifies both `msg.sender=manifest.anchorV4` and `tx.origin=reservedAnchorSystemSender`. AnchorV4 then calls `TerminalDomainRegistrarV2.activateDomain`; the latter authenticates the fresh Store/Quota/Bridge and Pool state, transfers no native liquidity, then atomically seals Store/Bridge, inbox route, Pool domain and accumulator registration before that block's InboxApplyRouterV2 batch. The custom transaction type is invalid under the legacy fork and its reserved sender has no ordinary signing key, so neither a legacy block nor an EOA can preactivate the release. Any first block that omits, duplicates, calls authority directly or through Bridge, substitutes an Anchor, or mismatches this transition is unprovable. All calls occur in one EVM transaction, so any failure rolls release and registration back together. Subsequent blocks in that candidate require the exact release seal and cannot repeat activation.

The proof has no input derived from the future cutover transaction or its block hash. The L1 coordinator instead compares the proof's exact base, generation/profile/manifest and queue snapshot with live state immediately before adoption. This removes an otherwise circular requirement to prove a commitment that can exist only after the proof is available.

Only after this proof verifies does one migration-adapter transaction freeze legacy proposal/ingress, authenticate the singleton queue and empty BridgeInboxAdapter, register the new Settlement, adopt the proved poststate as its sequence-zero canonical history cell with `canonicalizedAtBlock=block.number`, store the protocol-lifetime `firstV2BlockNumber`,

consume the launch target and make the router ACTIVE. It does not write L2 code or storage; it adopts the proved L2 poststate. Proof failure, bad component configuration or proving unavailability leaves the old authority intact. A separately delayed launch-cancel manifest is permissionlessly executable while QUIESCENT and restores legacy operation without changing the imported checkpoint; a canceled generation cannot later activate. L1 kind-1 ingress is enabled only after the authenticated L2 code/authorization proofs and router-history authorization have all succeeded. That atomic cutover enables kind-0 ingress, but V2 DIRECT send and kind-1 ingress remain disabled. The old `sendMessage` selector and all L2-to-L1 sends retain exact V1 signals and events for every caller.

After the activation block becomes canonical and final, anyone submits its version/sequence and bounded `MptProofV1` registrar proof to BridgeDomainRegistry. Only that confirmation plus 214 L1 blocks enables `sendMessageV2` and kind-1 ingress for the exact tuple. No V2 Vault selector exists. Thus no source credit can target an unregistered or merely proposed L2 endpoint.

Every later Settlement/profile migration uses a protocol-generated two-phase cutover. The delayed manifest arms a specific cutover generation and changes the gate from ACTIVE to ARMED. This is one protocol-lifetime gate word owned by the non-proxy `ActiveSettlementRouter`, not a caller-supplied quiescence Boolean or replaceable contract. Its exact state is

```
(uint64 generation, uint64 activeProtocolVersion,
 uint64 targetProtocolVersion, bytes32 targetManifestHash, uint8 phase)
```

where phase is ACTIVE, ARMED or READY. The `activeSettlementStateV1()` getter above derives `activeSettlement` from the append-only registration for `activeProtocolVersion`; no second mutable current-address slot exists. Only ProtocolVersionManager may arm `generation=current+1` for the exact current version, greater target version and active delayed manifest. Before arm, the manager checks every target address, runtime/configuration hash, object identity and L1 component that is knowable without executing the target L2 transition; stale, skipped, public or mismatched calls reject atomically. Settlement, registry, data-session, reward/preparation and both forced-ingress adapters all read that same router-owned word. Arming atomically closes new normal contexts/candidates, sessions, data posts, builder reservations and reward work; ingress returns SYNCED before retaining a deposit or queue write. Every canonical new-work entrypoint rechecks phase ACTIVE after `sync()`. The session open/post/seal sidecars are the sole exceptions: none calls sync, each checks current target and ACTIVE atomically, and the payable open/post calls revert with value on failure. Every ARMED cleanup and READY write returns a non-continuation status. Reaching READY sets `changed=true`, so no call can fall through and reopen normal work. Already-mature claim-only withdrawals remain callable.

ARMED and READY are reversible because authority has not moved. A separate cancel manifest binds the exact generation, old version, target version and target manifest hash and receives the same governance delay as an upgrade. After that manifest matures, `abortMigration` is permissionlessly executable. It requires the old Settlement and queue still authoritative, marks that generation canceled, clears the target fields and returns the old version to ACTIVE without restoring deleted sessions or an already-settled candidate. The same transaction runs one bounded old-version `sync()`: a due head or stale tip opens ordinary recovery, otherwise normal work may open again. A canceled generation can never cut over, and the next arm uses the next generation. Thus a bad deployment, unavailable proof or

abandoned upgrade cannot strand the protocol in ARMED or READY. Cancellation changes no reservation, canonical, queue, claim or liability state.

The old Settlement finishes only the already-open boundary. An open normal window may commit its already-recorded best at its deadline; a newly due head cancels it. A current recovery proof may commit until that round's existing `expiresAt`; after expiry `sync()` cancels the round and is forbidden to increment its revision or open another episode. If neither is open, arm clears a merely armed normal context immediately. At and after that boundary, admissions remain closed and every cleanup call scans exactly eight consecutive physical cells from the authenticated cursor. Empty and occupied cells both consume a step; every scanned LIVE cell becomes REFUND regardless of expiry, and no owner or sink is called. At arm, the generation freezes one objective refund deadline as saturating `uint64` addition of the 86,400-second claim window to the later of `armedAt` and the already-open normal deadline or recovery `expiresAt`. Every cell converted by that generation uses this same deadline; a keeper cannot extend it by delaying one scan. While any normal or recovery boundary remains open, no ARMED LIVE cell may be converted or claimed. A scan never converts and forfeits the same cell.

The manager's arm transaction writes the current Settlement's exact generation and this objective refund deadline before invoking any internal `sync()` that can observe the router's new ARMED phase. That write, the router phase change and the leading sync share one EVM revert domain. Thus an arm with LIVE cells and no open boundary can immediately perform its first migration scan, while any injected failure restores the pre-arm gate, deadline, cursor, cells and counters together; an ARMED scan can never observe an unset generation deadline.

The cursor advances by eight and makes `changed=true` even if the scan found no LIVE cell. Thus 128 uninterrupted migration scans cover every cell from any starting cursor; public refund maintenance cannot interleave and steer the cursor while ARMED. READY requires the source-local `liveCount` zero, not `refundCount` or `occupiedCount` zero. Claims may clear REFUND cells concurrently without changing the cursor. Once READY or FROZEN, the historical source's permissionless maintenance scans only overdue REFUND cells, and the unpausable claim/surplus surfaces remain live. Abort never restores a converted REFUND, claimed cell or forfeiture; unscanned LIVE cells remain live when ACTIVE returns, and a later generation cannot double-refund an old cell. BuilderRegistry, ScheduleOracle, generation records, tranche rings and reservations are protocol-lifetime objects shared by identity across every Settlement. Existing `RESERVED` tranches remain byte-for-byte `RESERVED`; migration never counts, scans, refunds, converts or releases them. Arming only closes creation of new reservations until ACTIVE returns. Ordinary expiry, replacement, exit, slashing and evidence-safe `RESERVED->LIABLE->RELEASED` transitions continue under their existing rules and are not migration cleanup. This keeps all already-funded current/future schedules usable after cutover and prevents a version change from manufacturing either a refund or a liability. READY is derived solely from zero live-session count, no normal boundary and no recovery; no calldata supplies those facts. The process preserves all schedule and slash/evidence state and leaves matured reward/refund claims callable on permanent claim-only surfaces. It then enters MIGRATION_READY before another `sync()` decision. A due queue head is not drained or discarded; old adapters receive SYNCED while their caller fee remains in the adapter pull-claim ledger and retry after activation.

READY is written only by the exact cross-contract handshake below:

```
migrationReadinessV1() returns
```

```
  (bytes4 magic,uint64 generation,uint64 activeProtocolVersion,
   uint64 targetProtocolVersion,bytes32 targetManifestHash,
   uint8 boundaryState,bool localArmComplete)
markMigrationReadyV1(uint64 generation) returns (bytes4 magic)
```

The readiness magic is `0x4d525331` (ASCII `MRS1`); boundary states are exactly NONE=0, NORMAL=1 and RECOVERY=2; and its returndata is exactly 224 bytes with canonical padding. RECOVERY is derived exactly when canonical mode is RECOVERY and its exact current round exists, with no normal marker. NORMAL is derived when there is no recovery and any canonical normal boundary marker is live, including `normalArmBlockNumber`, `normalDeadline` or `normalBest`. NONE is derived only when no recovery record and none of those normal markers exists. A conflicting mode/record/marker combination makes the view revert; it can never be reported as NONE. The view derives, rather than stores, `localArmComplete=true` exactly when the shared gate is ARMED or READY for the returned tuple, `boundaryState=NONE`, the same Settlement's local LIVE count is zero, `seatMigrationLocalGeneration=generation`, the stored exact seat-arm tuple/receipt is present and matches every returned field, `migrationRefundGeneration=generation`, the frozen migration refund deadline is nonzero, and seat closure has completed. The arm callback's stored receipt phase remains ARMED permanently; the current shared gate may instead be ARMED or READY, so receipt validation compares generation, active and target versions, and target manifest hash but never compares phase. The arm callback's leading sync always uses `allowReady=false`; it writes the refund tuple before that sync and writes the seat-arm receipt/local generation only after all local closure succeeds. Consequently even an empty source remains ARMED after the arm callback and can become READY only on a later permissionless sync. The mark success magic is `0x4d524459` (ASCII `MRDY`) encoded as one exact 32-byte ABI word; the other 28 bytes are zero.

Only the exact currently active Settlement may call `markMigrationReadyV1` and only while the Router is ARMED for that supplied generation. Before its single READY write, the Router makes separate 100,000-gas `STATICCALL`s to the caller's `migrationReadinessV1()` and `dataSessionAccountingV1()` views. Revert, OOG, wrong magic, wrong exact length, noncanonical padding or any tuple mismatch reverts. The readiness view must match the Router's generation, active/target versions and target manifest, report NONE and true; the accounting view must report LIVE count zero, NOT_ENTERED guard and the registration's exact DataSession configuration hash. The Router checked-adds the returned live and refund liabilities and requires `BALANCE(sourceSettlement)` at least that total. Only then does the Router write READY and return MRDY. Settlement treats any failed call or wrong return as a revert; a successful sync returns a non-continuation changed status. Activation repeats both exact reads from that same source while READY, before any proof adoption or authority write, and requires the identical closed/zero tuple and balance/liability lower bound. Abort writes no READY state and leaves the source as the unique current Settlement. Thus no per-session global counter or caller assertion is part of the safety argument.

While READY, any prover constructs an immutable fork-local first-V2 candidate from the frozen old Canonical and live queue snapshot. It carries the exact target release/profile, complete bounded Inbox rows and raw proof bytes. The proof executes the exact release-activation call as tx0 before tx1 InboxApply, authenticates the fresh Store/Quota/Bridge and lifetime rows, and binds both raw EIP-2718 transaction hashes. The statement binds old full core/sequence, migration generation, Queue root/count/cursor/range/descriptor commitment, candidate and beneficiary, target profile/manifest, full output core, deployment

observation, exact tx0/tx1 hashes and every release effect. Proof generation does not mutate L1 or canonical L2 state.

This migration candidate is exactly one block and is subject to the release- activation geometry in the custom-system-transaction section: tx0 is index 0, tx1 is index 1, the uniquely classified maximum forced kind-0 FIFO prefix (if any) follows, and the transaction list ends there. Its discretionary count and canonical discretionary `bodyRoot` are both zero; no private tail, builder-selected coinbase, blob or second block is legal. Every included kind-0 raw transaction is recovered from its canonical L1 Queue-admission transaction calldata, checked against the leaf hash and queue index; that published L1 calldata, not a prover-private witness, is the replay preimage. The circuit derives and binds the complete transaction root, receipt root, header/block hash and execution-output commitment. The migration statement's `candidateDigest` commits that exact single-block candidate and `outputCanonicalHash/outputCore.tipHash` commits its resulting block; substitution or omission changes the verifier input. If a required canonical L1 raw preimage is unavailable to the prover, migration cannot land and the delayed cancellation path restores the old protocol instead of adopting an unreplayable state.

The L1 call uses the exact five-argument activation ABI frozen above. Calldata supplies only transition kind/generation/source and target versions/source sequence, candidate digest, output core, beneficiary, anchor pair, force cutoff, pre-Inbox plus-one value, the complete registered manifest, canonical Inbox rows, the bounded launch header when applicable and bounded proof. Router derives the base core/hash, statement digest, execution profile, Queue address/ code/config/root/count/start/end and descriptor commitment, verifier descriptor, tx0/tx1 preimages and hashes, target Settlement, domains/Bridges, registration identities and activation receipt from current storage plus those exact bytes. None is a second caller field. The call contains no Router, verifier or L2 object, capability, seal, authority handle or validity Boolean. Future landing block and timestamp are absent. Router rereads the exact READY tuple and all live L1 identities, reconstructs the statement and raw tx0/tx1 preimages, and invokes only the execution-profile-pinned immutable verifier by bounded STATICCALL before any write. Success requires exact descriptor/code/config/key/ schema, proof bounds and exactly 32 bytes equal to the reconstructed statement hash; short/long return, revert, OOG or substitution leaves all state unchanged.

Only after verification does L1 atomically freeze old Settlement, move the singleton Queue authority/cursor and beneficiary claim, register the new Settlement history, write the proved output core/sequence and activation receipt, consume the target and set the Router ACTIVE. It updates only L1 Canonical/History/Queue/receipt state. It never reads or writes the target Protocol, InboxApply, Registrar, Store, Pool, Bridge or any other live L2 object. The execution node selects the immutable fork-local poststate only after observing this exact successful L1 transition for the same candidate, core, version, sequence and Queue range. Losing, abandoned, pre-cutover and reorged transitions cannot mutate canonical L2 state; a local replay failure is a node resync condition and cannot revert a valid L1 transition.

Any L1 late fault rolls all L1 state back. A canceled generation or changed source sequence invalidates the proof; later landing remains valid only while the complete bound tuple is unchanged. Kind-0 preimage loss retains its bounded expiry/discard path. Existing builder reservations and historical V1 bridge signals keep their original semantics; migration enumerates or converts neither.

13 Parameters and enforced geometry

Parameter	Initial value	Enforced relation
L2 slot / schedule window	1 s / 384 slots	Aligned by genesis timestamp.
L2 height / timestamp	`uint48` checkpoint bound	Launch import is strictly below `UINT48_MAX`; every later header is checked consecutive and at most that maximum. Timestamp equals genesis plus signed slot.
`N_MAX`, `Q_MAX`	64 / 76	Six addresses fill 384 tickets.
`BOND_MAX`	$2^{192} - 1$	`bond*(Q_MAX+1)<2^256`.
Entry/exit/reservation ahead; moves	8 / 268 / 16 windows; 4 per window	Liability residence < 268; capacity 4×268.
`D_SNAP`, `F_FINAL`	8 / 2 L1 epochs	Strictly exceeds scan + finality + seal margin + lookahead.
`H_LOOK`	768 L2 slots	Guaranteed by snapshot geometry with one-window slack.
`G_MAX`	3,600 L2 slots	Tier-1 parent gap; equals final-lag trigger and fits within the 12-window proof cap.
`W_SETTLE_SECONDS`	1,200 s	Timestamp deadline, never L1 block count.
`DELTA_TIP`	1,200 s	At least submit inclusion + skew = 144 s after the frozen escape slot.
`DELTA_FINAL_LAG`	3,600 s	At least activation + escape offset + submit + skew = 2,164 s.
`F_L1`; `T_DEPTH_MAX`	64 L1 blocks; 900 s	Stored recovery-anchor depth and assumed maximum time to reach it.
`P_PROVE_MAX`	900 s	Exact alias of `recovery.proofTimeMaxSeconds`; used by session OPEN and recovery.
`ESCAPE_OFFSET`	1,900 s	At least depth-time bound + proof bound + margin.
`FORCE_DELAY`	1,500 s	At least settlement window plus `T_INCLUDE_MAX_SECONDS=120`.
Aggregator seat geometry	1 primary + 3 standbys; 4 waiting cells; 1 stage	`pendingCount+stagedCount<=4`; every canonical seat/duty pass is fixed at four cells.
Seat runway/tenure	UNCALIBRATED	`SEAT_RUNWAY>=MIN_PRIMARY_TENURE+HANDOVER_EXECUTION_BUFFER+SLA_TAIL`; the handover buffer covers delay, grace and L1 inclusion.

Seat premium/bond	UNCALIBRATED	Claim delay covers reorg stability; SLA bond covers maximum-ask closed-tail grief and the modeled collusive-promotion subsidy. An uncalibrated profile rejects.
Forced prefix	64 items; 1,048,576 B; 20m gas	Each item at most 131,072 B and 5m accounted gas; descriptors and one boundary remain durable.
Candidate caps	4,096 blocks; 12 windows; 1 anchor; 16 sessions; 2,100 records	Also 256 forced items, 4 MiB, 80m accounted gas.
Queue / range proof	depth 64 / at most 257 hashes	`queueCount<UINT64_MAX`; final leaf unused; canonical contiguous range proof.
Same-L1 credit read	fixed-size record / bounded staticcall	Launch source equals settlement Ethereum; registry runtime, return length, record, source generation and Bridge liability are checked directly and atomically.
Bridge domain release	at most 64 new entries/version; append-only	Only atomic activation of a delayed immutable manifest registers entries; the historical registry has no global cap, delete, redirect or reinterpretation.
Source generation support	one immutable active-profile entry	The frozen endpoint tuple is usable only after `SUPPORT_FINALITY_BLOCKS`; no automatic implementation boundary exists.
V2 launch asset scope	DIRECT ETH only	ERC20/ERC721/ERC1155 and every Vault flow remain V1; a future asset kind cannot reinterpret DIRECT.
Kind-0 validity / kind-1 enqueue	604,800 / 604,800 s	Bounds missing bytes before expiry or source cancellation.
`BRIDGE_PROCESS_TTL_SECONDS`	2,592,000 s	Starts at the immutable L2 pin; afterward anyone transitions NEW/RETRIABLE to FAILED without Message bytes.
Terminal accumulator/proof	depth 64 / 64 siblings / 2,048 B	Checked `uint64count<UINT64_MAX`; leaf commits index, domain, Bridge, credit and terminal; storage is 64 frontier words plus root/count, and a complete leaf event supports trustless proof verification with explicit archive-availability dependence.

Settlement canonical history	256 version-sequence cells	$256 > \lceil (T_{include} + REORG\_MARGIN)/12 \rceil + 2$ with at most one canonical write per L1 block; the exact version/sequence tag prevents sparse L2 heights from choosing or churning a cell, and frozen versions never overwrite again.
Registration MPT proof	66 nodes/path; 600 B/node; 80,000 B; 8m gas	Canonical RLP and exact account/storage paths under the routed canonical state root; all bounds reject before an unbounded allocation or call.
L1 migration activation	64 rows; 2,048-B launch header; at most 131,072-B proof	Let `ceil32(x)=32*ceil(x/32)` and `maximumActivationCalldataBytes= 4+82*32+(32+32*64+64*(5*32+ 32))+(32+ceil32(2048))+(32+ ceil32(maximumProofBytes))`. This is the largest reachable successful call: GENESIS may carry 64 empty- descriptor kind-0 rows plus the 2,048-byte header; VERSION_MIGRATION has an empty header and its 13m activation budget permits at most two 541-byte kind-1 rows, which is 960 bytes smaller overall. `worstCaseActivationIntrinsicGas= 21000+16*maximumActivationCalldataBytes`. Activation requires $\lceil 1.30\,(G_{intrinsic} + worstCaseActivationAdoptionGas) \rceil \leq supportedL1BlockGasLimit$. The adoption measurement already includes a verifier that consumes the full configured limit.
Body chunks	9 / 1,142,748 B	Per-body chunks / maximum discretionary bytes.
`DATA_TTL`	86,400 s	Greater than candidate, settlement, proof and reorg path.
Data refund claim window	86,400 s	Same-cell REFUND grace; at least inclusion plus reorg margin.
`REORG_MARGIN`, `EVIDENCE_DELAY`	1,800 / 86,400 s	Data resubmission and finite liability retention.
`SUPPORT_FINALITY_BLOCKS`	214 L1 blocks	Begins only after the domain's finalized canonical L2 registrar record is proven to L1; a frozen endpoint/domain cannot create credits earlier.
Live windows / sessions	268 / 1,024	Fixed rings with safe-eviction predicates and bounded GC.

EIP history / max arm age	8,191 / 255 blocks	$255 + \lceil (W_{settle} + CLOCK\_SKEW + 384 + EVIDENCE\_DELAY + REORG\_MARGIN)/12 \rceil + 2 < 8,191$; greater than every normal-context evidence and schedule path.

Launch tooling evaluates every inequality in both seconds and its native clock unit. Governance cannot set a value that violates a relation.

14 Production gates and verdict

The state-machine architecture is fixed enough for a prototype, but the repository is *not* an implementation-ready consensus specification. Its initial executable profile release bundle is absent. Before that status can change, one reviewed commit must contain all of the following normative artifacts and two independent implementations must reproduce every hash and execution result:

- canonical CBOR schema, exact profile bytes and `executionProfileHash`, plus a rejecting decoder corpus;
- an immutable manifest binding protocol version, chain IDs, activation, profile hash, verifier address/runtime hash and verifier-key hash;
- exact predeploy and immutable topology, including both fresh V2 Bridge bundles, CREATE2 factory/salt/init-code derivation, Bridge-unique Registry and QuotaManager accounts and the non-proxy/no-owner/no-upgrade rule for every V2 endpoint; the 232-byte Settlement deployment descriptor, published target init-code artifact and CREATE2 derivation, internal DataSession custody/configuration and rejection of every other payable Settlement path, `targetRegistrationHash` inclusion in delayed arm/cancel/activation/receipt records, exact configuration-read ABI and fresh-PREACTIVE rejection corpus; one exact Router-only target-canonical adoption selector, calldata/return commitment, gas stipend and negative corpus, plus a Router-wide activation lifecycle guard. Router cannot write a separate target Settlement account's storage directly; until this callback and ordering are frozen, the current activation prose is not implementable; BuilderRegistry and ScheduleOracle runtime/configuration hashes and retained object identities; terminal runtime code hashes, constructor immutables and relevant storage selectors for AnchorV4, InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2, TerminalDomainRegistrarV2, TerminalAccumulatorV2, ActiveSettlementRouter, ForcedQueue (including its depth, capacity, empty leaf, descriptor schema, router binding, accounted-liability lower bound, forced-ETH surplus behavior and `forcedQueueConfigHash`), both immutable ingress adapter runtimes/configurations and their append-only router registrations, L1/L2 legacy SignalService, fresh V2 Bridges, Bridge-unique QuotaManagers and NativeLiquidityPoolV2, including proof that no V2 upgrade selector, owner, reinitializer or delegate target is reachable, plus the exact chain placement and L1-direct/L2-direct/finalized-manifest validation trace for every infrastructure row;
- exact nonzero source/destination domain namespaces and vectors, append-only BridgeDomainRegistry authorization, BridgeCreditRegistry direct ABI/runtime/record layout, immutable source generations, Bridge aggregate and per-credit liability/status/pull-

credit layouts, per-Settlement sequence-tagged canonical-history rings and fixed-depth TerminalSignalVerifier vector grammar, 64-word accumulator frontier/event layout and append gas bound, byte-exact Bridge-kernel and destination-infrastructure descriptors, registrar/adapter/store one-shot bindings, finalized L2 registration proof, stamped two-call ingress ABI, proof-first router activation ABI/invariants, exact delayed-cancel manifest/transition, direct-call limits and valid/invalid fixed-return corpus;

- exact unsigned system transaction type and reserved sender addresses; chain-ID equality, caller/origin injection, initial warm set, intrinsic gas, refund quotient, zero GASPRICE, nonce/balance non-transition and exact typed transaction/receipt trie values; Store/Bridge release seal and cutover proof; system nonces; the one DIRECT-only additive BridgeV2 selector and bounded immutable inbox-store batch ABI, selectors and storage; fee handling, gas ceilings and all fork and header recurrence rules, including byte-exact first activation and steady-state type-`0x7f` transaction/receipt/header vectors; and
- complete positive and negative vectors for parent, prestate, input, transactions, receipts, headers and poststate, as listed in §12, including domain/source-generation changes, unauthorized clones and foreign local-domain replay, enqueue/cancel and capacity races, PREACTIVE rejection, legacy preactivation attempts, first-block release activation, direct-record absence, router-SYNCED refund/retry, old-domain routing, permanent-pin independence from legacy upgrades, processing expiry, DONE release, FAILED recall, legacy SignalService pause, zero ordinary ETH quota, pooled-balance drain attempts, rejected Vault callers, endpoint authorization, exact old-selector V1 compatibility, explicit V2 opt-in, terminal-root/count public-output binding, canonical-sequence collisions, historical proof after later appends, direct-sender/Bridge/foreign-Anchor release attempts, reserved-system-sender-to-Anchor activation, old-domain terminal append after new-domain activation, mixed old/new-domain global inbox batches and domain/Bridge proof substitutions, multi-credit batches and every V1/V2 replay path, manifest/profile dependency-cycle rejection, chain-ID overflow/local mismatch, fresh/partial/reused endpoint state, singleton queue proofs at and across the 2^{32} boundary, forced-ETH surplus, stale ingress stamps, failed target proofs/components, ARMED/READY cancellation, shared-gate identity and two complete successive migration generations.

Those values are outputs of real compiled contracts, fork code and a real circuit/verifier key. Inventing placeholders in LaTeX or Python would make the profile hash look precise while leaving clients unable to construct the same escape block. This is missing normative consensus, not an empirical launch gate. Consequently a sound implementation-ready artifact cannot be completed by document-only revision; implementation and circuit work must now produce the bundle. Once it exists and is re-audited, production deployment remains blocked until every additional gate below produces a versioned artifact and passes:

1. **Proof performance.** Independent tier-1, tier-2 and worst-case tier-3 proofs, including 4,096 blocks and 2,100 data records, finish within 900 seconds on the published minimum hardware. Peak RAM, recursion setup and failure recovery are measured; a miss requires changing parameters and another design review.
2. **Contract gas.** `sealWindow`, `postData`, normal submit, `sync`, direct source-credit enqueue, recovery submit, slash and garbage collection fit current L1 gas limits with 30% margin. Fuzzing includes empty registry, zero seat, full data session, full queue prefix and maximum history age.
3. **Cryptographic conformance.** Solidity, Go, Rust and circuit code match the Keccak/EIP-712

golden vectors, direct same-L1 Bridge registry ABI and runtime-hash checks, KZG Fiat–Shamir transcript and exact body manifest. Two independent implementations generate every vector.

4. **State-machine verification.** Stateful fuzzing models EVM rollback, same-block ordering, reorg replay, stale versions, activation/close coincidence, message consume-and-discard and storage reclamation. The executable models in Appendix D remain green.
5. **Economics.** Auction bond, per-window tranche, forced-envelope deposit, storage bond and proof credits are calibrated under blob/L1 fee spikes. The published liveness claim explicitly becomes conditional above fee caps.
6. **Operations.** At least two independent prover stacks and three independently administered canonical archives, plus public data-session mirrors, schedule sealers and recovery bots, complete a shadow-fork outage exercise. One archive serves a clean node, after deleting its state, body, receipt *and code* databases, from only the latest finalized package while the other two are removed. The exercise also removes every builder and seat, forces a user transaction, reorgs the first recovery attempt, and still reaches finality. Loss and restore drills verify that unavailable prestate is never misreported as bounded recovery. For every supported source domain, an analogous drill creates an old immutable credit record, removes every message-archive copy, exercises source cancellation, then recreates a message, orphans its direct enqueue transaction and replays it from the surviving record. It also exercises two separately registered frozen source generations, attempts and fails to deploy a wrong CREATE2 bundle or mutate an immutable V2 account, pauses legacy SignalService and zeros ordinary quotas, replays a pin through a second funded Bridge, expires an unprocessed pin and proves that a superseding profile cannot remove an old domain or endpoint entry. The exercise also deletes one terminal-log archive, rebuilds a historical 64-sibling proof from a second independently operated archive, and verifies it against a retained Settlement root/count.
7. a Foundry activation harness using the exact release bytecode and profile-maximum proof. It covers the largest reachable successful genesis call (64 empty-descriptor kind-0 rows plus a 2,048-byte header), the version- migration 62-kind-0/two-kind-1 cold path with empty header, and the configured verifier consuming its complete stipend. It measures calldata intrinsic gas separately and the complete cold Router execution as one non-overlapping bound, while also reporting decode/read/getter/verifier/call-overhead/suffix components for diagnosis; it succeeds below `supportedL1BlockGasLimit`, and demonstrates at least 30% margin by the enforced integer inequality. Any compiler, fork or storage-layout change reruns and republishes that measurement before release;
8. **External review.** Separate proof-system, Solidity, bridge and economic audits close all critical/high findings; the migration rehearsal and emergency governance path are in scope.

Verdict. *The earlier fallback architecture was not implementation-ready: it depended on a scheduled builder, could revert activation inside a rejected submission, made payment a commit precondition, and could not bind whole-window data with a six-blob cap. This revision removes those dependencies, fixes a single consensus path, binds bridge credits to their source application, and defines bytecode-complete recovery packages. Its bridge integration now uses fresh immutable non-proxy V2 bundles, keeps legacy callback applications on V1, binds the local destination, and makes V2 terminal proof/refund paths transitively pause- and quota-independent. The remaining known critical control-plane defect is explicit: Router cannot directly install canonical state in a separate target Settlement account, while the exact target-adoption callback, activation guard and rollback ordering have not yet been frozen. The design is therefore* not implementation-ready *even before the absent executable profile bundle is considered. The next design-only round must close that sur-*

face and the target-registration codecs; only then may the artifact return to the document-only implementation boundary instead of falsely promoting test fixture hashes into production consensus.

A Normative commitment encodings

All integers in packed commitments are unsigned big-endian at the named width; addresses are 20 bytes, hashes are 32 bytes, and ASCII domains have no length prefix or trailing NUL. `H` is Ethereum legacy Keccak-256. No `abi.encodePacked` call may contain a dynamic field. Raw transaction bytes are represented by `H(rawTx)` plus an explicit `u32(byteLength)`.

The canonical core/base, candidate, schedule list, sealed-session list and execution outputs use these exact encodings:

```
coreHash = H("slot-chain-core-v3" || u64(l2BlockNumber) || tipHash ||
  u64(tipSlot) || stateRoot || u64(messageCursor) || winningDataCommitment ||
  u256(nextBaseFee) || u64(nextExcessBlobGas) || terminalRoot ||
  u64(terminalCount))
baseCanonicalHash = H("slot-chain-canonical-v2" || coreHash ||
  u64(canonicalizedAtBlock))
H("slot-chain-candidate-v2" || baseCanonicalHash || u16(n) ||
  (u64(slot) || blockStructHash || blockHash || bodyRoot ||
   dataManifestRoot || u64(messageEnd))*)
H("slot-chain-schedule-list-v1" || u8(count) ||
  (u64(window) || entryRoot || seed)*)
H("slot-chain-session-list-v1" || u8(count) ||
  (sessionId || u16(recordCount) || root)*)
winningDataCommitment = H("slot-chain-winning-data-v1" ||
  candidateCommitment || sessionRefsCommitment)
H("slot-chain-outputs-v2" || stateRoot || transactionsRoot || receiptsRoot ||
  logsBloomHash || withdrawalsRoot || terminalRoot || u64(terminalCount))
```

Here $n \leq 4096$; schedule windows are ascending and sessions are ascending by ID. `blockStructHash` is the EIP-712 struct hash of the exact signed `SlotChainBlock` fields in §5.1; tier-3 recovery uses the same struct fields without a signature. These hashes are respectively the statement's `baseCanonicalHash`, `candidateCommitment`, `scheduleRootsCommitment`, `sessionRefsCommitment` and `executionOutputsCommitment`. Settlement must recompute the displayed `winningDataCommitment` before verifier dispatch and again before the canonical write; unsorted or duplicate lists reject.

The 64-cell registry leaf at index i is

```
H("slot-chain-registry-leaf-v1" || u8(i) || u8(present) ||
  address20 || u192(bond) || u64(registrationIndex) ||
  u64(effectiveL2Slot) || bytes32(trancheRoot) || u64(tombstonedAtL2Slot))
```

An empty cell has `present=0` and every following byte zero. A present cell has `present=1`; `UINT64_MAX` means not tombstoned. At tree height $h = 0, \ldots, 5$, the parent is:

```
H("slot-chain-registry-node-v1" || u8(h) || left || right)
```

There is no padding because there are exactly 64 leaves.

Admission position $i < 1,136$ is:

```
H("slot-chain-admission-leaf-v1" || u16(i) || u8(present) ||
```

```
  u8(location) || address20 || u192(bond) || u64(registrationIndex) ||
  u64(effectiveL2Slot) || u64(tombstonedAtL2Slot))
```

Location is 1 for active and 2 for liability; an empty leaf has present and all following fields zero. Positions $1,136 \ldots 2,047$ and unused positions are canonical empty leaves. Its eleven levels use `H("slot-chain-admission-node-v1"||u8(height)||left||right)`. `admissionVersion` increments before publishing the corresponding root; the pair is stored together and never reconstructed from a newer root. Tranche roots occur only in the active registry and liability storage records, not in this stable signer-identity commitment.

The omission-free ranked-entry commitment for window W has exactly 64 leaves:

```
H("slot-chain-entry-leaf-v1" || u8(rank) || u8(present) || address20 ||
  u192(bond) || u64(registrationIndex) || u64(effectiveL2Slot) ||
  u64(tombstonedAtL2Slot) || bytes32(trancheLeafHash))
```

followed by six levels of `H("slot-chain-entry-node-v1"||u8(height)||left||right)`. Empty ranks have all fields zero. The schedule circuit derives all 384 tickets from this root and seed; a schedule list cannot substitute a different set.

Each registry cell's tranche ring is a 512-leaf tree. Leaf index j is:

```
H("slot-chain-tranche-leaf-v1" || u16(j) || u64(window) || u8(state) ||
  u192(amount) || u64(liableUntil))
```

An empty leaf has `window=UINT64_MAX`, state `EMPTY=0` and remaining fields zero. Other states are `FREE=1`, `RESERVED=2`, `LIABLE=3`, `RELEASED=4` and `SLASHED=5`. A ring-index collision may be overwritten only from RELEASED or SLASHED after its absolute release predicate; the write installs the new exact window and FREE/RESERVED state atomically. `liableUntil=0` for EMPTY/FREE; reservation sets it exactly to `evidenceDeadline(window)+REORG_MARGIN_SECONDS`, and LIABILITY preserves that value. RELEASED/SLASHED preserve it until safe overwrite. Its nine node levels use `H("slot-chain-tranche-node-v1"||u8(height)||left||right)`. The snapshot supplies one inclusion proof for index `Wmod512` for every active cell, including canonical empty and `FREE` leaves.

Kind-0 forced leaf i is legacy Keccak over this exact packed sequence:

```
"slot-chain-force-user-v2" || u64(i) ||
address20(sender) || u64(nonce) || u256(l2ChainId) || bytes32(rawTxHash) ||
u32(byteLength) || u64(gasLimit) || u64(accountedGas) ||
u256(maxFee) || u64(validUntil) || address20(refundAddress) ||
u64(enqueuedAt) || u64(dueAt) || u256(deposit)
```

A kind-1 leaf is:

```
"slot-chain-force-bridge-v11" || u64(i) || msgHash ||
u256(srcChainId) || sourceDomainId || u64(srcEpoch) || address20(srcBridge) ||
bridgeExecutionHash || u64(emittedAtBlock) || destinationDomainId ||
u256(destChainId) || u64(enqueueBy) ||
address20(sender) || address20(srcOwner) || address20(destOwner) ||
u256(value) || u64(fee) || u64(liquidityFee) || calldataHash || u8(refundMode) ||
address20(refundVault) ||
```

```
refundCapsuleHash || escrowId ||
u32(byteLength) || u64(accountedGas) || address20(refundAddress) ||
u64(enqueuedAt) || u64(dueAt) || u256(deposit)
```

Its application has no expiry; no `validUntil` field exists. Launch V11 requires `refundMode=DIRECT`, `refundVault=address(0)` and `refundCapsuleHash=bytes32(0)`; these retained fixed-width projection fields cannot be reinterpreted as an asset mode.

For `forcedDescriptorCommitment`, `descriptorBytes` is exactly the corresponding leaf sequence beginning immediately after `u64(i)`; the list already supplies index and kind, so neither leaf ASCII domain nor index is duplicated inside it. Its fixed byte length is 220 for kind 0 and 541 for kind 1. The list encoding and optional first excluded boundary are defined in §8. The circuit prepends the correct leaf domain and index to reconstruct every Merkle leaf; any other length rejects.

The DIRECT-only acyclic `canonicalBridgeKernelDescriptor` is exactly 115 bytes:

```
u16(kernelVersion=2) || u8(DIRECT=1) ||
u64(MAX_BRIDGE_ENQUEUE_DELAY=604800) ||
u64(BRIDGE_PROCESS_TTL_SECONDS=2592000) ||
bytes32(bridgeV2AbiHash) || bytes32(statusTransitionHash) ||
bytes32(custodyRulesHash)
```

The three final hashes commit byte-exact selector/tuple/event grammar, status transition table and liability/reserve/pause rules in the release bundle. None may contain a source/destination domain, infrastructure hash or full `executionProfileHash`. The kernel hash is

```
bridgeKernelProfileHash =
  H("slot-chain-bridge-kernel-profile-v2" || u32(115) ||
    canonicalBridgeKernelDescriptor)
```

`canonicalSourceBridgeDescriptor` has this exact 752-byte packed grammar:

```
address20(factory) || bytes32(factoryRuntimeHash) ||
bytes32(factoryConfigurationHash) || bytes32(bundleSalt) ||
bytes32(bundleInitCodeHash) || address20(bundleDeployer) ||
bytes32(bundleDeployerRuntimeHash) ||
address20(legacyV1Bridge) || address20(sourceBridge) ||
bytes32(bridgeRuntimeHash) || bytes32(bridgeConfigHash) ||
bytes32(storageLayoutHash) || address20(creditRegistry) ||
bytes32(registryRuntimeHash) || bytes32(registryConfigHash) ||
address20(quotaManager) || bytes32(quotaRuntimeHash) ||
bytes32(quotaConfigHash) || address20(supportRegistry) ||
bytes32(supportRegistryRuntimeHash) ||
bytes32(supportRegistryConfigurationHash) ||
address20(terminalVerifier) || bytes32(terminalVerifierRuntimeHash) ||
bytes32(terminalVerifierConfigHash) || address20(pauser) ||
address20(signalService) || bytes32(bridgeKernelProfileHash) || u64(srcEpoch)
```

Define, over the literal ASCII domains shown,

```
create2(factory,salt,initHash) = low20(keccak256(
  0xff || address20(factory) || bytes32(salt) || bytes32(initHash)))
createChild(bundleDeployer,nonce) = low20(keccak256(
  0xd6 || 0x94 || address20(bundleDeployer) || u8(nonce)))
bundleDeployer = create2(factory,bundleSalt,bundleInitCodeHash)
sourceBridge = createChild(bundleDeployer,1)
creditRegistry = createChild(bundleDeployer,2)
quotaManager = createChild(bundleDeployer,3)
```

Here `0xd6` and `0x94` are the canonical RLP prefixes for the 23-byte `[address,smallNonce]` preimage. The bundle init code is the release artifact that commits the generic constructor logic plus every child creation-code/configuration primitive. Its constructor computes the three addresses from `address(this)`, performs exactly those three CREATEs at nonces 1, 2 and 3, verifies their returned addresses/configurations and returns the pinned inert nonproxy bundle-deployer runtime; any child failure reverts all three. Thus peer addresses are runtime-derived rather than embedded in the CREATE2 preimage, and a configuration change changes the bundle init hash and the entire address set without a hash fixed point or post-deployment configuration race.

Every address/hash and epoch is nonzero. The factory, bundle deployer, legacy Bridge, SourceBridge, Registry and QuotaManager are pairwise distinct; the bundle and all three children equal those derivations and have the exact current runtime and configuration hashes. Factory code/configuration likewise equals the first three descriptor fields, and the support Registry's code/configuration equals its three descriptor fields. After each runtime-hash check, every configuration-bearing source object (factory, SourceBridge, credit Registry, QuotaManager, support Registry and terminal verifier) is read only through `componentConfigHashV2()` with exactly four calldata bytes, a 50,000-gas `STATICCALL` and exactly one 32-byte return. Its selector is `0xf6c0f7d2`. Fault, empty/short/trailing return or mismatch rejects; no opaque factory receipt substitutes for these direct checks. The support registry stores the whole manifest-authorized descriptor and computes:

```
bridgeExecutionHash = H("slot-chain-source-bridge-execution-v4" ||
  u32(752) || canonicalSourceBridgeDescriptor)
```

The descriptor's `terminalVerifier`, runtime hash and configuration hash are exactly infrastructure component row 3. Its one configuration is the 21-byte `address20(activeSettlementRouter)||u8(64)` grammar below and its getter returns that row's `configHash`. It does not fold the BridgeDomainRegistry, its own address/runtime, Router runtime/configuration or any live support record into a second verifier configuration identity; those authorizations remain independent SourceBridge/Registry checks. The credit registry exposes this fixed descriptor for the source generation; there is no block-indexed executor. Direct admission checks the immutable runtime and record on the same L1. No source state root, beacon walk or caller-chosen topology proof exists in that path.

`canonicalDestinationBridgeDescriptor` has this exact 248-byte packed grammar:

```
address20(bridge) || bytes32(runtimeHash) || bytes32(configurationHash) ||
bytes32(storageLayoutHash) || bytes32(bridgeKernelProfileHash) ||
address20(inboxCreditStoreV2) || address20(terminalAccumulatorV2) ||
address20(terminalDomainRegistrarV2) || address20(quotaManager) ||
address20(nativeLiquidityPoolV2)
```

Every field is nonzero and `bridge=destinationBridge`. The account itself must be fresh immutable non-proxy code with the exact runtime/config/layout; no topology byte, implementation slot, reuse branch or delegated getter exists. It commits destination custody, terminal handshake, quota and Pool binding separately from the source descriptor. The hash is:

```
destinationBridgeExecutionHash =
  H("slot-chain-destination-bridge-execution-v3" || u32(248) ||
    canonicalDestinationBridgeDescriptor)
```

`canonicalDestinationInfrastructureDescriptor` has exactly ten component records in this order:

```
BridgeInboxAdapter, ActiveSettlementRouter, TerminalSignalVerifier,
InboxApplyRouterV2, InboxCreditStoreV2, ProtocolReleaseAuthorityV2,
TerminalDomainRegistrarV2, TerminalAccumulatorV2, NativeLiquidityPoolV2,
destination Bridge
```

Each record is

```
address20(component) || bytes32(runtimeHash) || bytes32(configHash)
```

so the descriptor is exactly 840 bytes. For kind $k = 1, \ldots, 10$ in that order,

```
configHash = H("slot-chain-component-config-v1" || u8(k) ||
               u16(byteLength) || configBytes)
```

Kinds 1–10 expose the fixed getter

```
componentConfigHashV2() view returns (bytes32)
```

A checking side requires exactly 32 bytes equal to this value under the manifest-pinned runtime hash. Empty, short, trailing or malformed returndata rejects. L1 checks rows 1–3 directly and L2 checks rows 4–10 directly; neither substitutes a Boolean result for the fixed call and hash comparison. The circuit also authenticates the fresh row-10 account/runtime/config/layout and empty private state directly under the imported state root; the getter is not a substitute for those observations. The ten `configBytes` lengths are exactly 80, 136, 21, 73, 60, 52, 80, 21, 52 and 164 bytes, respectively, and their contents are:

1. `address20(bridgeCreditRegistry) || address20(sourceBridge) || address20(activeSettlementRouter) || address20(protocolVersionManager)`
2. `address20(protocolVersionManager) || address20(forcedQueue) || bytes32(forcedQueueRuntimeHash) || bytes32(forcedQueueConfigHash) || bytes32(routerNamespace)`
3. `address20(activeSettlementRouter) || u8(64)`
4. `address20(terminalDomainRegistrarV2) || address20(inboxSystemSender) || bytes32(routerNamespace) || u8(64)`
5. `address20(inboxApplyRouterV2) || address20(destinationBridge) || address20(terminalDomainRegistrarV2)`
6. `address20(anchorSystemSender) || bytes32(manifestNamespace)`

7. `address20(protocolReleaseAuthorityV2) || address20(inboxApplyRouterV2) || address20(terminalAccumulatorV2) || address20(nativeLiquidityPoolV2)`
8. `address20(terminalDomainRegistrarV2) || u8(64)`
9. `address20(terminalDomainRegistrarV2) || bytes32(poolNamespace)`
10. `bytes32(storageLayoutHash) || bytes32(bridgeKernelProfileHash) || address20(inboxCreditStoreV2) || address20(terminalAccumulatorV2) || address20(terminalDomainRegistrarV2) || address20(quotaManager) || address20(nativeLiquidityPoolV2)`

The runtime semantics for kind 1 fix its stated initializer authority and bind the protocol-lifetime Queue code/configuration into every release. Kind 4 is protocol-lifetime and preserves its cursor and existing routes while requiring the new route absent. Kind 5 requires a fresh endpoint domain and empty pin state. Kinds 6–9 are protocol-lifetime and preserve all old append-only release/domain/writer/ticket/reservation entries while requiring only the new release/domain registrations absent. Kind 10 requires fresh immutable code/configuration, empty private state, full initial quota and `v2Active=false`; exact reuse is forbidden. Tx0 seals kinds 5 and 10 and registers the new kind-9 Pool domain atomically. It transfers no ETH. Kind 4 contains only protocol-lifetime constants and never an endpoint Store or Bridge; endpoint-specific targets enter only through the registrar's append-only route. `configBytes` excludes the derived `destinationDomainId` itself to avoid self-reference. Every component address/runtime/config hash is nonzero and equals the separately named domain/profile field. The infrastructure commitment is:

```
destinationInfrastructureHash =
  H("slot-chain-destination-infrastructure-v3" || u32(840) ||
    canonicalDestinationInfrastructureDescriptor)
```

Rows 1–3 name L1 contracts and rows 4–10 name L2 contracts. At manifest activation, L1 ProtocolVersionManager directly checks rows 1–3 against L1 code/configuration and commits all ten rows. The finalized manifest proof authenticates those L1 rows to ProtocolReleaseAuthorityV2; TerminalDomainRegistrarV2 directly checks only rows 4–10 against L2 code/configuration. Both sides recompute the same 840-byte descriptor and hash. No contract is required or permitted to treat a cross-chain address as locally inspectable code.

Each release profile also publishes the complete bounded `ProfileIngressAuthorizationV2[]` from which the manifest's `ingressAuthorizationRoot` is recomputed. Its Solidity type is fixed:

```
struct ProfileIngressAuthorizationV2 {
  uint8 kind; address adapter;
  bytes32 adapterRuntimeHash; bytes32 adapterConfigurationHash;
  address activeSettlementRouter;
  bytes32 routerRuntimeHash; bytes32 routerConfigurationHash;
  address forcedQueue;
  bytes32 queueRuntimeHash; bytes32 queueConfigurationHash;
  bytes32 sourceDomainId; uint64 sourceRegistrationEpoch;
  bytes32 sourceBridgeExecutionHash;
  uint256 destinationChainId; bytes32 destinationDomainId;
  address destinationBridge; bytes32 destinationBridgeExecutionHash;
```

```
  bytes32 destinationInfrastructureHash;
  uint256 fixedIngressWei; uint256 executionWeiPerAccountedGas;
  uint256 proofWeiPerAccountedGas; uint256 permanentWeiPerByte;
  uint256 maximumAcceptedFeeWei;
}
```

`INGRESS_AUTHORIZATION_TYPEHASH` is the Keccak hash of the exact canonical Solidity signature above with the field names and order shown. `authorizationId=keccak256(abi.encode(TYPEHASH,rowfields...))`. Kind 0 requires its three source fields and all four destination-topology fields (domain, Bridge, Bridge execution hash and infrastructure hash) to be zero but binds the nonzero destination chain, Router, Queue and exact kind-0 adapter. Kind 1 requires all source and destination fields nonzero and equal to the separately committed source Bridge and destination infrastructure descriptors; its adapter configuration independently binds the exact source Registry/Bridge and Router/Queue constructor graph. Both kinds use the same nonzero five-word fee schedule.

There are between one and 64 rows. Launch requires exactly one kind-0 row and exactly one kind-1 row; multiple rows of either kind reject even when their adapters differ. Remaining capacity is reserved for a future explicitly tagged kind and cannot duplicate or reinterpret launch roles. The decoder rejects an unknown kind, noncanonical ABI padding, duplicate adapter, duplicate authorization ID or a row inconsistent with the profile graph. IDs are sorted in ascending unsigned `bytes32` order and

```
idsHash = keccak256(sortedAuthorizationId[0] || ... ||
                    sortedAuthorizationId[count-1])
ingressAuthorizationRoot = keccak256(abi.encode(
  INGRESS_AUTHORIZATION_ROOT_TYPEHASH, uint16(count), idsHash))
```

where the root typehash is Keccak of `"IngressAuthorizationRootV2(uint16count,bytes32idsHash)"`. Concatenation contains only the fixed 32-byte IDs and has no length prefix; the typed root supplies the count. ProtocolVersionManager recomputes this value from canonical profile bytes before registration and stores an append-only `authorizationId->row` and `adapter->authorizationId` index. No caller-supplied root, row-validity Boolean or post-cutover governance bind is accepted.

The Queue exposes this exact constructor/configuration commitment:

```
descriptorSchemaHash = H("slot-chain-force-descriptor-schema-v11")
queueConfigBytes = address20(activeSettlementRouter) || u8(64) ||
                   u64(UINT64_MAX) ||
                   H("slot-chain-force-empty-v2") || descriptorSchemaHash
forcedQueueConfigHash =
  H("slot-chain-forced-queue-config-v1" || u16(93) || queueConfigBytes)
```

Its account runtime hash and this config hash are the two row-2 fields above. The depth-64 vector empty leaf is `H("slot-chain-force-empty-v2")`. A parent at height $h = 0,\ldots,63$ is `H("slot-chain-force-node-v2"||u8(h)||left||right)`. The queue commitment is `H("slot-chain-force-root-v2"||u64(count)||treeRoot)`; leaves at indices `count..2^64-1` are empty. The canonical range proof recursively visits the smallest tree covering `[start,end]` (including a boundary leaf when `end<count`); it emits in DFS order

exactly one hash for each maximal outside subtree. The verifier rejects extra/missing nodes, a nonempty padding leaf, or a reconstructed root/count mismatch. Append at old index i starts with the new leaf at height zero, repeatedly hashes `frontier[h]` on the left while bit h of i is one, stores the carry in the first zero-bit frontier position, increments count, and reconstructs all higher right subtrees from `FORCE_EMPTY[h]`. Genesis frontier words are exactly zero. After increment, the fold is

```
node = FORCE_EMPTY[0]
for h = 0..63:
  node = bit(count,h)
    ? H(D_NODE || u8(h) || frontier[h] || node)
    : H(D_NODE || u8(h) || node || FORCE_EMPTY[h])
```

Only set-bit frontier words are meaningful; stale zero-bit words are ignored. At old count `UINT64_MAX-1`, append writes height zero and produces final count `UINT64_MAX`; old count `UINT64_MAX` rejects, so leaf index `UINT64_MAX` is unused and no height-64 frontier exists. This is the same root as the full vector and uses exactly 64 frontier words. Every leaf and descriptor list uses `u64(index)`; no queue index is narrowed to 32 bits.

The terminal vector uses depth 64. Its empty leaf is `H("slot-chain-terminal-empty-v2")`; a parent at height $h = 0, \ldots, 63$ is

```
H("slot-chain-terminal-node-v2" || u8(h) || left || right)
```

and leaf index i is

```
H("slot-chain-terminal-leaf-v2" || u64(i) || destinationDomainId ||
  address20(destBridge) || creditId || u8(terminal) || liquiditySettlementHash)
```

with `terminal=1` for DONE or 2 for FAILED. FAILED requires `liquiditySettlementHash= bytes32(0)`. DONE requires the nonzero exact `H("slot-chain-liquidity-settlement-v1"|| ||ticketId||address20(l1Recipient)||u256(value+fee))` authenticated by the pool reservation; substitution of any ticket, recipient or amount changes the leaf. The committed root is

```
H("slot-chain-terminal-root-v2" || u64(count) || treeRoot)
```

The accumulator stores only `frontier[0..63]`, wrapped root and checked `uint64count`. To append old count i, it starts with the leaf, combines `frontier[h]` on the left while bit h of i is one, stores the carry in the first zero-bit position, increments count, and reconstructs the tree root:

```
node = TERMINAL_EMPTY[0]
for h = 0..63:
  node = bit(count,h)
    ? H(D_NODE || u8(h) || frontier[h] || node)
    : H(D_NODE || u8(h) || node || TERMINAL_EMPTY[h])
```

Frontier words at zero bits are stale and ignored. The complete canonical leaf event is the proof-generation data source; any party reconstructs exactly 64 siblings in ascending height,

and the L1 verifier uses bit h of the index for left/right order. It requires `index<count`, status DONE/FAILED, exactly 64 siblings and exact wrapped root/count. Append rejects old `count=UINT64_MAX`. Carry-depth and full-fold gas remain subject to the mandatory Foundry calibration in §14.

A data-chunk leaf is:

```
H("slot-chain-data-leaf-v1" || bytes32(sessionId) || u16(index) ||
  bytes32(versionedHash) || bytes32(fullBodyRoot) || u16(blockOrdinal) ||
  u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
  bytes32(chunkRoot) || address20(publisher) ||
  u64(validUntil) || u256(z) || u256(y))
```

Each cell stores fixed `bytes32peaks[12]` and `uint16count<=2100`. Slot h is meaningful exactly when bit h of count is one; zero-bit slots may be stale. Append at old count n uses leaf index n and carry=leaf. While bit h of n is one it sets `carry=H("slot-chain-data-node-v1"||u8(h)||peaks[h]||carry)`; it stores the carry at the first zero height, then increments count. Old count 2,100 rejects.

Logical peaks left-to-right have strictly decreasing height. The single root preimage bags them rightmost-to-leftmost, which means the repeated tuples are encoded at set heights in strictly *ascending* order:

```
H("slot-chain-data-bag-v1" || u16(count) || u8(popcount(count)) ||
  (u8(h) || peaks[h]) for h=0..11 ascending where bit(count,h)=1)
```

The empty MMR uses the same preimage with zero count and zero peaks. An inclusion proof requires `index<count`, derives the unique target mountain and base from count's descending set bits, and accepts exactly target-height siblings bottom-up. Sibling orientation is derived from the local leaf index, never supplied as a caller bit. It then accepts exactly `popcount(count)-1` other peaks in the ascending bag order above. A wrong height, duplicate, missing or extra sibling/peak, inconsistent count or unused proof element rejects.

A manifest leaf is:

```
H("slot-chain-manifest-leaf-v1" || u16(position) ||
  u16(blockOrdinal) || sessionId || u16(recordIndex) ||
  u16(chunkIndex) || u16(chunkCount) || u32(chunkByteLength) ||
  fullBodyRoot || chunkRoot)
```

Its next-power-of-two tree uses

```
H("slot-chain-manifest-node-v1" || u8(height) || left || right)
H("slot-chain-manifest-root-v1" || u16(count) || treeRoot)
```

for the node and final root respectively; padding uses the domain-separated empty leaf. For zero entries, `treeRoot=H("slot-chain-manifest-empty-v1")`. The zero-discretionary- transaction body is the ordinary body encoding `u32(0)` and therefore `H("slot-chain-body-v1"||u32(4)||u32(0))`. Tier 3 requires exactly those empty body/manifest values and the zero-count session-list commitment. This tree is instantiated separately for every block. `position` starts at zero inside that block; `blockOrdinal` remains the candidate-wide block index. A two-block candidate therefore commits two signed manifest roots and never hashes their leaves into one shared tree. The InboxApply disposition commitment is:

```
H("slot-chain-dispositions-v1" || u64(start) || u64(end) ||
  u16(count) ||
  (u64(queueIndex)||u8(disposition)||u32(txIndex)||resultHash)*)
```

Here `UINT32_MAX` means no Ethereum transaction.

The recovery ID is the encoding in §7. Schedule leaf/root, seed and tie hashes are as in §4; body and blob encodings are as in §6. `commitment-model.py` is the normative executable fixture. Its initial vectors are:

```
TYPEHASH      ee6a8c8e31e8245cd527869508f6e464d6084893991203876f734d1855aed87c
registryRoot  0bf297d7b9b6a5529a319a06cb08484923a89bab15d51f8baeaf5c30bebdf3fd
admissionRoot 3bf2dcaf78292c832108e29205bf99cc2d22137a0545e4528d8da7309d4b482b
entryRoot     acee83a690b868a4a7960c55a9f7228f91cad26b704e24106d4db87e9c7a8f34
forcedRoot    a54e9f797ffe7f04dd5ca7df4c858edf02ce45a81808522633fc9cee8fe72e57
sourceDomain  3ca03af3984e7b2ece804a84fe1f80b20d1ca0c710a03b4483adfabd63a35376
destDomain    97e53e2233611efb9ee4dbb5a8d97a108b4d33ec43711b3d7250bd0c33863cf0
bridgeKernel  a23f9994e8d1c475500768b67cf2b2d1f7a0f367df6f44bac5ccf1fa12bc1338
bridgeExec    fa1e986cef0ff6d8474aeb73698573487e761635dd5999ff1cce625ddf223dd3
destBrExec    78c541f426d06c2d96470d6e9e226dc9186d7632c15c587a3199d0db2766c6c4
destInfra     1dd64cb0e4fe42cf2f54a7daa9936854c654ef1f43f5339d664a6d07c10d27d1
poolRowAlt    af9e17cc8c95267b679e9ae20ec16363a8fa06ceb4e56d63e40bd6e671a05647
mvConfigType  0acb8f9e39dd43a4208edc38c8925bbd3433e72eb46f9e5fac584bd14a970b98
mvConfigHash  969a782016029488dab2179a992f96b0331759058a03d0105c6a157de09e206d
mvDescType    6f1be44c261607b4c2345985bfb5a081aabd621b9168982df9b7820c7dedebd6
mvDescHash    3717fd77ba3254853408d20623f95df4c88b5e6abc20cf436797b7155a5fb965
mvConfigSel   476b9aef
relTypehash   603555ff1d82bc9012b0e0c8a36df28e154b16cbd89be4eed9d01228c502965b
activateSel   28f73572
relManifest   b19d98a5c4494c4b37d5b84da44b1217fb66284376f6fd8b377e22b9dfa1f869
destReg       7e05192d70ce3bc37609173005691b2cfac0d7965a58969e342c3a849ec98c62
migStmtType   1c99db64554200a004b213159ac780ebbde4379622f59ad73f8f503afcbb2fe1
deployType    bcfcbbdc64e08793950be4f8cd516833c105b8aff92ed7f5482e7aa122df709e
components    1ea18ec884a6b6589f5ffc6b26ed65ba1c2e1b1c2a236d01b56870a10599517c
genesisDeploy d280ae426b49d9b76957272ec052ade6c5fb3689cb0422422a73cd6ed5d1f71f
versionDeploy 050582303e9ccf3b2caed29c1f534b20072933a6d5438389638d0d70ffcbb007
legacyCkpt    6ab84b0c77035309ac300830aa1c31d95f68c44b697d97a6836f7f3f54bf0708
genesisStmt   437db4c7add4914d0663065bb6637e858a41af59b98299347a983f717bc92df4
versionStmt   bfb307caed765013a6dc050dd0bfe21d7e86702cdfa6cb2714b567dc1b923b74
regStmtType   c049f967468e58f1a5c9b9e1a147dfc233695ae69c5d4a95ec4ffb49b5687da0
regRouteType  4368ad9403b46ef3830e21af8cddcaedbd444c8c57bc3414ebc6fdd250e1e6da
regRoute      7ff8e1d53c3ddb482fd789fa3b29c92cdc5fd42aed2151b7f684c70c5c40f7c6
verifyRegSel  33639818
regStmtHash   57421732f8694364afe5d47f05bd562b536c6469e7f891ae2c1b9ba61359f819
regProof      50ac70c83c4d85e9e0790d2413e35216b0c490814ee435879d0ce27e4a12e5e5
regCfgType    38f7fbc63e45f650bd5cbaffba0d81d5ca69f21ebdddfa74280d7e6eb5c319d6
regCfgHash    b266045c553f010d052a847ea18459bb268cbaacde008574e8cf4c738453911f
regDescType   9533e2f6ac9bcf6830306a9ef5d14e21a06008f9ceb59208d09a8d7ab1e6100f
regDescHash   99491cce6d0ea603e0b9862bd3a2509953ea05e50e9243f8086a19aa92b2f1bf
regCfgSel     59bfe418
```

```
verifyRegCall 5254ee30ab9239869c34664d1cc4c6945615c21dc8434bd76d438c418d92ae36
verifyRegLen  516
srcCtxType    6069dff5f628f94ceff984d5ce3ef62019eaeb4efd7e0627c45a14677bd13c70
srcCtxHash    c76e3ae4a8d80df05787562d494ca2b66681d840c7bc6a870a8349bb125daf27
dstCtxType    b8170dbf684e1fc4dd4dae8fb78ad24984cc0aa0c03cbd1e29b1b2eb5728eefb
dstCtxHash    019ecba422e0615a07edc6e2ba112d112e35ed77e9004489025ac6a8b5764e52
ingAuthType   a2dccbd60c366ade3f5af12af640f3453bbb4106405ef8da5dd52484db8f2a82
kind0Auth     31e0887e3c9b8e063d322fccda4f8d01d2be91a8c8f0ee20d8cd9984ee972aaf
kind1Auth     1cee9bf0f46a9f58aa5e47ee5fc2c0d1cb865f1cd70ce1353aa25497c38b272e
ingRootType   c7b11126d8d1984cc17cbc108be2a1be0ef9c4e8fd519fa9033c949962f1b042
ingRoot       e925350f82d47fa044d352e8e65a68f4641304abc0996a20985ea1ee8479ffac
sendMsgSel    9211d7e9
enqCreditSel  81805d6b
msgTuple      0e85a708462e96cbaca7158a1534011a25137c3b7aada7f381e4fc5b3afbe40d
msgData       f08683775f4a25dfef721c487073fb77026d45ac57e423424290e47af9fd2835
sendMsgCall   9099cce58d3f36714ee59f37510261394e31583f8bd55da4551b653b5a060e12
sendMsgLen    516
enqCreditCall 02db9c77025d4c2bae1d3f79849020e6d8711c4b6b6c54ec45ce9181e48cfeb4
msgPreLen     576
normalizedMsg f3010702b7b7bf10b6dbfe396a4c7ab07e7c560c28ffdbf6ff8d01d9a96ea4c7
enqueueTxSel  9f06b1b4
enqueueTxCall 3e802e6f72e50c267cc18d256c2863d04446eb8f2d47bc474bbcb9c68876d19a
enqueueTxLen  196
syncSel       6c880b72
syncStamp     1fd01b194948c635358fbb51b4a5f32f8ceab4dc4153e0230215f8afc94ee434
syncChanged   ef662a629ce07c9ed715124d8141a6e430d0a3065f8ce8074a7ea95e8751f184
appendSel     1927261d
appendK0      caabdaadee6df8cea48fb88dacad863e6e24e13f5b0c06f99408d5c71611788a
appendK0Len   388
appendK1      a1460e7adedb5855522f00a851f9025b96534ab3c6c83f513fd7841382f915de
appendK1Len   708
queuedReturn  76f821bd39721ec0e26efc55d7b667d20aab74992e0feae7d7755e386ecd694d
inboxApplySel 6b326168
inboxApply    c55aea0f32c9ad41c0999db1f4b7ba9375bb1780c9e6e28f10f182b44eba37fd
inboxApplyLen 1092
markBatchSel  a92f72cd
markBatch     30a0a90e1ff8add20c78619d04749502540c0f2b3d405da3d71bc2ff1bd4b01f
inboxMagic    49425632
routeCfgSel   4b64fa11
verifyPinSel  28933d28
getPinSel     31e85ab1
liqQuoteSel   43dc48e0
reserveSel    45a933f1
attemptSel    b3e5c861
attemptCall   2d2c0279d18df6ac41cc77e6c9e4d3190a6110ae27e8f8f0358837aef66db5dd
attemptLen    1060
finalFailSel  745dcb69
targetErrSel  f9cc2b44
```

```
appendTermSel abc194f5
termCommitSel 2c984c97
termStateSel  998c57ed
ticketId      69e7a081b31736d338fa57ad67287ae534499bd1f800bc08272703f08b9a6ec9
policyType    d702a337b74fc40bfc746fb1aeeaa705e60a95947bfc3076c76222703205b4b1
policyHash    5eb7c00399d64d91e416d5a3dbe75187c39dfc2a4645b867864b5fd9e649f3e3
policySel     b2d0e286
hookSel       7f07c947
policyMagic   49505632
l2ReceiptId   6315904b1838529ad9b66ec508ef3e2ec6e1985db23f3a20bd0b0a4b41517c27
l1ReceiptId   9c73be9f9f317489fc99cc07ef6824f48a52b371637f1a8bead0e58e7dd5555c
activateVSel  14c37693
genesisFixed  273742c984e16e7a4eac27323bbfface0e334cb90da2640dc482683114ac0c01
versionFixed  8ef8c50c81fa6748590f2affe7ac89e9d0adb4a6644d288edfa5dd2923dfc490
genesisCall   7b790b4826a04aae87637711542ba55dc17e0f4f2189a1d761406d55b5d6a0a2
genesisLen    3876
versionCall   18591525f0a1de1e05bda1dc26d8d975ce682f5c8add2fa8805a2d99f2792eee
versionLen    3812
maxGenesis    3081eac3864cafad843262459d25e63909e5b891399129bf5517da2a0b745139
maxGenesisLen 150180
maxVersion    856b4e553a4819cda95fc6971945dc818b8a7e259318dc92b79f99d3880b9284
maxVersionLen 149220
regBase       20b3dfc457e3cecf32b0c047177351f0814e426c1548e87b79f58830655810c3
regSlot       dfa6283b763bbadeb604401a78e2fefeddb72000addcdb94ed2e3de5cc69846b
regTrie       200031adff46d90b1cd5c67ff8e31098235d1dddb08ec98b0d20f5f8660c0ac8
relBase       a0b7a29a75032f37561036cd3741e7b375213309367f37b5ffec4ad55cf6154f
relSlot       719bb73ba856aeab1b203e322bfefc6d84a4c41a3222bcf1634b1b44e5b9aba8
relTrie       dac8109059d03da2ad16ac3acc50d2e58897b8c3a7f6889ae77bfb20737e87a2
routeConfig   2256dd6e98891531a80b2ee16b67a7f1d77db7e17d8523f9b5ab3fa540e9c4a3
queueConfig   72e27c19ebfab08e1fb27feeff50609c7c4bb69f0570a7bdb72eda7736c50f4e
dataSessCfg   83e7e252277ea66b70b59a490421d21454032a3fe64107c7390131f83a487b76
dsOpenSel     7bda4d11
dsPostSel     a1cc526a
dsSealSel     340e11fb
dsMaintSel    1e7a916a
dsClaimSel    fdd2b0db
dsSweepSel    9d083a2b
dsCellSel     011efada
dsByIdSel     eeaad0bb
dsAcctSel     e2a62969
activeSetSel  4a95c306
migReadySel   b36c83ce
markReadySel  e0c25827
activeSetMagic 41535231
migReadyMagic 4d525331
markReadyMagic 4d524459
dsOpenTopic   a81132592bd8a549a0bfc83415ab47fbca586f0b48fff5f6cb5bfae0a9fd1f68
dsRecordTopic 30ee2de166c53a480d028e5b94d4f8759dbd84b5f7b6af1f23e0c5889ea17f8c
```

```
dsSealTopic   f6d45ab0ecc3348b36ac48bc769f7391962fc2fa240c1c6bb43269ed3780078e
dsRefundTopic 086de7ad4d27cf66f63e9dc6ecb09c0c3122146dd43e7f480f3a875070eb984e
dsClaimTopic  69fe7d8d95811e3a02a4ad4b0d1a3e1360b683a31f7cb30b4c69546b258232fb
dsForfeitTopic f93f771339298d1fb502bd2c556fc18130b95d44599f35109de39d8d5731f957
dsSweepTopic  3a5f2fa0ab342d3b79fc2aa40be5e1296009e8fb31f86c3577c51839dd159a86
dsMaintTopic  920669b9670911aa86cd718dceebaa1372d224ca0fdac50a63dc1d45a53e1e89
bridgeLeaf    956ac8402556d0a89ca016866d374e6f69fa3fd205435aef24135287c44b9b76
feeLeafAlt    24b35a966effbd11f3ac01e523adcc1bd534144f65ec2a51e0d0082ba782394f
creditId      44c7f6349cfc851f3d08f76268d0bf63986b3c2d66ad1de8aa22c3a684e922cb
feeCreditAlt  14094883c8f217eb0db2c615ec87bff6f96295dfe5fd67963e399f767f0a7630
escrowId      c12559a405b00da5acac02621ba6f341a8d4d6adf17594254ed061d739980f6d
inboxSlot     27723fe890a8d23e0b7d0b14880f03f3c34cc646f9600a83919840f6a0bd5b89
terminalDone  1c89cc4ea4844a4d892d59a0cede83ad74b12724acd7214bc0c11cb9403e7e33
liquiditySet  625ff42ed879b94a7499fecb7abc07988b348300d1c4e9cc1f7596968eaf2f19
settleLeafAlt 8db1f7f9e7fa1c5075075286eed748933d59e22759bd0f3011a91950ad9d67b7
terminalFail  1391f7851ecffe7093f154cc399ac59891b7aa7766b32375df9a53c6d84b1d5f
terminalRoot2 121e62e05147e60c2db287e8e7d4de42fff6cdec8f51d53f6f67214457c97581
emptyTermRoot c5da197edc2f03c7023cc6afe137ccb77d01fc56514d322b6ba66a149315bcb0
bridgeResult  20b827ea751b10e09d74228c914983753c5b11def4ea5db33ba98223020161c2
manifestRoot  417be737a57e38eb410f2d6e65c77ee19d5c314cdaf432067861c6a36c6a990f
normalContext 5b93691b397d8ab377682acee7cf6cc77ff2726cd83a8a7b725084f5d6f468bd
migrationData 2c36740d76ae6192335d4c603f42edace094b33e8f54e959b40241a94c1f6deb
winningData   4ae34aa9efb842528d353b175f94191d01cfd168b5ad828f64a0e7972a2ca9e3
statementHash 5e3f9ad95b6ccfa825d4d860a724cdd4acde1951a0f7ba8649acaedd73eda2a0
recoveryId    fd0552b28542fa3e236c86807695f3d5a4bc0436add285daa8506d9a79511b15
bodyRoot      0f4e161a46c8b18c2a86f23a0a4e7169a838a12af8b389f65e97b547a99707e9
```

Release CI must reproduce every vector in Solidity, Go, Rust and the circuit.

B Normative transition ordering

Every external candidate entry point implements the following structure:

```
function armNormalContext() returns (Status) {
    if (mode == PREACTIVE) return REJECTED_PREACTIVE;
    if (sync()) return SYNCED;
    if (migrationGate.phase != ACTIVE) return MIGRATION_ARMED;
    if (mode == NORMAL_ARMED && block.number > armBlockNumber + 255) {
        armBlockNumber = block.number;
        emit NormalContextRearmed(armBlockNumber);
        return REARMED;
    }
    if (mode != NORMAL_IDLE) return IGNORED;
    armBlockNumber = block.number; // no caller-selected hash or anchor
    mode = NORMAL_ARMED;
    return ARMED;
}
```

```
function activateNormalContext(armHeaderRlp) returns (Status) {
    if (sync()) return SYNCED;
    if (migrationGate.phase != ACTIVE) return MIGRATION_ARMED;
    if (mode != NORMAL_ARMED) return IGNORED;
    age = block.number - armBlockNumber;
    require(age > 0 && age <= 255);
    armBlockHash = blockhash(armBlockNumber);
    require(keccak256(armHeaderRlp) == armBlockHash);
    freeze base, stable admission pair and parsed arm header;
    freeze minimum slot, deadline and landing horizon;
    mode = NORMAL_OPEN;
    return ACTIVATED;
}

function submit(input) returns (Status) {
    if (mode == PREACTIVE) return REJECTED_PREACTIVE;
    if (sync()) return SYNCED;     // successful transaction, never revert
    if (migrationGate.phase != ACTIVE && mode != RECOVERY)
        return MIGRATION_ARMED;
    if (mode == NORMAL_OPEN) return submitNormal(input);
    if (mode != RECOVERY) return REJECTED_NO_CONTEXT;
    return submitRecovery(input);
}

function sync() returns (bool changed) {
    if (migrationGate.phase == ARMED) {
        if (mode == RECOVERY && now > round.expiresAt) {
            cancelCurrentRecoveryWithoutRolling();
            changed = true;
        } else if (isNormalMode(mode) && forcedHeadDue &&
                   normal boundary open) {
            cancel normal window;
            changed = true;
        } else if (normal deadline matured) {
            settleOrCancelAlreadyRecordedBest();
            changed = true;
        }
        if (no boundary/recovery) {
            inspectExactly8AndConvertLiveToSameCellRefund();
            changed = true; // cursor progress counts even when all are empty
        }
        // Existing protocol-lifetime reservations are never migration work.
        if (allowReady && localLiveCount == 0 && no boundary/recovery &&
            seatMigrationLocalGeneration == migrationGate.generation &&
            exactSeatArmReceiptMatchesGate &&
            migrationRefundGeneration == migrationGate.generation &&
            migrationRefundClaimDeadline > 0 && seatClosureComplete) {
            magic = ActiveSettlementRouter.markMigrationReadyV1(
```

```
                migrationGate.generation);
            require(magic == MRDY); // Router re-STATICCALLs readiness/accounting
            changed = true;
        }
        return changed; // no context opens or rolls while armed
    }
    if (migrationGate.phase == READY) return true;
    if (mode == RECOVERY && now > round.expiresAt) {
        rollRound(); // same episode/base/cause and duty; new revision/target
        return true;
    }
    if (mode in {NORMAL_ARMED,NORMAL_OPEN} && forcedHeadDue &&
        (no deadline || normal window immature)) {
        cancel normal window;
    } else if (normal deadline matured) {
        if (best exists) {
            liveBoundaryDueAt = ForcedQueue.dueAt(best.endCursor);
            dataLive = now + REORG_MARGIN_SECONDS <= best.minDataExpiry;
        }
        if (best exists && liveBoundaryDueAt > now && dataLive)
            commit(normalBest);
        else cancel normal window;
    }
    if (isNormalMode(mode) &&
        (finalLagBreach || forcedHeadDue)) {
        openRecoveryEpisodeAndRound();
        if (finalLagBreach) run bounded duty/failover pass once;
        return true;
    }
    return changed;
}
```

No transfer, reward calculation, unbounded loop or caller-controlled external call occurs in canonical settlement. Optional reward materialization is a later transaction in a separate module.

C Rejected alternatives

1. **Heaviest fallback.** Rejected because an observer cannot prove it has seen the globally heaviest off-chain tail; transparent races can prevent a stable target. Recovery is episode-bound first-valid.
2. **Scheduled-builder-only fallback.** Rejected because the same cartel that stalls normal production also controls recovery. Tier 3 is deterministic and unsigned.
3. **Promote the pre-activation normal best.** Rejected because it was accepted under different cutoff/version semantics. Only a mature best that covers the required forced prefix closes; immature state is canceled.
4. **Atomic reward-funded commit.** Rejected because an empty seat or vault becomes a con-

sensus deadlock. Reward materialization is a separate non-authoritative transaction.

5. **One same-transaction blob set.** Rejected because Ethereum exposes at most a handful of blobs per transaction while a candidate spans thousands of blocks. Publisher-owned authenticated sessions separate posting from proof.
6. **One KZG opening at a prover-chosen point.** Rejected because it does not soundly tie declared bodies to the blob. The challenge commits to both blob hash and body root, and the circuit recomputes the evaluation and body root.
7. **Arithmetic snapshot slot without EIP-4788 parent semantics.** Rejected. A bounded forward carrier search authenticates the parent source or fails closed.
8. **Bond as insurance.** Rejected without an on-chain reservation ledger. The tranche is deterrence only.
9. **Block-indexed delegatecall Bridge executors.** Rejected because an executor shares a proxy's storage context and can overwrite implementation, liability or status. V2 instead uses fresh immutable non-proxy Bridge bundles; new semantics require a fresh endpoint and domain.
10. **Legacy SignalService as the V2 terminal verifier.** Rejected because its verification entry points are guardian-pausable and versioned. The V2 pin store is separate and immutable; the terminal verifier proves a stable application-level vector leaf against a proved canonical root.
11. **Implicit V2 behind the legacy selector.** Rejected because it silently changes events, signals, relayer/status APIs and EOA behavior, while a vault-wide interface probe cannot choose V1 per asset. `sendMessage` is always exact V1; launch V2 exposes one separate DIRECT-only selector and rejects every Vault caller.
12. **External checkpoint copier or L2-height-indexed publication ring.** Rejected because a caller can front-run or omit the copy, while a candidate may advance 4,096 L2 blocks and collide modulo 256 in one L1 publication. Each immutable Settlement writes its full canonical tuple internally under a checked version/sequence; no external publication call exists.
13. **Frozen EVM account/storage terminal proof.** Rejected because a future state-proof format or verifier defect can strand already-queued credits. Canonical settlement instead carries a profile-independent depth-64 terminal vector root/count, while governance remains only the pre-existing validity- verifier safety root.

D Executable consensus models

`lookahead-model.py` reports 36 assertions over legacy Keccak, EIP-4788 carrier/parent selection, immutable seal deadlines, post-seal tombstones, partial/empty registry sealing, eligibility-before-ranking, capped D'Hondt and feasible placement, including absolute clock conversion, execution-block finality depth, version-independent protocol-lifetime seed and complete-schedule vectors.

`settlement-window-model.py` reports 186 assertions over PREACTIVE/migration, proof-first launch/cutover, delayed exact-target abort, early-return semantics, force-due precedence, renewable recovery, no-seat escape, exact slot/header time, durable-descriptor prefix predicates, immutable anchor chronology, O(1) due boundaries, late close, sealed data sessions, frozen admission pairs, bounded replacement/liability generations, tranche release units, stamped direct bridge enqueue, forced-ETH-tolerant fee custody and capacity races, persistent SYNCED

refund/retry, enqueue/cancel ordering, source-generation/domain support, immutable destination pins, processing expiry, DIRECT ETH liability, quota-independent refunds, immutable authorization versus mutable liability, fresh V2 Bridge bundles, exact legacy-selector V1 compatibility and explicit DIRECT-only V2 opt-in, immutable pin storage, finalized registrar authorization, authenticated router switching, local destination binding, frontier/event terminal proofs, pause-independent escape, L2 release sealing, replay and timing geometry. Cryptographic booleans in that model are not cryptographic evidence.

`commitment-model.py` reports 234 golden vectors and 441 executable assertion sites for exact EIP-712 domain/struct/digest, canonical and statement hashes, registry/admission/entry/tranche commitments, forced descriptors, vector/range proofs, session MMR and per-block manifests. It also covers source/destination domains, acyclic Bridge-kernel/source-bundle, complete destination-Bridge and ten-component infrastructure descriptors, bounded chain IDs, acyclic profile-to-manifest dependencies and authenticated release manifests, the acyclic migration/registration-verifier configuration/descriptor hashes, exact configuration-getter selectors, MessageV1 and ContextV2, ingress and Store/Bridge/Pool/accumulator/policy interfaces, the static migration public input and five-argument L1 activation ABI, terminal frontier/root vectors, historical 64-sibling verification and complete credit leaves, credit/escrow/inbox IDs, terminal leaves/vector roots and atomic ordered batches, dispositions, recovery ID, Fiat–Shamir input, body chunks and blob codec. The valid KZG-opening, signature-recovery, independent-implementation and proof gates remain mandatory.

E Security invariants checklist

1. Canonical state changes only after one valid proof and only from its stored base tuple.
2. A sync transition cannot be rolled back by validation of the same call's candidate.
3. No payment, seat or standby is required for canonical commit.
4. Recovery candidates bind one live episode/revision, the current base tuple, one stored prior-block anchor at required depth and one frozen queue target. Expired rounds are permissionlessly renewable without creating a second duty, fault disposition or seat penalty.
5. A due forced head advances in the next canonical block; appends omitted by a frozen recovery round cannot become due until that round expires. Prefix count, bytes and accounted gas are independently bounded.
6. Tier 3 has no builder signature, discretionary transaction, arbitrary coinbase or blob body.
7. Every block header timestamp equals genesis plus its signed slot. Every tier-1/2 block has a valid scheduled signature under the frozen admission version/root pair, strict slot progress and the candidate's one authenticated anchor no later than its L2 time.
8. Every discretionary body byte is covered exactly once by a unique, unexpired authenticated data record.
9. Schedule inputs share one authenticated execution payload and use exact Keccak encodings.
10. A tombstoned generation cannot re-enter while retained; safe later address reuse receives a fresh registration index only after every evidence deadline. Each window has an independent slashable tranche and finite release state.
11. Candidate, window, data-session, Merkle range proof, queue count/bytes/gas, history and storage-retention work is bounded by consensus constants and safe eviction.
12. L1 reorg replay removes canonical state, penalties and credits from orphaned transactions together.
13. An armed later migration shares one generation gate across ingress, Settlement and transient ledgers, opens no new context, ends the current boundary without recovery renewal and reaches READY af-

ter at most 128 bounded cleanup calls. While old authority remains intact, a target proof executes the exact release activation from that frozen base; cutover verifies it before freezing the old Settlement, then atomically adopts its canonical/bridge output, switches the one active queue authority and returns the same router-owned gate READY-to-ACTIVE with no pending release. A separately delayed exact-target cancel restores old ACTIVE operation from ARMED or READY if no cutover occurred. The immutable non-proxy ForcedQueue, registry, schedule oracle and L1 migration gate are shared by object identity across both Settlements; incumbent reservations remain RESERVED throughout. At launch, V2 L2 entry points remain disabled under legacy rules; the first custom Anchor transaction is proved before cutover and atomically registers and seals the fresh destination bundle and route in the adopted L2 poststate. Source V2 waits for a finalized proof of that record plus 214 L1 blocks.

14. A kind-1 credit is enqueued only by direct same-L1 comparison with an append-only CreditRecord, immutable source generation, source-approved destination domain and live liability at the immutable source Bridge endpoint. Enqueue and queue deposit are atomic; a reorg removes both observation and append. Launch V2 admits DIRECT ETH only; the retained legacy refund-mode, refund-vault and capsule fields are fixed to their canonical DIRECT zero values and cannot select another path. Missing Message bytes cause bounded source cancellation before enqueue or destination expiry after the immutable pin. The exact destination domain, Bridge and terminal accumulator leaf bind recall; the complete canonical append-event history reconstructs every old proof and is retained by at least two independently operated, replaceable archives; every destination action checks its local domain and `address(this)`, and no L2 nonempty source-proof path exists. Terminal proof and reserved refund paths have no transitive guardian-pause or ordinary-quota dependency.
15. Each V2 credit keeps `value+fee+liquidityFee` as exact source liability until a pull succeeds. A destination LP reservation is backed by one non-transferable ticket; Pool balance covers aggregate unreserved plus reserved amounts, and its checked per-domain count equals the number of live reservations. Only the external only-self attempt frame may consume it. Every target/tail failure reverts consumption, while FAILED/expiry releases it and DONE commits the actual Pool-returned settlement tuple. Retired Bridge reclaim requires this domain count, pull liability and all pin/terminal gates to be zero and moves only unaccounted surplus.
16. A migration proof adopts exactly one replayable activation block with zero discretionary body. Its system transactions are reconstructed from activation calldata and every included kind-0 transaction from canonical L1 admission calldata; the proof binds complete transaction/receipt roots, header, block hash and output core. Missing preimages cause proof failure/cancellation, not adoption of a private poststate.